

AgentCore -- Universal Agentic AI Platform

Version 2.0 | The Voice and Intelligence OS for Every Customer-Facing Business

March 2026 | HelpRevX Technologies Pvt Ltd | Confidential

TABLE OF CONTENTS

- 1. Executive Summary
 - 1.1 What AgentCore Is
 - 1.2 The Core Problem It Solves (Universal)
 - 1.3 Products in the HelpRevX Portfolio (Built on AgentCore)
 - 1.4 Target Market (Primary + Platform Markets)
 - 1.5 The Universal Pain (Why Every Industry Needs This)
 - 1.6 Why This Matters Now
 - 1.7 Document Purpose
 - 1.8 Version History (V1 -> V2 Changes)
- 2. Market Analysis & Pain Points
 - 2.1 The Universal Customer Service Crisis (2025 Data)
 - 2.2 Market Size and Growth
 - 2.3 Industry-Specific Pain Points (12 Industries)
 - 2.4 Competitor Analysis (3 Tiers: Startup + Enterprise Legacy + Enterprise AI)
 - Tier 1: Retell, Vapi, Bland, ElevenLabs, Synthflow
 - Tier 2: Genesys Cloud CX, Five9, NICE CXone
 - Tier 3: PolyAI, Cognigy, Salesforce Agentforce, AWS Connect, Zendesk, Intercom, Freshdesk
 - 2.5 What ALL Competitors Lack (Unified Comparison Table)
- 2A. Enterprise-Grade Requirements Checklist
 - 2A.1 Security Certifications (SOC 2 Type II, HIPAA BAA, PCI-DSS 4.0, GDPR)
 - 2A.2 Workforce Engagement Management (WEM)
 - 2A.3 Supervisor & Live Call Management
 - 2A.4 Conversation Intelligence & Analytics (A/B Testing, Predictive Routing, Knowledge Base)
 - 2A.5 SLA Framework & Uptime Guarantees
 - 2A.6 Enterprise Implementation Timeline Guarantee
- 2B. Product Market Fit Analysis
 - 2B.1 Market Segmentation & Ideal Customer Profile (ICP)
 - 2B.2 Competitive Positioning Matrix
 - 2B.3 Go-To-Market Strategy (SMB vs Mid-Market vs Enterprise)
 - 2B.4 Product Market Fit Evidence & Validation
 - 2B.5 Competitive Gap Closure Roadmap

- 3. Product Definition -- All 24 Engines
 - 3.1 The Platform Definition
 - 3.2 Three Deployment Modes
 - 3.3 Engine Groups Overview
 - 3.4 Group A -- Communication Engines (4)
 - 3.5 Group B -- Intelligence Engines (3)
 - 3.6 Group C -- CRM Lifecycle Engines (8)
 - 3.7 Group D -- Operations Engines (6)
 - 3.8 Group E -- Vertical Extension Engines (2)
 - 3.9 Feature Comparison Matrix
 - 3.10 All 24 Engines Manifest Table
- 4. Complete Customer Journey -- Universal
 - 4.1 The Universal Journey
 - 4.2 Journey Mapped Across 5 Industries
 - 4.3 Detailed Flow with Engine Mapping
 - 4.4 Parallel Tracks
 - 4.5 Cancellation Branch
- 5. Technical Architecture
 - 5.1 The Hybrid Pipeline (Cascaded + Speech-to-Speech)
 - 5.2 The Multi-Agent Architecture (LangGraph)
 - 5.3 The 4-Layer Memory System (Detailed Architecture)
 - 5.4 AWS Infrastructure (Self-Hosted)
 - 5.5 Language Architecture (Updated with Sarvam + ElevenLabs API Routing)
 - 5.6 The Database Schema (Full SQL)
- 6. STT/TTS/LLM Stack (V2 Updated)
 - 6.1 The Decision: ElevenLabs API + Sarvam API
 - 6.2 Complete Component Selection Table
 - 6.3 Intelligent Language Routing
 - 6.4 Sarvam AI Complete Reference
 - 6.5 ElevenLabs API Direct Pricing
 - 6.6 Multilingual Chat Agent (Updated)
- 7. Vertical Deployments -- All Industries
 - 7.1 ItinerarySuite -- Travel CRM
 - 7.2 Ytrawise -- Travel Public Portal
 - 7.3 Aayiq -- Expense Management
 - 7.4 SpendPilot -- Spend Analytics
 - 7.5 PhotoAdda -- Photography CRM
 - 7.6 HelpRevX -- Internal Platform Operations
 - 7.7 Financial Services (BFSI)
 - 7.8 Healthcare
 - 7.9 Real Estate
 - 7.10 Legal Services
 - 7.11 Education & Coaching
 - 7.12 Fitness & Wellness
 - 7.13 Automotive

- 7.14 Hospitality & Food
- 7.15 Retail & E-Commerce
- 7.16 Home Services
- 8. AgentCore as a Standalone Product (Go-to-Market)
 - 8.1 Three Revenue Models
 - 8.2 Who Buys AgentCore (All 15 Industries)
 - 8.3 TAM by Industry
 - 8.4 Detailed Pricing Table
 - 8.5 Revenue Projections (Multi-Industry)
 - 8.6 The Moat (Why Hard to Replicate)
- 9. Complete Cost Model (V2 Updated)
 - 9.1 Per-Minute Cost for Each Language Path
 - 9.2 Weighted Average
 - 9.3 Monthly Costs at 5 Scale Tiers
 - 9.4 AWS Hosting Costs Breakdown
 - 9.5 Cost Per Tenant (at 100 Tenants)
 - 9.6 Gross Margin Analysis
 - 9.7 Sarvam Savings vs V1 Estimates
 - 9.8 Cost by Industry Use Case
- 10. Compliance & Governance
 - 10.1 DPDP Act 2023 Requirements
 - 10.2 TRAI Compliance
 - 10.3 Data Residency
 - 10.4 Kill-Switch Framework
 - 10.5 Audit Trail
 - 10.6 HIPAA Eligibility
 - 10.7 Industry-Specific Compliance Notes
- 11. Implementation Phases (2.5-Month Timeline)
 - 11.1 Phase 1: Core Voice + CRM + Proposal Engine (Month 1)
 - 11.2 Phase 2: Intelligence + S2S + Chat Multilingual + Invoice Engine (Month 2)
 - 11.3 Phase 3: Full Platform + All New Engines + Load Test (Month 3)
- 12. Appendix -- Data Tables
 - 12.1 Channel x Object Matrix (Universal -- All Industries)
 - 12.2 All 24 Engines Comparison Table (V1 vs V2)
 - 12.3 Language Support Matrix (All Providers, All Languages)
 - 12.4 Component Cost Summary
 - 12.5 V1 vs V2 Changes Log
 - 12.6 API Response Time Targets
 - 12.7 Scaling Reference
- 13. Claude Implementation Prompts
 - 13.1 How to Use These Prompts
 - 13.2 Phase 0 -- Audit Existing Codebase
 - 13.3 Phase 1 -- Database Foundation
 - 13.4 Phase 2 -- LiveKit Voice Agent Service

- 13.5 Phase 3 -- Inbound Call Dispatch API
- 13.6 Phase 4 -- WhatsApp Engine
- 13.7 Phase 5 -- Memory Service & Lead Engine
- 13.8 Phase 6 -- Proactive Outbound Engine
- 13.9 Phase 7 -- Real-Time Agent Dashboard
- 13.10 Phase 8 -- Document Intelligence
- 13.11 Phase 9 -- AWS Infrastructure as Code
- 13.12 Phase 10 -- Integration & End-to-End Testing
- 13.13 Phase 11 -- Proposal Engine (NEW)
- 13.14 Phase 12 -- Advanced Chat Agent (Live Agent + Callback)
- 13.15 Master Environment Variables
- 13.16 File Tree of All New Files
- 14. Enterprise-Grade Implementation Phases (Addendum) [Phase 13 Prompts]
 - 14.1 Phase 13.0 -- Vanta / SOC 2 Compliance Automation
 - 14.2 Phase 13.1 -- HIPAA BAA Implementation
 - 14.3 Phase 13.2 -- PCI-DSS Compliance for Payment Flows
 - 14.4 Phase 13.3 -- Workforce Engagement Management (WEM) Module
 - 14.5 Phase 13.4 -- Conversation A/B Testing Framework
 - 14.6 Phase 13.5 -- Predictive Routing Engine
 - 14.7 Phase 13.6 -- Knowledge Base Management Engine
 - 14.8 Updated Phase Summary Table (All Phases 0-13)

Section 1: Executive Summary

1.1 What AgentCore Is

AgentCore is a self-hosted, universal agentic AI platform that gives any customer-facing business an intelligent voice, chat, and WhatsApp brain -- without replacing their existing CRM, without vendor lock-in, and without per-minute markups.

AgentCore is not built for one industry. It is built for every industry where humans talk to customers. It is the voice and intelligence infrastructure layer on which HelpRevX Technologies builds ItinerarySuite (travel), PhotoAdda (photography), Aaiiq (expense management), SpendPilot (spend analytics), Ytrawise (travel portal), and HelpRevX internal operations. And it is the layer that any other CRM company, enterprise IT team, or contact center operator can license to add voice AI capabilities to their own products.

AgentCore is built on LiveKit Agents, Deepgram Nova-3, GPT-4.1, OpenAI Realtime API, ElevenLabs Flash v2.5 API, Sarvam AI, and LangGraph multi-agent orchestration. It is not a chatbot. It is not a simple IVR replacement. It is a full-stack intelligence platform that combines real-time voice conversation, multi-agent orchestration, 4-layer persistent memory, document intelligence, proactive outbound engagement, WhatsApp commerce, and deep CRM integration into a single, self-hosted deployment.

The Two-Sentence Pitch:

"AgentCore is the voice brain for every customer-facing business. Build it once, power every industry."

This single platform powers all six HelpRevX products while simultaneously being licensable to external CRM builders, contact center operators, and enterprise IT teams across Financial Services, Healthcare, Real Estate, Legal, Education, Fitness, Automotive, Hospitality, Retail, Logistics, Home Services, Event Management, and more -- creating a compounding moat where every new vertical deployment makes the platform smarter, more battle-tested, and more valuable.

1.2 The Core Problem It Solves (Universal)

A tour operator, a photographer, a banker, a doctor, a lawyer, a gym owner, a car dealer, a hotel manager -- ALL have the same problem:

- Calls go unanswered after business hours. Potential customers call, nobody answers, they call a competitor.
- Agents manually type notes. Every conversation requires a human to listen, understand, and transcribe key details into a CRM.
- Customers repeat themselves on every interaction. A patient who called yesterday about test results calls today and must re-explain everything from scratch.
- No proactive outreach. A policy renewal is due in 15 days, but nobody calls the customer because agents are too busy with inbound queries.
- No multilingual support. A Tamil-speaking patient calls a Bangalore clinic and gets an English-only receptionist who cannot help.
- Quotes and estimates are sent too slowly. A customer asks for a photography package quote, and it takes 4 hours for a human to assemble and send it. By then, they have already booked elsewhere.
- Follow-up is inconsistent or forgotten. A gym member stops attending for 3 weeks. Nobody notices. Nobody calls. The member lapses.

AgentCore solves ALL of these for ANY business in ANY industry, in ANY of 30+ languages, across voice, chat, and WhatsApp simultaneously.

1.3 Products in the HelpRevX Portfolio (Built on AgentCore)

AgentCore serves as the shared intelligence layer for the following products:

Product	Vertical	Role of AgentCore
ItinerarySuite	Travel CRM	Lead qualification, proposal generation, package quoting, visa assistance, booking support
Ytrawise	Travel Public Portal	Consumer-facing voice search, destination recommendations, booking
Aayiq	Expense Management	Expense capture via voice, policy compliance, approval workflows
SpendPilot	Spend Analytics	Spend summary queries, anomaly flagging, budget alerts

Product	Vertical	Role of AgentCore
PhotoAdda	Photography CRM	Studio booking, package quoting, trial session scheduling
HelpRevX	B2B Supply Platform	Supplier status checks, tenant escalation, platform operations

These six products demonstrate AgentCore's versatility across radically different business domains -- from travel to photography to finance. But these are just the first six. The platform is architecturally designed to support unlimited verticals through its configurable tool registration system, tenant-level customization, and industry-agnostic engine design.

1.4 Target Market

PRIMARY MARKET (Direct HelpRevX Products)

These are products built and sold directly by HelpRevX, powered by AgentCore:

- **ItinerarySuite:** Tour operators, DMCs (Destination Management Companies), travel agencies, B2B travel platforms, travel portals
- **PhotoAdda:** Professional photographers, photo studios, event photographers, wedding photographers, commercial photography businesses
- **Aayiq:** SMEs, startups, and corporates needing voice-first expense management and receipt capture
- **SpendPilot:** Finance teams, CFOs, procurement teams needing conversational spend analytics

PLATFORM MARKET (AgentCore Licensed to Other Builders)

These are industries where AgentCore is sold as a platform product to CRM builders, IT teams, and operators who build their own applications on top:

FINANCIAL SERVICES (BFSI):

Banks, NBFCs (Non-Banking Financial Companies), insurance companies, mutual fund distributors, lending platforms, wealth management firms, stock brokers, payment companies - any financial service where customers call to query account status, apply for products, check loan status, file insurance claims, or get investment advice. India's BFSI sector handles over 2 billion customer service calls annually. AgentCore can automate 60-80% of these.

HEALTHCARE:

Clinics, hospitals, diagnostic laboratories, pharmacies, telemedicine platforms, dental clinics, veterinary clinics, physiotherapy centers, mental health practices, Ayurvedic and wellness centers -- any health service where patients book appointments, query lab results, request prescription refills, or follow up after procedures. India has 1.4 million registered doctors and over 70,000 hospitals. Patient communication is almost entirely manual.

REAL ESTATE:

Property agents, real estate developers, housing project sales teams, commercial leasing

companies, property management firms, co-living operators, PG/hostel operators -- any real estate business handling property enquiries, scheduling site visits, managing tenant communications, or processing rental applications. 80% of property enquiries come outside business hours.

LEGAL SERVICES:

Law firms, corporate legal departments, independent advocates, notary services, legal-tech platforms, mediation services, IP firms, tax consultants -- any legal practice taking client queries, scheduling consultations, managing case status updates, or collecting documentation. Most law firms in India have zero automated client communication.

EDUCATION & COACHING:

Coaching centers (IIT-JEE, NEET, UPSC, CAT), test prep companies, online tutors, K-12 schools, colleges, universities, EdTech platforms, vocational training institutes, language schools -- any education business handling admission enquiries, fee payment reminders, course completion guidance, or parent queries. Admission enquiry volume spikes seasonally, overwhelming staff.

FITNESS & WELLNESS:

Gyms, yoga studios, CrossFit boxes, personal trainers, physiotherapy clinics, spas, salons, beauty studios, Pilates studios, swimming academies, martial arts dojos -- any wellness business booking sessions, managing memberships, tracking attendance, and retaining members. Member retention is the biggest challenge; AI can detect inactive members and re-engage them proactively.

AUTOMOTIVE:

Car dealerships (new and used), authorized service centers, independent workshops, fleet management companies, EV charging networks, car rental companies, two-wheeler dealerships, auto parts retailers -- any automotive business handling test drive scheduling, service reminders, vehicle availability queries, or insurance and finance conversations during purchase.

HOSPITALITY & FOOD:

Hotels, resorts, homestays, boutique properties, restaurants, cloud kitchens, catering companies, banquet halls, food delivery aggregators, cafes, bars, QSRs (Quick Service Restaurants) -- any hospitality business managing reservations, room upgrades, local experience recommendations, menu queries, or post-stay feedback collection.

RETAIL & E-COMMERCE:

D2C (Direct-to-Consumer) brands, physical retail stores, multi-brand retailers, online marketplaces, subscription box companies, wholesale distributors -- any retail business handling order status queries, return initiation, product recommendations, abandoned cart follow-ups, or loyalty program management. Order status queries alone constitute 40-60% of all support calls.

LOGISTICS & DELIVERY:

Courier companies, freight brokers, last-mile delivery operators, warehousing companies, 3PL (Third-Party Logistics) providers, moving companies, cold chain operators -- any logistics

business tracking shipments, scheduling pickups, handling delivery complaints, or managing B2B shipping documentation queries.

HOME SERVICES:

HVAC technicians, plumbers, electricians, home cleaning services, pest control companies, renovation contractors, painting services, appliance repair, locksmith services, landscaping companies -- any home service business taking job bookings, scheduling technicians, providing service quotes, or handling customer complaints.

EVENT MANAGEMENT:

Wedding planners, corporate event companies, venue management businesses, entertainment booking agencies, exhibition organizers, conference management firms -- any event business handling enquiries, sending proposals, managing bookings, and coordinating logistics.

NGOs & GOVERNMENT:

Citizen service centers, social organizations, government helplines, municipal corporations, public grievance cells, charitable trusts -- any public service entity handling citizen queries, grievance registration, scheme information, or document processing.

1.5 The Universal Pain (Why Every Industry Needs This)

Every customer-facing business, regardless of industry, suffers from the same set of problems. The table below maps 8 universal pain points across 12 industries to demonstrate that AgentCore's value proposition is truly horizontal:

Pain Point	Travel	Photography	BFSI	Healthcare	Real Estate	Legal
Calls missed after hours	[YES]	[YES]	[YES]	[YES]	[YES]	[YES]
Manual note-taking	[YES]	[YES]	[YES]	[YES]	[YES]	[YES]
Customer repeats context	[YES]	[YES]	[YES]	[YES]	[YES]	[YES]
No proactive outreach	[YES]	[YES]	[YES]	[YES]	[YES]	[YES]
No multilingual support	[YES]	[YES]	[YES]	[YES]	[YES]	[YES]
Slow quote/estimate	[YES]	[YES]	[YES]	[YES]	[YES]	[YES]
Inconsistent follow-up	[YES]	[YES]	[YES]	[YES]	[YES]	[YES]
No unified memory	[YES]	[YES]	[YES]	[YES]	[YES]	[YES]
Pain Point	Education	Fitness	Automotive	Hospitality	Retail	Logistics
-----	-----	-----	-----	-----	-----	-----
Calls missed after hours	[YES]	[YES]	[YES]	[YES]	[YES]	[YES]

Pain Point	Travel	Photography	BFSI	Healthcare	Real Estate	Legal
Manual note-taking	[YES]	[YES]	[YES]	[YES]	[YES]	[YES]
Customer repeats context	[YES]	[YES]	[YES]	[YES]	[YES]	[YES]
No proactive outreach	[YES]	[YES]	[YES]	[YES]	[YES]	[YES]
No multilingual support	[YES]	[YES]	[YES]	[YES]	[YES]	[YES]
Slow quote/estimate	[YES]	[YES]	[YES]	[YES]	[YES]	[YES]
Inconsistent follow-up	[YES]	[YES]	[YES]	[YES]	[YES]	[YES]
No unified memory	[YES]	[YES]	[YES]	[YES]	[YES]	[YES]

The result is clear: [YES] across every single cell. Every industry has the same problems. AgentCore solves them all with one platform.

1.6 Why This Matters Now

The voice AI market is at an inflection point. Legacy IVR systems ("Press 1, Press 2") have conditioned consumers to hate phone-based automation. Meanwhile, modern voice AI powered by large language models can deliver conversations indistinguishable from human agents -- but only if the platform is architected correctly. Most competitors offer voice-only solutions with no CRM integration, no memory, no WhatsApp channel, and no ability to self-host. AgentCore fills every one of these gaps.

Key market timing factors:

- India has 800M+ smartphone users by 2026, with 90% of new internet users preferring non-English content
- WhatsApp dominance: 500M+ users in India, making it the primary business communication channel
- Government digital push: UPI, DigiLocker, ABDM (Ayushman Bharat Digital Mission), Aadhaar-enabled services
- Rising labor costs in traditional BPO/call center operations (40-55% annual turnover in India)
- LLM capabilities have reached the threshold where voice AI can genuinely resolve customer issues, not just deflect them
- Sarvam AI and India-specific AI infrastructure maturing rapidly, making Indian language AI viable at scale

1.7 Document Purpose

This document serves as the definitive technical, business, and implementation reference for AgentCore. It is intended for:

- Internal engineering teams building and extending the platform
- Product leadership making roadmap decisions
- Investors evaluating the technology and market opportunity
- Potential B2B customers across ALL industries evaluating AgentCore for their products
- Compliance and legal teams assessing regulatory readiness
- Sales teams pitching AgentCore to CRM companies, enterprises, and contact centers in any vertical

1.8 Version History (V1 -> V2 Changes)

Area	V1 (14 engines)	V2 (24 engines)
Total Engines	14 core services	24 engines (10 new)
Industry Focus	Travel-centric	Universal – 15+ industries
Proposal Engine	Missing entirely	[NEW] Full proposal generation, tracking, approval
Invoice Engine	Missing	[NEW] GST-compliant invoicing, partial payment
Refund/Cancel Engine	Missing	[NEW] Complete post-booking lifecycle
Approval Engine	Missing	[NEW] Multi-step approval workflows
Campaign Engine	Missing	[NEW] Segment-based marketing automation
NPS/Feedback Engine	Embedded in Proactive	[NEW] Standalone engine with insights
Supplier Mgmt Engine	Missing	[NEW] Catalog, rates, contracts, performance
Itinerary Builder	Missing	[NEW] Day-by-day planner, templates, AI assist
Expense Capture	Missing	[NEW] Receipt OCR, policy check, voice capture
ChatAgent	English-only	[UPDATED] Full multilingual with Sarvam routing
STT/TTS Stack	ElevenLabs Platform	[UPDATED] Direct API: ElevenLabs Flash + Sarvam
LLM for Indian	GPT-4.1-mini only	[UPDATED] Sarvam-M (FREE) for regional context
Cost per minute	\$0.023-0.057	\$0.039-0.077 per path; weighted avg \$0.050
Customer Journey	Lead -> Quote -> Booking	Lead -> Proposal -> Approval -> Quote -> Booking -> Invoice -> NPS
Vertical Deployments	6 HelpRevX products	6 HelpRevX + 10 industry templates
Target Market	Travel + Photography + Finance	15 industries with detailed deployment specs

Section 2: Market Analysis & Pain Points

2.1 The Universal Customer Service Crisis (2025 Data)

The customer service industry is facing a crisis of trust and effectiveness. Despite billions invested in automation, customer satisfaction is declining -- not because AI is inherently bad, but because current implementations are poorly designed. This crisis is not limited to one industry. It affects every business that interacts with customers.

Consumer Frustration is at an All-Time High

- 75% of consumers experienced frustration with AI-driven customer service in 2025, even when responses were fast. Speed without resolution creates more frustration, not less. (Source: Glance CX Trends Report 2025)
- 68% of consumers say "Getting a complete resolution" is the most important factor in support interactions -- ahead of speed, friendliness, or channel availability. (Source: Glance CX Trends Report 2025)
- 34% say AI customer support "made things harder" -- these are systems that actively damage the customer relationship. (Source: Glance CX Trends Report 2025)

Channel Fragmentation Destroys Context

- Only 7% of customers say they rarely or never have to repeat themselves when switching channels (phone to chat, chat to email, etc.). The other 93% are re-explaining their issue every single time. (Source: Glance CX Trends Report 2025)
- This is not a technology problem -- it is an architecture problem. Systems are siloed. CRM data does not flow to voice. Voice context does not persist to chat. AgentCore's 4-layer memory system solves this by maintaining a unified customer memory across every channel and interaction.

Human Support Removal Kills Loyalty

- Nearly 90% of consumers report reduced loyalty when human support is removed entirely. (Source: Glance CX Trends Report 2025)
- 64% of consumers prefer no AI in customer service -- but this statistic reflects experience with OLD IVR systems and poorly designed chatbots, not modern voice AI. When consumers interact with well-designed voice AI that resolves their issues, preference shifts dramatically.
- The solution is not "no AI" -- it is "better AI with seamless human escalation." AgentCore's Handoff/Escalation Agent provides exactly this: AI handles routine queries with full context, and seamlessly transfers to human agents when needed, passing along complete conversation history, customer memory, and recommended actions.

Call Center Operations are Unsustainable

- Call center agent annual turnover rates hit 40-45% in 2025, with high-stress centers (financial services, healthcare) reaching 55-60%. In India specifically, turnover ranges from 45-55%. (Source: Insignia Resources 2025 Industry Report)
- 6% average call abandonment rate during normal times jumps to 15-20% during peak periods -- these are potential customers who simply give up. (Source: industry benchmarks)
- The average cost to recruit, hire, and train a single call center agent ranges from \$5,000-\$9,000 in India and \$10,000-\$20,000 in the US. With 45% annual turnover, companies are spending this repeatedly, every year.
- Agent burnout is the primary driver: repetitive queries, angry customers, rigid scripts, poor tools, and no career growth path.

Legacy IVR Systems are Fundamentally Broken

- Traditional IVR systems force customers through rigid decision trees ("Press 1 for billing, Press 2 for support, Press 3 for..."). These trees rarely match the customer's actual intent.
- No context awareness: the IVR does not know who is calling, what their history is, or what they likely need.
- No natural language understanding: customers must conform to the system's menu structure, not the other way around.
- No cross-channel continuity: a customer who starts on the website chat and then calls in starts from zero.
- No learning: IVR trees are static. They do not improve from interactions.

2.2 Market Size and Growth

India Conversational AI Market

The India conversational AI market represents a massive and rapidly growing opportunity:

- **2024 Market Size:** USD 455.4 million (INR 38.10 billion). (Source: Grand View Research)
- **2030 Projected Size:** USD 1,846.0 million (INR 152.31 billion). (Source: Grand View Research / Research and Markets)
- **CAGR (2025-2030):** 26.3%. (Source: Grand View Research)
- India's voice AI market is projected to reach INR 15,000 crore by 2030. (Source: NASSCOM)

Growth drivers include:

- 800M+ smartphone users in India by 2026
- Vernacular language preference: 90% of new internet users in India prefer non-English content
- WhatsApp as the dominant communication channel: 500M+ users in India
- Government push for digital services (UPI, DigiLocker, Aadhaar-enabled services)

- Rising labor costs in traditional BPO/call center operations

Global Voice AI Market

- 2025 Market Size: USD 416 billion. (Source: MarketsandMarkets)
- 2031 Projected Size: USD 2071 billion. (Source: MarketsandMarkets)
- CAGR (2025-2031): 30.7%. (Source: MarketsandMarkets)

The APIs, SDKs, and developer tools segment is expected to register the highest CAGR of 34.7% -- this is exactly the segment AgentCore targets as a platform product.

Industry-Specific Voice AI Market Sizing

Industry	India Market (Voice AI, 2025)	India Market (Voice AI, 2030 Est.)	CAGR	Key Driver
BFSI	INR 2,500 Cr	INR 8,500 Cr	28%	Regulatory compliance, call volume
Healthcare	INR 800 Cr	INR 3,200 Cr	32%	ABDM digitization, telemedicine
Retail & E-Commerce	INR 1,200 Cr	INR 4,500 Cr	30%	D2C explosion, order support
Education	INR 600 Cr	INR 2,200 Cr	29%	EdTech growth, admission volume
Real Estate	INR 400 Cr	INR 1,500 Cr	30%	Digital property platforms
Hospitality	INR 500 Cr	INR 1,800 Cr	29%	Post-COVID travel rebound
Automotive	INR 350 Cr	INR 1,300 Cr	30%	EV transition, digital dealers
Logistics	INR 450 Cr	INR 1,700 Cr	30%	E-commerce delivery volume
Legal	INR 150 Cr	INR 600 Cr	32%	Legal-tech adoption
Fitness & Wellness	INR 200 Cr	INR 750 Cr	30%	Health awareness, retention
Home Services	INR 300 Cr	INR 1,100 Cr	29%	Platform economy growth
Travel & Tourism	INR 550 Cr	INR 2,000 Cr	29%	Outbound travel growth
Total Addressable	INR 8,000 Cr	INR 29,150 Cr	29%	

2.3 Industry-Specific Pain Points

FINANCIAL SERVICES (BFSI)

India's BFSI sector processes over 2 billion customer service interactions annually. The pain points are acute:

- **KYC verification calls require manual agent reading checklist items:** Every account opening, policy purchase, or loan application requires a KYC verification call where an agent reads from a checklist. This is 100% scriptable and automatable by AI.
- **Policy renewal reminders go unmade:** Insurance companies have thousands of policies expiring each month. Agents are too busy with inbound queries to make outbound renewal reminder calls. Result: policies lapse, revenue lost.
- **Loan application status queries flood call centers:** "What is the status of my loan application?" constitutes 30-40% of all inbound calls at NBFCs and banks. Each call requires an agent to look up the status in the LOS (Loan Origination System) -- a 3-5 minute call that AI can handle in 30 seconds.
- **Insurance claim updates require agent lookup in 3 different systems:** A customer calling about a claim status forces the agent to check the Claims Management System, the policy database, and the payment system. AI can query all three simultaneously.
- **Mutual fund SIP queries require financial product knowledge agents do not always have:** "Should I increase my SIP?" or "What is my current portfolio value?" requires product knowledge that junior agents lack, leading to incorrect advice or unnecessary escalations.
- **Compliance risk: agents sometimes give wrong financial advice on calls:** Without AI guardrails, human agents can inadvertently provide non-compliant financial advice, exposing the company to regulatory penalties.
- **After-hours queries lost:** A customer wanting to report a lost credit card at 11 PM reaches voicemail. By morning, fraudulent transactions may have occurred.

HEALTHCARE

India has 1.4 million registered doctors, 70,000+ hospitals, and 1.5 million pharmacies. Patient communication is almost entirely manual:

- **Appointment booking requires human agent for every slot:** A clinic receptionist spends 60-70% of their time answering phone calls, most of which are appointment booking requests. This is the single most automatable task in healthcare.
- **Prescription refill calls are 100% repetitive and manual:** "I need a refill of my blood pressure medication" requires the agent to pull up the patient record, verify the prescription, and confirm the pharmacy. AI can do this in under a minute.
- **Lab result delivery calls need agent to read from PDF -- slow, error-prone:** "Are my blood test results ready?" requires staff to check the LIS (Lab Information System), open the PDF, and read values over the phone. AI can retrieve, interpret, and communicate results with appropriate medical context.
- **Patient follow-up after procedure is rarely done (no time):** Doctors want post-operative follow-up calls made to patients, but clinic staff are overwhelmed. Result: complications go

undetected, patient satisfaction drops.

- **Multilingual challenge: patients speak regional languages:** A Tamil-speaking patient in Chennai calling a multi-specialty hospital may reach an English-speaking receptionist who cannot help. AI with Sarvam's 22 Indian languages solves this instantly.
- **After-hours emergency triage:** Medical emergencies happen at 2 AM. No staff available. AI can perform basic triage, provide first-aid guidance, and connect to on-call doctors.
- **ABDM (Ayushman Bharat Digital Mission) integration need:** Government push for digital health records requires automated data capture and reporting.

REAL ESTATE

India's real estate market is valued at INR 12 lakh crore (2025) with millions of property transactions annually:

- **80% of property enquiries come outside business hours:** Prospective buyers browse property portals at night and on weekends. By Monday, the lead is cold.
- **Site visit scheduling requires back-and-forth with multiple agents:** Scheduling a site visit for a housing project involves coordinating with the sales agent, checking availability, confirming the customer's preferred time, and sending reminders. AI can handle the entire flow.
- **Property details vary per unit -- agents often do not know specifics:** "What is the carpet area of flat 503 in Tower B?" requires looking up specific unit details. AI connected to the inventory database has instant, accurate answers.
- **Tenant renewal reminders not done consistently:** Property management companies with hundreds of tenants rarely make proactive renewal calls. Result: tenants leave without notice.
- **Document collection for rental/purchase agreements is entirely manual:** Collecting Aadhaar, PAN, address proof, salary slips -- all done via email chains that span weeks. WhatsApp-based AI document collection can compress this to days.
- **RERA (Real Estate Regulatory Authority) compliance:** Mandated disclosures and transparency requirements that AI can ensure are communicated consistently.

LEGAL SERVICES

India has 1.7 million registered advocates, most operating with minimal technology:

- **First consultation scheduling requires paralegal time for basic intake:** Before meeting a lawyer, clients need to describe their issue, provide basic documents, and schedule a time. This intake process consumes paralegal hours.
- **Case status queries from clients are repetitive and time-consuming:** "What is happening with my case?" is the most common client question. Lawyers or paralegals must look up the case file and provide an update -- a task perfect for AI.
- **Client intake: collecting basic facts of case manually before meeting:** A lawyer meeting a new client for a property dispute needs facts: property address, parties involved, timeline of

events, documents available. AI can collect this via voice or WhatsApp before the meeting.

- **Document checklist communication to clients is manual email:** "For your divorce petition, please bring: marriage certificate, address proof, income proof, children's details..." -- this is a template that AI sends and tracks.
- **Fee estimate conversations are sensitive and require calibrated response:** Quoting legal fees requires understanding the complexity of the case, the court jurisdiction, and the expected timeline. AI can provide preliminary estimates based on parameters.
- **Bar Council compliance:** Recording and disclosure requirements that must be followed in every client interaction.

EDUCATION & COACHING

India's education sector serves 250 million students across schools, colleges, and coaching centers:

- **Admission enquiry volume spikes seasonal:** IIT-JEE coaching centers receive 10x normal enquiry volume in March-April. Staff is overwhelmed. 60% of enquiry calls go unanswered.
- **Fee payment reminders require manual calls to hundreds of students:** Every month, coaching centers and schools must remind parents about fee payments. This is 100% automatable.
- **Course completion and next-step guidance needs personalization:** "I just finished the NEET foundation course. What should I do next?" requires understanding the student's performance, goals, and available programs.
- **Parent queries repeat daily:** Schools receive the same questions hundreds of times: exam dates, PTM schedule, bus route, fee receipt, uniform availability. AI handles all of these.
- **Scholarship and loan guidance requires knowledge agent does not always have:** Students asking about scholarship eligibility, education loan options, or government schemes need accurate, up-to-date information.
- **Demo class scheduling:** Coaching centers rely on trial/demo classes for conversion. Scheduling these efficiently via AI increases conversion rates.

FITNESS & WELLNESS

India's fitness industry is growing at 20%+ annually with over 30,000 gyms and studios:

- **Trial class booking requires agent availability:** A potential member calls to book a trial class. The front desk is busy with existing members. Call goes unanswered. Customer joins a competitor gym.
- **Renewal reminders not made consistently -- members lapse:** Gym memberships expire silently. Without proactive renewal calls, members lapse and revenue drops 15-25%.
- **Attendance pattern detection:** AI can detect that a member who was attending 4x/week has dropped to 1x/week and proactively reach out before they quit.
- **Package upsell:** trainers do not always pitch upgrades: A member on a basic plan who is consistently attending should be upsold to personal training. Trainers are busy training,

not selling.

- **Feedback collection after sessions not done:** Post-class feedback from yoga or group fitness sessions is rarely collected, missing valuable retention signals.
- **Class schedule communication:** "What time is the 6 AM batch?" -- repeated dozens of times daily.

AUTOMOTIVE

India sells 4+ million cars and 18+ million two-wheelers annually:

- **Test drive scheduling requires dealer agent availability:** A customer wants to test drive the new Tata Nexon this Saturday. The dealer's phone is busy. Customer goes to Hyundai instead.
- **Service reminder calls are repetitive -- perfect for AI:** "Your vehicle is due for its 10,000 km service. Would you like to book a slot?" -- this call is made thousands of times monthly by every dealership.
- **Vehicle availability queries: agents do not always know current stock:** "Do you have the XUV700 AX7 in red?" requires checking real-time inventory. Agents often give wrong information.
- **Insurance and finance queries during purchase conversation:** Car buyers have dozens of questions about loan EMI, insurance options, and extended warranty. AI with calculator tools can answer all of these in real-time.
- **Post-purchase follow-up for service package upsell not done:** After car delivery, the perfect time to pitch extended warranty or service packages. But salespeople have moved on to the next sale.

HOSPITALITY & FOOD

India's hospitality industry serves 10+ million tourists monthly:

- **Reservation queries come at all hours:** International travelers in different time zones, domestic travelers browsing at night. Hotels miss 30% of direct booking calls.
- **Room upgrade upsell not pitched consistently:** A guest booked a Deluxe room, but a Suite is available at 30% premium. The front desk rarely offers the upgrade proactively.
- **Local experience recommendations require local knowledge:** "What should we do tomorrow?" -- concierge-level recommendations that AI can provide based on guest preferences and local partnerships.
- **Check-in instructions, parking, F&B -- repetitive queries:** "What time is checkout?", "Is breakfast included?", "Where do I park?" -- repeated hundreds of times, easily handled by AI.
- **Feedback collection after stay -- rarely done proactively:** Post-checkout feedback calls can rescue low ratings and drive Google Reviews for high ones.
- **Restaurant reservation management:** Table booking, menu queries, dietary restriction handling, special occasion arrangements -- all automatable.

RETAIL & E-COMMERCE

India's e-commerce market processes over 5 billion orders annually:

- **Order status queries are 40-60% of all support calls -- 100% automatable:** "Where is my order?" is the single most common customer service query across all e-commerce and D2C brands. AI connected to the OMS (Order Management System) answers this instantly.
- **Return initiation process requires agent walkthrough:** "I want to return this product" requires verifying the order, checking return policy, generating a return label, and scheduling pickup. AI handles the entire flow.
- **Product recommendation calls need product catalog knowledge:** "I'm looking for a moisturizer for oily skin" requires product expertise. AI with semantic search across the product catalog provides expert recommendations.
- **Abandoned cart follow-up calls not made at the right time:** A customer abandoned checkout 2 hours ago. A well-timed call offering free shipping converts 15-20% of abandoned carts.
- **Loyalty program queries (points, redemption) are repetitive:** "How many points do I have?", "Can I use points for this purchase?" -- fully automatable queries.

LOGISTICS & DELIVERY

India's logistics industry handles 5+ billion shipments annually:

- **Delivery status is the number-one inbound call type -- fully automatable:** "Where is my package?" constitutes 50-70% of all inbound calls at courier companies. Each call costs INR 15-25 when handled by a human. AI cost: INR 2-3.
- **Pickup scheduling requires agent coordination:** "I need a pickup from my office tomorrow between 2-4 PM" requires checking serviceable area, slot availability, and confirming with the customer.
- **Complaint about delayed delivery requires empathy plus system lookup:** AI must acknowledge frustration, look up the actual delivery status, provide a revised ETA, and offer resolution options.
- **B2B shipping documentation queries require operations knowledge:** "What documents do I need for shipping chemicals to Dubai?" requires regulatory knowledge that most agents do not have.
- **Address mismatch resolution is a frequent manual pain:** "The delivery boy says the address is wrong" -- AI can call the customer, verify the address, and update the system in real-time.

2.4 Competitor Analysis

The competitive landscape for AgentCore spans three distinct tiers: the startup voice AI tier (Retell, Vapi, Bland, ElevenLabs, Synthflow), the enterprise contact center legacy tier (Genesys, Five9, NICE CXone), and the next-generation enterprise AI tier (PolyAI, Cognigy, Salesforce Agentforce, AWS Connect). AgentCore must be benchmarked against all three — because customers evaluating us will comparison-shop across all three. Understanding precisely what

each tier does well, and where each fails, is essential to positioning AgentCore as the only platform that eliminates the tradeoffs.

TIER 1: Startup Voice AI Platforms

Retell AI

What they do well:

- Clean developer experience — quick to integrate for English-language outbound calls
- Good latency optimization (sub-500ms end-to-end in ideal conditions)
- Pre-built integrations with common LLMs (GPT-4, Claude, Gemini)
- Visual call flow builder for non-technical users
- Transparent per-minute pricing (\$0.07-0.15/min) with no setup fees
- Webhook-based integration with third-party CRMs
- Active developer community and documentation

Critical gaps (how AgentCore resolves each):

- **No Indian language depth** — English only with surface-level multilingual. AgentCore resolves: 22 Indian languages via Sarvam Saaras V3 + Bulbul V3, native Hinglish code-switching, dialect detection, and language routing per caller.
- **No CRM integration layer** — voice exists in isolation from business data. AgentCore resolves: 8 CRM lifecycle engines (Lead, Proposal, Quote, Case, Booking, Invoice, Refund) with full bi-directional sync.
- **No WhatsApp** — India's dominant communication channel is completely absent. AgentCore resolves: Full WhatsApp Commerce Engine with Flows v2, in-chat UPI payments, document exchange, carousels.
- **No persistent memory** — every call starts from zero context. AgentCore resolves: 4-layer memory (Redis short-term + PostgreSQL episodic + pgvector semantic + procedural rules).
- **Vendor-hosted only** — customer data leaves the company's infrastructure. AgentCore resolves: full self-hosted deployment on customer's AWS with no data egress.
- **No multi-agent orchestration** — single agent architecture cannot handle complex multi-step workflows. AgentCore resolves: LangGraph Supervisor + 7 parallel specialist agents per call.
- **No document intelligence** — cannot process passports, visas, invoices, or KYC documents. AgentCore resolves: Document Intelligence Engine with OCR, validation, classification.
- **No proactive outbound** — purely reactive, cannot initiate intelligent campaigns. AgentCore resolves: Proactive Outbound Engine with triggers, segments, personalized messaging.
- **No compliance for regulated verticals** — zero HIPAA, BFSI, Legal, or RERA support. AgentCore resolves: full compliance engine per vertical with audit trails, consent management, kill-switch.

- **No vertical templates** — every deployment built from scratch. AgentCore resolves: 15 industry templates with pre-built tools, workflows, and compliance rules.

Vapi

What they do well:

- Maximum flexibility — bring your own LLM, STT, TTS (modular architecture)
- Largest developer community in the voice AI space
- Reasonable base orchestration rate (\$0.05/min before component costs)
- Supports custom LLM endpoints for self-hosted models
- Squad (multi-agent) feature for basic handoffs
- Built-in analytics dashboard
- Extensive webhook support for custom integrations
- Good documentation with code examples

Critical gaps (how AgentCore resolves each):

- **True all-in cost is \$0.13-0.31/min** — base rate hides STT + LLM + TTS + telephony costs across 4-6 separate provider invoices. AgentCore resolves: unified billing at \$0.039-0.077/min — 42-75% cheaper on a like-for-like basis.
- **6-7 second latency in real-world deployments** — optimal latency requires expert-level tuning that most customers never achieve. AgentCore resolves: pre-optimized LiveKit pipeline with sub-800ms end-to-end latency out of the box.
- **4-6 separate provider invoices** — STT (Deepgram), LLM (OpenAI), TTS (ElevenLabs), telephony (Twilio), orchestration (Vapi) each billed separately. AgentCore resolves: single invoice, single platform, single SLA.
- **No Indian language focus** — no Hinglish understanding, no regional language routing, no Sarvam integration. AgentCore resolves: dedicated Indian language stack at \$0.040/min (cheapest available).
- **\$1,000/month add-on for HIPAA compliance** — compliance is an afterthought bolt-on. AgentCore resolves: compliance-first design with HIPAA, DPDP, TRAI, BFSI, RERA built in from day one.
- **Weak enterprise support** — community-dependent, no SLA guarantees for production deployments. AgentCore resolves: dedicated support with defined SLA tiers (999% uptime for Growth, 99.99% for Enterprise).
- **No CRM intelligence, memory, or WhatsApp commerce** — same gaps as Retell. AgentCore resolves: full platform stack across all three.
- **No vertical templates** — each industry deployment is a custom project. AgentCore resolves: 15 pre-built vertical templates.

Bland AI

What they do well:

- Strong outbound dialing capabilities — predictive, progressive, and preview dialer modes
- Good for single-prompt, linear call flows at high volume
- Competitive base pricing (\$0.09/min connected)
- Retry logic and answering machine detection
- Basic call analytics and recording
- Reasonably fast for simple scripted flows

Critical gaps (how AgentCore resolves each):

- **1-2 second barge-in delay** — conversations feel robotic and unnatural when AI doesn't detect user speech quickly. AgentCore resolves: Silero VAD with 10ms local detection, sub-50ms barge-in response.
- **Open-source LLM by default** — less reasoning capability for complex, multi-intent conversations. AgentCore resolves: GPT-4.1 / Claude Sonnet with automatic fallback, or Sarvam-M for Indian deployments.
- **Spam and carrier issues** — calls flagged as spam, reducing answer rates to 15-30%. AgentCore resolves: Exotel (India) with carrier-grade caller ID reputation, Twilio (global) with registered brand.
- **Outage history** — production reliability concerns for enterprise customers. AgentCore resolves: multi-AZ AWS deployment with ECS auto-scaling and Circuit Breaker pattern.
- **No CRM, memory, WhatsApp, or Indian language support** — fundamental platform gaps. AgentCore resolves: full 24-engine platform.
- **No compliance frameworks for regulated industries** — cannot serve BFSI, Healthcare, Legal. AgentCore resolves: industry-specific compliance per tenant in the compliance engine.

ElevenLabs Conversational AI

What they do well:

- Industry-leading voice quality and naturalness — the gold standard for TTS
- 74 language support with emotional tone control
- Exceptional voice cloning capabilities
- Strong developer API ecosystem
- Good latency for TTS generation (350-500ms)
- Reliable uptime and API stability
- Extensive voice library with instant clone and professional clone options
- Real-time streaming API with low time-to-first-audio

Critical gaps (how AgentCore resolves each):

- **\$0.08-0.10/min platform pricing with no CRM integration** — you pay premium for voice quality but get zero business logic. AgentCore resolves: uses ElevenLabs Flash v2.5 API directly (not the platform) at embedded cost, wrapped in full 24-engine platform.
- **No memory system** — no customer context carried across calls. AgentCore resolves: 4-layer memory so every interaction knows the customer's history.
- **No WhatsApp, no omnichannel** — voice-only product with no path to meeting customers on their channel of choice. AgentCore resolves: full omnichannel across voice, chat, WhatsApp, email.
- **No Indian market context** — no understanding of Indian business workflows, regional language depth, or Indian compliance. AgentCore resolves: Indian-first design with Sarvam stack, DPDP/TRAI compliance.
- **No multi-agent orchestration** — cannot handle complex conditional business logic. AgentCore resolves: LangGraph with 7 specialist agents.
- **Voice-only strategy** — fundamental limitation as customers demand omnichannel AI. AgentCore resolves: unified memory and orchestration across all channels.
- **No vertical-specific logic** — cannot handle insurance claims, medical appointments, or loan inquiries without heavy custom development. AgentCore resolves: 15 industry vertical templates.

Synthflow

What they do well:

- Visual no-code flow builder — fastest path to a working voice AI for non-technical teams
- Good for simple, linear intake flows (lead capture, appointment booking)
- Quick setup — first call in under 30 minutes
- White-label reseller program
- Pre-built integrations with HubSpot, GoHighLevel, Calendly, Zapier
- Reasonable pricing for simple use cases

Critical gaps (how AgentCore resolves each):

- **No code-level control** — visual builder cannot implement complex conditional business logic. AgentCore resolves: Python + LangGraph codebase with full programmatic control.
- **Not self-hostable** — all data processed on Synthflow's servers, no data sovereignty. AgentCore resolves: full self-hosted deployment on customer AWS.
- **No Indian language support** — entirely English-focused. AgentCore resolves: 22 Indian languages + Hinglish.
- **No CRM depth** — surface-level integrations, no bi-directional business object sync. AgentCore resolves: 8 deep CRM lifecycle engines.

- **Cannot serve regulated industries** — no audit trail, no compliance engine, no kill-switch, no HIPAA. AgentCore resolves: compliance-first design from the ground up.
- **Visual builder caps complexity** — cannot handle multi-step verification, document processing, or cross-system orchestration. AgentCore resolves: full multi-agent orchestrator handles arbitrarily complex workflows.

TIER 2: Enterprise Legacy Contact Center Platforms

This tier is the \$15B+ market that AgentCore's enterprise and mid-market customers currently use. Understanding these platforms is critical: any enterprise prospect evaluating AgentCore is also evaluating Genesys, NICE, or Five9. We must know exactly where they excel and exactly where they fail — because our wins come from those failure points.

Genesys Cloud CX

What they do well:

- Named #1 in 3 of 5 Gartner Magic Quadrant contact center use cases (2024-2025)
- Unified omnichannel from single platform: voice, chat, email, SMS, social media, WhatsApp, messaging
- Workforce Engagement Management (WEM): agent scheduling, quality scoring, gamification, coaching
- Predictive routing: ML-based prediction of best agent match based on customer history, agent skills, sentiment
- Native QA/scoring: supervisors score calls against defined rubrics, automated scoring for AI interactions
- Real-time supervisor dashboards: live barge-in, whisper coaching, silent monitor, queue management
- Proven enterprise scale: 6,000+ customers globally including Fortune 500
- Genesys AI (DX): bot orchestration, predictive engagement, interaction analytics
- Open API ecosystem (AppFoundry): 350+ marketplace integrations
- SOC 2 Type II, ISO 27001, PCI-DSS, HIPAA BAA available
- Cloud-native architecture with 99.99% uptime SLA for enterprise tiers

Critical gaps (how AgentCore resolves each):

- **\$150-300/agent/month pricing** — at 50 agents, that is \$7,500-15,000/month before AI add-ons. AgentCore resolves: usage-based API pricing (per minute + per seat) that scales from ₹5,000/month (SMB) to ₹5L/month (large enterprise) — no per-seat ceiling.
- **3-6 month implementation timelines** — large enterprises need professional services engagements lasting months. AgentCore resolves: 15 pre-built vertical templates + Claude-assisted implementation phases reduce deployment to 10-12 weeks.

- **No native LLM integration depth** — Genesys AI bolt-on, not core architecture. LLM reasoning is shallow. AgentCore resolves: LLM-native from the ground up — every decision, every response generated by Claude/GPT-4.1/Sarvam-M.
- **No Indian language depth** — Genesys supports English globally but Indian regional language support is limited and expensive to configure. AgentCore resolves: Sarvam stack with 22 Indian languages natively.
- **Not self-hostable** — all data on Genesys cloud. Healthcare, BFSI, Legal customers in India require on-premise or own-cloud deployment. AgentCore resolves: full self-hosted on customer AWS.
- **No CRM-native intelligence** — Genesys integrates with CRMs (Salesforce, ServiceNow) but does not own the CRM layer. AgentCore resolves: 24 engines covering the full CRM lifecycle, not just the conversation.
- **No WhatsApp commerce** — Genesys supports WhatsApp for messaging but not in-chat payments, Flows v2, or commerce flows. AgentCore resolves: Full WhatsApp Commerce Engine with UPI in-chat payment.
- **\$30,000-100,000+ annual contracts** — out of reach for SMBs and mid-market in India. AgentCore resolves: India-first pricing starting at ₹5,000/month.
- **No proactive AI outbound** — Genesys outbound is rule-based dialer, not AI-triggered intelligence. AgentCore resolves: Proactive Outbound Engine with AI-triggered messaging based on CRM events, sentiment, and behavioral signals.

Five9

What they do well:

- Strongest outbound dialer capabilities in the industry: predictive, progressive, preview, and power dialer modes
- Genius AI: AI-powered agent assist, real-time transcription, next-best-action suggestions
- Strong Salesforce integration — deep bi-directional sync, embedded in Salesforce UI
- Workforce management (WFM): scheduling, forecasting, adherence tracking
- IVR visual designer with advanced conditional routing
- Good analytics: historical and real-time, custom dashboards
- Digital channel support: email, chat, SMS, social
- Reliable uptime with 99.99% SLA for enterprise
- Compliance: SOC 2 Type II, PCI-DSS, HIPAA BAA

Critical gaps (how AgentCore resolves each):

- **\$150-200/agent/month** — same pricing tier as Genesys — prohibitive for Indian market. AgentCore resolves: usage-based pricing, no per-seat cost.
- **US/North America focus** — Indian language support minimal, no DPDP/TRAI compliance, no Indian telephony optimization. AgentCore resolves: Indian-first design with Exotel,

Sarvam, DPDP compliance.

- **Cloud-only, not self-hostable** — critical failure point for Indian regulated industries. AgentCore resolves: full AWS self-hosted deployment.
- **Genius AI is a bolt-on** — AI is layered on top of a legacy IVR architecture, not native. Complex AI reasoning requires significant custom configuration. AgentCore resolves: LLM-native architecture from the ground up.
- **No WhatsApp commerce, no Indian payment integration** — significant gap for Indian market. AgentCore resolves: WhatsApp UPI commerce.
- **No 4-layer memory** — conversations are contextual within the session but no long-term customer memory. AgentCore resolves: persistent memory across sessions, channels, and time.

NICE CXone

What they do well:

- CXone is the most comprehensive cloud contact center platform in terms of feature breadth
- Enlighten AI: deep conversation analytics, coaching, quality management at scale
- Native Enlighten AutoSummary: automatic call summary generation post-call
- Performance management: agent gamification, coaching, KPI dashboards
- Workforce management: scheduling, adherence, forecasting with AI-assisted planning
- Strong omnichannel: voice, email, chat, messaging, social with unified agent desktop
- Native knowledge base: integrated article suggestions during live interactions
- Compliance: SOC 2 Type II, ISO 27001, PCI-DSS Level 1, HIPAA, FedRAMP
- 99.99% uptime SLA with DR across multiple regions
- Open platform (CXone DEV) with REST APIs and marketplace

Critical gaps (how AgentCore resolves each):

- **\$100-250/agent/month with complex add-on pricing** — Enlighten AI, WFM, QM each sold separately. Total cost for a 50-agent center: \$15,000-30,000/month. AgentCore resolves: all features bundled at one predictable per-minute + per-seat price.
- **18-24 month full deployment timelines** — NICE implementations are multi-year IT projects. AgentCore resolves: 10-12 week deployment with Claude-assisted phases.
- **Cloud-only — no self-hosted option** — same fundamental gap for regulated Indian industries. AgentCore resolves: full self-hosted AWS deployment.
- **No LLM-native reasoning** — Enlighten AI is analytical (transcription + classification), not generative. AI cannot take actions, only surface insights. AgentCore resolves: generative AI that takes actions — books appointments, generates proposals, processes refunds.

- **No Indian language depth, no DPDP compliance** — same gap as Genesys/Five9. AgentCore resolves: Indian-first architecture.
- **No CRM ownership** — NICE integrates with CRMs but does not provide one. AgentCore resolves: full CRM lifecycle built in.

TIER 3: Next-Generation Enterprise AI Platforms

This is AgentCore's most important competitive tier for mid-2026 positioning — AI-native platforms that have raised \$100M+ and are targeting the same customer base.

PolyAI

What they do well:

- \$150K+ enterprise contracts — proven ability to sell large deals
- Industry-leading containment rates: 80-90% of calls fully resolved by AI without human escalation
- 45 languages with true dialect handling (not just language detection — accent-aware models)
- Managed service model: PolyAI builds and manages the voice agent for the customer (not DIY)
- Enterprise-grade security: SOC 2 Type II, ISO 27001, GDPR-compliant
- Fortune 500 deployments: Marriott, Five Guys, Caesars Entertainment
- Human-like conversation quality — natural turn-taking, interruption handling, silence management
- Dedicated customer success team for every deployment
- Proven ROI metrics: 40-60% call handling cost reduction case studies
- Real-time analytics dashboard for operations teams
- CRM integrations: Salesforce, ServiceNow, Oracle, SAP via pre-built connectors

Critical gaps (how AgentCore resolves each):

- **Minimum contract \$150K/year** — completely out of reach for SMBs and mid-market. AgentCore resolves: API-first pricing starting at ₹5,000/month, scaling to enterprise contracts.
- **Managed service only** — no self-service or developer access. Customer depends entirely on PolyAI for changes. AgentCore resolves: full API access, code-level control, Claude-assisted self-development.
- **No CRM engine ownership** — PolyAI integrates with CRMs but does not build or own the CRM layer. AgentCore resolves: 24 engines with full CRM lifecycle management.
- **Voice-only** — no WhatsApp, no chat, no email intelligence. AgentCore resolves: full omnichannel across all 4 channels with unified memory.

- **No Indian language depth** — "45 languages" covers major world languages but Indian regional languages (Tamil, Telugu, Marathi, Bengali, Gujarati) are not a core capability. AgentCore resolves: Sarvam-native 22 Indian languages.
- **No self-hosted option** — data sovereignty problem for Indian BFSI and Healthcare. AgentCore resolves: full AWS self-hosted deployment.
- **3-6 month deployment timeline** — managed service model requires intensive setup. AgentCore resolves: developer-friendly API with vertical templates for faster go-live.
- **No proactive outbound AI** — purely inbound-focused. AgentCore resolves: Proactive Outbound Engine.

Cognigy

What they do well:

- Built for Fortune 2000 — enterprise-grade from ground up
- Cognigy.AI: full conversational AI platform for voice + chat + messaging
- No-code + code hybrid: visual flow builder for non-technical teams, full code access for complex logic
- Deep backend system integrations: SAP, Salesforce, ServiceNow, Microsoft Dynamics via pre-built connectors
- Cognigy Insights: advanced conversation analytics, NLU performance tracking, custom dashboards
- Agent Copilot: real-time AI assist for human agents during live conversations
- Comprehensive enterprise security: ISO 27001, SOC 2 Type II, PCI-DSS, HIPAA, GDPR
- Multi-deployment: cloud (AWS, Azure, GCP) or on-premise
- 24/7 enterprise support with SLA guarantees
- NLU training tools: intent management, entity extraction, test suites
- Cognigy.AI Marketplace: pre-built skills for common industry use cases
- Agentic AI (2025 feature): autonomous task execution with tool-use capability
- Multi-language: 100+ languages with full NLU support

Critical gaps (how AgentCore resolves each):

- **\$50,000-500,000/year enterprise contracts** — prohibitive for Indian mid-market and SMB. AgentCore resolves: India-first pricing model.
- **6-18 month implementation projects** — full Cognigy deployment is a significant IT project requiring dedicated professional services. AgentCore resolves: 10-12 week deployment with Claude-assisted phases.
- **No CRM ownership** — Cognigy integrates with CRMs but is not a CRM. No Lead, Proposal, Quote, or Invoice engine. AgentCore resolves: 24 engines covering the complete CRM lifecycle.

- **No Indian language optimization** — 100+ languages but no deep Indian regional language capability (Sarvam Saaras V3 quality). AgentCore resolves: Sarvam-native stack with dialect-aware routing.
- **No WhatsApp commerce with Indian payments** — supports WhatsApp Business API but not Flows v2 or UPI in-chat payments. AgentCore resolves: full WhatsApp Commerce Engine.
- **No proactive outbound AI** — Cognigy is inbound/service-focused. AgentCore resolves: Proactive Outbound Engine.
- **4-layer memory not available** — session-level context, no long-term multi-layer memory. AgentCore resolves: full 4-layer memory architecture.

Salesforce Agentforce + Einstein (Service Cloud Voice)

What they do well:

- CRM-native AI: deepest possible integration with Salesforce data — every customer record, case history, purchase history directly accessible
- Einstein Conversation Insights: automatic call transcription, keyword tracking, coaching suggestions, sentiment analysis
- Service Cloud Voice: direct CTI integration with Amazon Connect for voice + Salesforce in one screen
- Agentforce (2025): autonomous AI agents that can take actions across Salesforce objects
- Einstein Next Best Action: real-time suggestion of offers or solutions during live interactions
- Massive ecosystem: 150,000+ Salesforce admins, Trailhead training, partner network
- Native ticketing: Cases with full lifecycle, SLA tracking, escalation rules
- Knowledge Base: articles, search, agent suggestions, customer self-service portal
- Reporting: native Salesforce reports + Tableau CRM for advanced analytics
- Compliance: SOC 2, ISO 27001, GDPR, HIPAA, FedRAMP for government

Critical gaps (how AgentCore resolves each):

- **Salesforce dependency** — Agentforce only works if the customer uses Salesforce. Customers on HubSpot, Zoho, or proprietary CRMs are excluded entirely. AgentCore resolves: CRM-agnostic, can integrate with any existing system or operate as the CRM.
- **Add-on pricing complexity** — Service Cloud (\$150/user/month) + Agentforce (\$2/conversation) + Einstein (\$50/user/month) + Amazon Connect (\$0.018/min) stack up to \$200-250/agent/month. AgentCore resolves: unified pricing.
- **Inbound-only** — Agentforce is optimized for inbound service resolution. No proactive AI outbound. AgentCore resolves: Proactive Outbound Engine.
- **3-6 month implementation** — Salesforce implementations require certified Salesforce partners, lengthy configuration. AgentCore resolves: 10-12 week deployment.

- **No Indian language support** — Salesforce AI is English-dominant. Indian regional language NLU is not supported. AgentCore resolves: Sarvam stack.
- **External knowledge blindspot** — Agentforce excels with Salesforce data but cannot dynamically pull from external knowledge sources (TBO flights, hospital HMS systems, banking LOS). AgentCore resolves: configurable tool registry with any external API.
- **No self-hosted option** — Salesforce is SaaS-only. AgentCore resolves: full self-hosted AWS deployment.

AWS Connect + Lex + Bedrock

What they do well:

- AWS-native: no data leaves AWS infrastructure, works within existing AWS IAM/VPC/security perimeter
- Pay-as-you-go pricing: \$0.018/min (Connect) + \$0.004/char (Lex) + \$0.001/1K tokens (Bedrock Haiku) — very low cost at extreme scale
- No per-seat pricing — usage-based only
- ML-powered routing: Contact Lens ML predicts best agent match based on sentiment, skills, history
- Contact Lens: real-time call analytics, sentiment, compliance issue detection
- Native Salesforce, Zendesk, ServiceNow integrations
- Amazon Q in Connect: AI agent assist with knowledge retrieval during live calls
- CloudFormation/Terraform deployable — infrastructure-as-code friendly
- 99.99% SLA on underlying services (EC2, ECS, Lambda)
- Bedrock integration: access to Claude, Llama, Titan, Mistral for custom AI logic

Critical gaps (how AgentCore resolves each):

- **Extreme complexity** — not a product, it's an assembly kit — connecting Connect + Lex + Bedrock + Contact Lens + Lambda + DynamoDB requires AWS expertise and significant engineering. 6-12 months to build a production-grade voice AI. AgentCore resolves: pre-built platform on AWS with all components assembled, tested, and optimized.
- **No WhatsApp commerce** — AWS does not support WhatsApp Business API natively. AgentCore resolves: full WhatsApp Commerce Engine.
- **No CRM lifecycle ownership** — same gap as Genesys/Five9. AgentCore resolves: 24 CRM lifecycle engines.
- **No proactive outbound AI intelligence** — Connect outbound is dialer-based, not AI-triggered. AgentCore resolves: Proactive Outbound Engine.
- **No Indian language optimization** — Lex supports Hindi but not 22 Indian languages at Sarvam quality. AgentCore resolves: Sarvam-native stack.
- **No 4-layer memory** — DynamoDB can store data but there is no pre-built multi-layer memory architecture. AgentCore resolves: full 4-layer memory system.

- **Setup and maintenance burden** — continuous AWS maintenance, cost optimization, and architecture updates require dedicated DevOps. AgentCore resolves: pre-packaged platform with documented Terraform modules.

Zendesk (Suite + AI)

What they do well:

- Industry-standard for enterprise ticketing and case management — the benchmark for structured support operations
- Omnichannel: email, phone (via Talk), chat, social media, WhatsApp (via third-party), in-app messaging — all under one platform
- Zendesk Explore: enterprise reporting with custom dashboards, SLA tracking, CSAT analysis, agent performance
- AI triage: automatic routing, intent detection, sentiment tagging, priority scoring
- Knowledge Base (Guide): comprehensive article management, customer self-service portal, AI-powered article suggestions
- SLA management: response time SLAs per ticket priority, breach alerting, escalation rules
- Marketplace: 1,500+ integrations including Salesforce, Slack, Jira, HubSpot
- Zendesk QA (Klaus acquisition): AI-powered quality review — automatic conversation scoring, coach assignments, QA workflows
- Workforce Management (Tymeshift acquisition): agent scheduling, adherence, forecasting
- Compliance: SOC 2 Type II, ISO 27001, PCI-DSS, HIPAA BAA, FedRAMP

Critical gaps (how AgentCore resolves each):

- **No native voice AI beyond basic IVR** — Zendesk Talk is a call center, not an AI voice agent. To add AI, you need third-party integrations. AgentCore resolves: native AI voice agent with human-level conversation capability.
- **\$55-115/agent/month** — Growth to Enterprise tiers. At scale: significant per-seat cost. AgentCore resolves: usage-based API pricing.
- **No Indian language voice support** — Talk supports English and major European languages. AgentCore resolves: 22 Indian languages.
- **No WhatsApp commerce / UPI payments** — Zendesk supports WhatsApp messaging (via Sunshine Conversations add-on) but not payment flows. AgentCore resolves: full WhatsApp Commerce Engine with in-chat UPI.
- **No proactive outbound AI** — Zendesk is reactive (ticket-based). AgentCore resolves: Proactive Outbound Engine with AI-triggered engagement.
- **No 4-layer persistent memory** — conversation history in tickets is not structured memory. AgentCore resolves: 4-layer memory with semantic search.
- **Cloud-only** — no self-hosted option for regulated Indian industries. AgentCore resolves: full self-hosted AWS deployment.

Intercom (Fin AI Agent)

What they do well:

- Industry-leading in-product contextual messaging — the benchmark for proactive customer engagement
- Fin AI Agent: \$0.99/resolution pricing — performance-based model (only pay when Fin fully resolves the issue)
- 45%+ resolution rates achieved in production across diverse industries
- Proactive campaigns: behavior-triggered in-app messages, product tours, banners, push notifications
- Messenger: branded in-app chat widget with extremely high open rates (vs email)
- Conversation routing: skill-based, round-robin, load-balanced with human handoff
- Native CRM: full contact timeline, custom attributes, company-level tracking
- Reporting: resolution rates, team performance, conversation volume by source
- Inbox efficiency: shared team inbox, collision detection, private notes
- Help center: article management, AI-powered suggestions within Fin
- Good integrations: Salesforce, HubSpot, Stripe, GitHub, Slack, 300+

Critical gaps (how AgentCore resolves each):

- **Chat-only, no voice AI** — Fin cannot handle phone calls. AgentCore resolves: full omnichannel voice + chat + WhatsApp + email with unified memory.
- **No Indian market depth** — Intercom is English-first, US/European SaaS market focus. No Hindi/Tamil/Telugu/Marathi support. AgentCore resolves: 22 Indian languages across all channels.
- **No WhatsApp commerce** — Intercom supports WhatsApp for messaging but not UPI payment flows. AgentCore resolves: full WhatsApp Commerce Engine.
- **\$0.99/resolution hides true cost** — at 10,000 resolutions/month that is \$9,900/month before base seat costs (\$29-132/seat). AgentCore resolves: transparent per-minute + per-seat pricing with no per-resolution fee.
- **No CRM lifecycle depth** — Intercom is a messaging/support tool, not a full CRM. No Proposal, Quote, Booking, Invoice, or Refund engine. AgentCore resolves: 8 CRM lifecycle engines.
- **No self-hosted option** — cloud SaaS only. AgentCore resolves: full self-hosted AWS deployment.
- **No compliance for regulated industries** — HIPAA and SOC 2 available but no BFSI (SEBI/RBI), no DPDP, no TRAI. AgentCore resolves: multi-framework compliance engine.

Freshdesk / Freshchat (Freshworks)

What they do well:

- Best value-for-money in the mid-market tier — \$15-79/agent/month vs Zendesk's \$55-115
- Freddy AI: AI-powered agent assist, auto-resolution, suggested responses, sentiment analysis
- Omnichannel: email, chat, phone (FreshCaller), WhatsApp, social media, in-app
- Strong ticketing with SLA management, canned responses, automation rules
- Freshdesk — Freshservice integration: IT and customer service in one ecosystem
- Good analytics: CSAT, resolution time, agent performance
- India-based company — understands Indian market context
- Growing WhatsApp Business API integration
- 1,000+ marketplace integrations
- Compliance: SOC 2 Type II, ISO 27001, GDPR, HIPAA available at higher tiers

Critical gaps (how AgentCore resolves each):

- **No deep Indian language voice AI** — FreshCaller handles calls but AI is English-dominant. AgentCore resolves: 22 Indian languages with Sarvam-native stack.
- **Freddy AI is shallow** — AI suggestions and auto-classification, not autonomous action-taking. AgentCore resolves: fully autonomous AI that books, proposes, invoices, and resolves without human intervention.
- **No proactive outbound AI intelligence** — campaigns are rule-based, not AI-triggered. AgentCore resolves: Proactive Outbound Engine.
- **No 4-layer persistent memory** — ticket history is not structured memory with semantic search. AgentCore resolves: 4-layer memory system.
- **No self-hosted option** — cloud SaaS only. AgentCore resolves: full self-hosted deployment.
- **No CRM lifecycle depth** — no Proposal, Quote, Booking, or Invoice engine. AgentCore resolves: 8 CRM lifecycle engines.
- **No WhatsApp UPI commerce** — WhatsApp messaging only, no payment flows. AgentCore resolves: full WhatsApp Commerce Engine.

Why No Competitor Can Serve All 15 Industries

The critical architectural requirement for a universal platform that serves Travel, BFSI, Healthcare, Real Estate, Legal, Education, Fitness, Automotive, Hospitality, Retail, Logistics, Home Services, Event Management, Photography, and NGOs — simultaneously — is:

1. **Configurable tool registry:** Each industry has different backend systems (HMS for hospitals, LOS for banks, PMS for hotels, TBO/TripJack for travel). The platform must dynamically register and route to industry-specific tools without code changes per vertical.

2. **Compliance engine:** BFSI needs SEBI/RBI/IRDAI compliance. Healthcare needs HIPAA/ABDM. Legal needs Bar Council rules. Real Estate needs RERA. The platform must enforce industry-specific compliance rules per tenant — not as bolt-ons but as core architecture.
3. **Memory with vertical context:** "Last time you called about your Bali trip" vs "Last time you called about your claim" vs "Last time you called about flat 503" — memory must understand industry-specific entities, not generic conversation history.
4. **WhatsApp flows per industry:** A travel booking flow looks different from a loan application flow, which looks different from an appointment booking flow. The WhatsApp commerce engine must be template-aware per industry.
5. **Self-hosted for data sovereignty:** Banks and hospitals will not send customer data to third-party clouds. This is non-negotiable in regulated Indian industries.
6. **Indian language depth:** 73% of India's 1.4 billion people communicate primarily in a language other than English. Any platform that cannot serve in Tamil, Telugu, Marathi, Hindi, Bengali, Gujarati, Kannada cannot serve 73% of the market.
7. **Cost architecture for Indian pricing:** Indian businesses cannot absorb \$0.10-0.30/min when INR voice cost must be competitive with human agents at ₹15,000/month.

No competitor has ALL seven. AgentCore has all seven.

2.5 What ALL Competitors Lack

Every competitor — startup or enterprise — shares fundamental gaps that AgentCore fills. The table below maps each gap across the full competitive spectrum:

Capability	Startup Tier (Retell/Vapi/Bland)	Enterprise Legacy (Genesys/Five9/NICE)	Enterprise AI (PolyAI/Cognigy)	AgentCore
Indian Language Depth (22 languages)	None	Minimal	Partial (not regional)	Full (Sarvam-native)
WhatsApp Commerce + UPI Payments	None	None	None	Full (Flows v2 + UPI)
4-Layer Persistent Memory	None	None	None	Redis + PostgreSQL + pgvector + Procedural
CRM Lifecycle Ownership (Lead→Invoice)	None	None	None	8 CRM lifecycle engines
Self-Hosted on Customer AWS	None or limited	None (cloud SaaS)	Partial (PolyAI: none; Cognigy: yes)	Full AWS self-hosted
Proactive AI Outbound	None	Basic dialer	None	AI-triggered intelligence

Capability	Startup Tier (Retell/Vapi/Bland)	Enterprise Legacy (Genesys/Five9/NICE)	Enterprise AI (PolyAI/Cognigy)	AgentCore
Document Intelligence (OCR/KYC)	None	None	None	Full OCR + validation
Multi-Agent Orchestration (7 agents/call)	None	None	None	LangGraph Supervisor + 7 workers
India-First Pricing (₹3-5/min)	No (USD only)	No (per-seat)	No (enterprise contracts)	Yes (INR pricing)
15-Industry Vertical Templates	None	None	Partial (Cognigy Marketplace)	Full 15-industry templates
DPDP Act 2023 Compliance	None	None	None	Built-in
TRAI Compliance (DND/disclosure)	None	None	None	Built-in
Unified Omnichannel Memory	None	Partial (session only)	None	Single memory across voice/chat/WhatsApp
Proposal + Quote + Invoice Engine	None	None	None	Full generation + tracking + approval
SMB to Enterprise Pricing Ladder	SMB only	Enterprise only	Enterprise only	₹5K/month to ₹10L/month
Cost per Minute (self-hosted)	\$0.07-0.31	\$0.018-0.025 (bare infra)	\$0.10-0.20 (managed)	\$0.039-0.077

Section 2A: Enterprise-Grade Requirements Checklist

This section defines what "enterprise-grade" means in the context of a contact center AI platform — and maps every requirement to AgentCore's current or planned implementation. This checklist is the standard by which Genesys, NICE CXone, Cognigy, and PolyAI are judged in enterprise procurement. AgentCore must clear every one of these bars.

2A.1 Security Certifications

SOC 2 Type II

What it is: System and Organization Controls 2, Type II audit — a rigorous third-party audit verifying that the platform's security controls (confidentiality, integrity, availability, processing integrity, and privacy) have operated effectively over a minimum 6-month observation period.

Why it matters: Required by 94% of enterprise B2B procurement processes. Without SOC 2 Type II, most Fortune 500 and large enterprise customers cannot legally onboard a SaaS vendor. Any healthcare, financial services, or legal customer will ask for the SOC 2 report as a precondition to contract.

Who has it: Genesys ✓, NICE CXone ✓, Cognigy ✓, PolyAI ✓, Zendesk ✓, Intercom ✓, Vapi ✗, Retell ✗, Bland ✗

AgentCore plan:

- Phase 1 (Months 1-6): Implement SOC 2-aligned security controls (access logging, encryption, change management, incident response)
- Phase 2 (Months 7-12): Engage SOC 2 audit firm (Drata, Vanta, or Secureframe for automated evidence collection)
- Phase 3 (Months 13-18): Achieve SOC 2 Type II certification
- Infrastructure readiness: AWS infrastructure is SOC 2 compliant — the audit covers AgentCore's software and operational controls layered on top
- Fast path: Use Vanta (automated compliance monitoring) from Day 1 to build evidence continuously — reduces audit preparation from 6 months to 6 weeks

Implementation in Claude Phases: Phase 9 (AWS Infrastructure) expanded to include Vanta integration, security control documentation, and compliance evidence collection automation.

HIPAA BAA (Business Associate Agreement)

What it is: Under US HIPAA (and applicable to any platform handling Protected Health Information of US patients), a Business Associate Agreement is a legal contract between the covered entity (healthcare provider) and the business associate (AgentCore) that establishes permitted uses of PHI and security obligations.

Why it matters: Every healthcare system, hospital, clinic, telemedicine platform, diagnostic lab, and pharmacy deploying AgentCore will require a signed HIPAA BAA before handling any patient information. Without it, the healthcare vertical is legally inaccessible in the US market.

Who has BAA available: AWS ✓, OpenAI ✓, Deepgram ✓, ElevenLabs ✓, Genesys ✓, NICE CXone ✓, Cognigy ✓, PolyAI ✓, Vapi ✓ (at \$1,000/month premium)

AgentCore plan:

- Obtain BAAs from all sub-processors: AWS (already available), OpenAI (available for Enterprise), Deepgram (available), ElevenLabs (available for Enterprise), Sarvam (India-based, DPDP compliant)
- Create AgentCore HIPAA BAA template (legal review required)
- Deploy HIPAA-compliant tenant infrastructure: dedicated RDS instance, dedicated S3 bucket, VPC isolation from other tenants, no shared compute
- PHI handling rules: flag PHI fields in conversation transcript, apply field-level encryption (AES-256), restrict access to authorized roles only, automatic redaction in logs
- Minimum necessary data principle: agents only receive PHI fields required for the specific task
- Audit log for every PHI access event: who accessed, what field, at what time
- Available as: dedicated HIPAA-compliant infrastructure tier (additional cost, separate from standard multi-tenant)

PCI-DSS 4.0 (Payment Card Industry Data Security Standard)

What it is: The Payment Card Industry Data Security Standard version 4.0, the mandatory security framework for any organization that stores, processes, or transmits cardholder data (credit/debit card numbers, CVV, expiry dates).

Why it matters: BFSI (banking, insurance, fintech), retail/e-commerce, hospitality (hotel payment), and any industry where customers pay by card over voice (IVR payment flows) requires PCI-DSS compliance. Without it, the platform cannot handle card payments in any industry vertical.

PCI-DSS for voice AI specifically:

- **DTMF masking:** When a customer enters card numbers via keypad, the DTMF tones must not be recorded or logged
- **CVV prohibition:** CVV/CVV2 codes must never be stored — even temporarily in memory, logs, or recording transcripts
- **Call recording pause:** Call recording must be paused (or DTMF tones suppressed) during card number collection
- **Scope reduction via tokenization:** Ideally, redirect card data to a PCI-compliant payment processor (Stripe, Razorpay, Paytm Business) without AgentCore ever touching raw card data
- **Network segmentation:** Payment processing components must be isolated in a separate network segment

AgentCore plan:

- **Default architecture:** never handle raw card data — redirect to Razorpay/Stripe tokenization API during payment steps

- DTMF masking: suppress DTMF tones from call recordings when payment step is active (LiveKit recording pause API)
- PCI scope reduction: integrate with Razorpay PCI-DSS Level 1 certified payment page (voice IVR payment via Razorpay IVR API)
- Transcript redaction: replace card number patterns with [CARD-XXXX] in all stored transcripts
- PCI DSS SAQ-A compliance (the lightest tier — applicable when card data collection is fully outsourced to a compliant processor)
- Available as: PCI-DSS compliant deployment option for BFSI, retail, and hospitality tenants

GDPR / CCPA Data Residency

What it is: The General Data Protection Regulation (GDPR, EU) and California Consumer Privacy Act (CCPA, US) impose strict requirements on how personal data of EU/US residents is processed, stored, and transferred.

Why it matters: Any AgentCore customer serving EU-resident customers must comply with GDPR. Failure results in fines of up to 4% of global annual revenue (GDPR) or \$7,500/intentional violation (CCPA). As AgentCore scales to global markets, EU and US regulatory compliance becomes a commercial necessity.

GDPR requirements for AgentCore:

- **Right to Access:** Customer can request all data AgentCore holds about them
- **Right to Erasure ("right to be forgotten"):** Customer can request deletion of all personal data across all 4 memory layers + all transcript storage
- **Data Portability:** Customer data exportable in machine-readable format (JSON/CSV)
- **Consent management:** Explicit consent required before data processing — call recording consent (already in TRAI section), WhatsApp message consent, data retention consent
- **Data Processing Agreements (DPAs):** Required with all sub-processors (AWS, OpenAI, Deepgram, ElevenLabs, Sarvam)
- **Data residency:** EU customer data must stay in EU region — AWS EU regions (Frankfurt eu-central-1, Dublin eu-west-1)
- **Breach notification:** 72-hour notification requirement to supervisory authority

AgentCore plan:

- **Multi-region AWS deployment:** India (ap-south-1 Mumbai), EU (eu-central-1 Frankfurt), US (us-east-1 N. Virginia) — data residency enforced at tenant level
- **Right to Erasure implementation:** `DELETE /api/customer/{id}/gdpr-erase` — cascading delete across Redis, PostgreSQL, pgvector, audit trail (except legally required retention), S3 recordings
- **Data export:** `GET /api/customer/{id}/data-export` — full JSON export of all customer data
- **Consent ledger:** immutable record of every consent event in the audit trail

- DPAs executed with all sub-processors
- Privacy by design: no personal data in logs by default — structured logging with PII masking

2A.2 Workforce Engagement Management (WEM)

Workforce Engagement Management is a critical feature category for any platform serving contact centers with 20+ human agents. Genesys, NICE CXone, and Zendesk have dedicated WEM modules. AgentCore must provide equivalent capability to compete for contact center deals.

Agent Scheduling & Workforce Management

What enterprise customers need:

- **Demand forecasting:** Predict call/chat volume for each 15-minute interval based on historical data, seasonal patterns, and campaign schedules
- **Agent scheduling:** Match scheduled agent headcount to forecasted demand, respecting shift preferences, skills, and labor rules
- **Adherence tracking:** Real-time comparison of scheduled vs actual agent state (available, on-call, break, training)
- **Intra-day management:** Re-forecast and re-schedule within the current day based on actual volume

AgentCore plan:

- Demand forecasting engine using historical call volume data from the AgentCore analytics database
- Schedule generation: optimization model (integer programming or heuristic) to match demand forecast with available agents
- Agent state tracking: real-time agent status in the supervisor dashboard
- Adherence alerts: supervisor notification when agent is out of adherence for >5 minutes
- Integration with HR systems (optional): sync agent schedules from BambooHR, Workday, or custom HR system via webhook

Quality Scoring & Conversation QA

What enterprise customers need:

- **Conversation scoring rubrics:** Configurable scorecards (e.g., "Did agent confirm booking details? 10pts. Did agent offer upsell? 5pts. Did agent resolve without escalation? 20pts.")
- **Auto-scoring:** AI automatically scores every AI-handled conversation against the rubric — no manual review required

- **Calibration sessions:** QA team reviews a sample of scored calls to validate AI scoring accuracy
- **Agent coaching:** Supervisors assign coaching tasks based on low-scoring conversations
- **Quality trends:** Track quality score over time per agent, per team, per vertical

AgentCore plan:

- **QA Scorecard Engine:** configurable rubric per tenant (JSON-defined criteria, weights, scoring logic)
- **Auto-scoring pipeline:** post-call evaluation — conversation transcript + tool call log evaluated against rubric by Claude
- **Human QA review:** supervisor UI shows auto-scored conversations, allows score override with notes
- **Coach assignment:** auto-assign coaching if score < threshold (configurable)
- **Quality analytics dashboard:** scores over time, top issues, team benchmarks
- **Agent self-review:** agents can review their own scored calls with annotations

Agent Gamification & Performance Management

What enterprise customers need:

- **Leaderboards:** Rankings by resolution rate, CSAT, response time, quality score
- **Badges and achievements:** Reward top performers, new skills mastered, perfect QA weeks
- **Performance targets:** Individual and team OKRs with real-time progress tracking
- **Recognition programs:** Public recognition for high performers in team dashboard
- **Manager coaching tools:** 1:1 performance review templates, coaching notes, goal tracking

AgentCore plan:

- **Agent performance hub in supervisor dashboard:** individual metrics, team ranking, trend visualization
- **Badge system:** configurable achievement criteria with automatic award
- **Monthly performance reports:** auto-generated PDF performance summary per agent
- **Team leaderboard:** configurable visibility (transparent or private)
- **Manager coaching notes:** structured 1:1 template linked to conversation examples

2A.3 Supervisor & Live Call Management

Live Monitoring Suite

Enterprise contact centers require real-time visibility and control over every active interaction. This is a non-negotiable feature for any deal with a contact center operations team.

Supervisor capabilities AgentCore must provide:

Feature	Description	Implementation
Live Queue Dashboard	Real-time view of all active calls, queue depth, wait times, agent states	WebSocket-based live dashboard, refresh every 2 seconds
Silent Monitor	Supervisor joins call audio in listen-only mode — agent and customer cannot hear supervisor	LiveKit listener participant with muted publisher
Whisper Coaching	Supervisor speaks only to agent (not customer) during live call	LiveKit separate audio track routed only to agent
Barge-in	Supervisor takes over the call — AI is paused, supervisor speaks to customer	LiveKit supervisor publishes to main room, AI agent muted
Call Intercept	Supervisor can pull a call from an AI agent or queue to handle directly	Intercept API endpoint with notification to agent dashboard
Live Transcript	Real-time streaming transcript visible in supervisor dashboard	Deepgram streaming → WebSocket → supervisor UI
Real-Time Sentiment	Sentiment score updated every 10 seconds during live call	Transformer-based classifier on streaming transcript
Alert Triggers	Auto-alert supervisor when: sentiment drops <30%, call duration >10min, customer says trigger phrase	Redis pub/sub alert system
Quality Flags	Supervisor flags a call for QA review during or after the call	Flag stored in conversation record, visible in QA queue

Technical implementation:

```
class SupervisorController:
    """
    Real-time supervisor control for live calls.
    Integrates with LiveKit room management API.
    """

    async def silent_monitor(self, call_id: str, supervisor_id: str):
        """Add supervisor as silent listener to active call."""
        room_name = f"call_{call_id}"
        # Create supervisor token with subscribe-only permissions
        supervisor_token = livekit_api.create_access_token(
            room=room_name,
            identity=f"supervisor_{supervisor_id}",
            can_publish=False, # Cannot speak
            can_subscribe=True # Can listen
        )
        return {"join_token": supervisor_token, "mode": "silent_monitor"}

    async def whisper_coach(self, call_id: str, supervisor_id: str):
```

```

"""Enable whisper coaching – supervisor audible to agent only."""
# Supervisor publishes to dedicated agent whisper track
# Customer receives main audio track only
supervisor_token = livekit_api.create_access_token(
    room=room_name,
    identity=f"supervisor_{supervisor_id}",
    can_publish=True,
    can_subscribe=True,
    metadata=json.dumps({"mode": "whisper", "visible_to": "agent_only"})
)
return {"join_token": supervisor_token, "mode": "whisper_coach"}

async def barge_in(self, call_id: str, supervisor_id: str):
    """Barge in – supervisor takes over call from AI."""
    # Mute AI agent, promote supervisor as main speaker
    await livekit_api.mute_participant(f"call_{call_id}", "ai_agent")
    supervisor_token = livekit_api.create_access_token(
        room=room_name,
        identity=f"supervisor_{supervisor_id}",
        can_publish=True,
        metadata=json.dumps({"mode": "barge_in"})
    )
    # Notify human agent dashboard of barge-in event
    await notify_agent_dashboard(call_id, "supervisor_barge_in", supervisor_id)
    return {"join_token": supervisor_token, "mode": "barge_in"}

```

2A.4 Conversation Intelligence & Analytics

A/B Testing Framework

What it is: The ability to run controlled experiments on AI prompts, conversation flows, greetings, and escalation logic — measuring containment rate, CSAT, resolution time, and conversion rate as primary metrics.

Why it matters: Without A/B testing, prompt optimization is guesswork. With A/B testing, teams can systematically improve containment rate from 60% to 80%+ over time, reduce escalation rate, and increase conversion on upsell offers.

Enterprise players with this: Genesys (Genesys DX experiments), NICE CXone (A/B testing in Digital First), Cognigy (Flow version control with traffic splitting)

AgentCore A/B Testing Framework:

```

class PromptExperiment:
    """
    A/B testing framework for AI conversation elements.
    Supports prompt variants, greeting variants, escalation threshold variants.
    """
    experiment_id: str
    name: str
    tenant_id: str

```

```

# Traffic split configuration
variants: List[ExperimentVariant] # Each with weight (0.0-1.0), must sum to 1.0

# Targeting rules
target_industries: List[str] # e.g., ["travel", "healthcare"]
target_call_types: List[str] # e.g., ["inbound", "outbound"]

# Metrics to track
primary_metric: str # "containment_rate" | "csat" | "resolution_time" | "conversion_rate"
secondary_metrics: List[str]

# Statistical significance threshold
min_sample_size: int # Minimum calls per variant before declaring winner
significance_level: float # Default 0.95 (95% confidence)

class ExperimentVariant:
    variant_id: str
    name: str
    weight: float # Traffic proportion (0.0-1.0)

    # What changes in this variant
    system_prompt_override: Optional[str]
    greeting_template_override: Optional[str]
    escalation_threshold_override: Optional[float]
    tools_enabled_override: Optional[List[str]]

```

Metrics tracked per variant:

- Containment rate (calls resolved without human escalation)
- Average handle time (AHT)
- CSAT score (post-call IVR or WhatsApp survey)
- First-contact resolution (FCR)
- Escalation rate
- Conversion rate (for sales flows)
- Sentiment score at call end

Statistical significance engine: Two-proportion z-test for containment rate; Welch's t-test for continuous metrics (AHT, sentiment). Auto-declare winner when significance level reached. Auto-notify experiment owner via dashboard + email.

Predictive Routing Engine

What it is: ML-based prediction of the best agent match for each incoming call — not just skill-based routing, but predicting which specific agent will achieve the best outcome (highest CSAT, fastest resolution, lowest repeat contact) for each specific customer.

Why enterprise customers need it: Genesys reports 15-30% improvement in CSAT and 20% reduction in handle time when predictive routing is used vs skill-based routing. It is a key differentiator in enterprise RFPs.

AgentCore Predictive Routing:

```
class PredictiveRouter:
    """
    ML-based prediction of optimal human agent for escalated calls.
    Used when AI agent decides to escalate to a human.
    """

    async def predict_best_agent(
        self,
        customer_id: str,
        conversation_context: ConversationContext,
        available_agents: List[AgentProfile]
    ) -> RankingResult:
        """
        Predict best agent match using:
        1. Customer history with each agent (if any prior interactions)
        2. Agent skill match score vs conversation intent
        3. Agent current load (calls in queue, avg handle time)
        4. Customer sentiment – if negative, route to senior agent
        5. Customer value tier – VIP customers to senior agents
        6. Language match (Tamil speaker → Tamil-proficient agent)
        """

        features = await self._extract_features(
            customer_id, conversation_context, available_agents
        )

        # Gradient boosting model trained on historical outcomes
        # Features: customer_sentiment, intent_category, customer_tier,
        #           agent_skill_score, agent_queue_depth, prior_interaction_outcome,
        #           language_match, time_of_day, day_of_week
        scores = self.gbm_model.predict(features)

        return RankingResult(
            ranked_agents=sorted(zip(available_agents, scores), key=lambda x: x[1], reverse=True),
            predicted_csat=scores.max(),
            routing_reason=self._explain_top_choice(features, scores)
        )
```

Knowledge Base Management

What it is: A structured repository of articles, FAQs, SOPs, and product documentation that the AI agent references during conversations — with tools for creating, versioning, and optimizing articles based on usage patterns.

Why it matters: Zendesk Guide, NICE CXone Knowledge, and Cognigy Knowledge AI are key competitive advantages in enterprise deals. Without a managed knowledge base, agents hallucinate answers or give outdated information. A well-managed knowledge base directly drives containment rate.

AgentCore Knowledge Base Engine:

Knowledge article structure:

```
class KnowledgeArticle:
    article_id: str
    tenant_id: str
    title: str
    content: str # Markdown
    category: str
    tags: List[str]
    industry: str

    # Versioning
    version: int
    published_at: datetime
    last_reviewed_at: datetime
    reviewed_by: str

    # Performance metrics
    retrieval_count: int # How many times retrieved during AI conversations
    helpful_count: int # How many times customer confirmed it resolved their issue
    unhelpful_count: int # How many times customer was not satisfied after retrieval
    helpfulness_rate: float # helpful / (helpful + unhelpful)

    # AI-optimized fields
    embedding_vector: List[float] # pgvector embedding for semantic search
    search_keywords: List[str] # Extracted keywords for keyword search
    auto_summary: str # Claude-generated one-paragraph summary
```

Gap detection — the most valuable capability:

- When the AI agent fails to find a relevant article for a customer query: log the query as an "unanswered question"
- Weekly report: "Top 20 questions customers asked that had no article — create these articles"
- Article suggestion quality: track which articles are retrieved but marked unhelpful by customers
- Content freshness: alert knowledge manager when article is >90 days old without review

Article management UI (supervisor dashboard):

- Create/edit articles with rich text editor
- AI-assisted drafting: Claude writes draft article from a topic prompt, human reviews and approves
- Version history with diff view
- Bulk import from existing help center (URL scraping)
- Analytics: top articles, gap report, helpfulness trends
- Approval workflow: draft → review → published states

2A.5 SLA Framework & Uptime Guarantees

Enterprise customers require contractual uptime guarantees and defined service level agreements for support response times. This is a standard section in any enterprise software procurement.

Platform Uptime SLA

Tier	Monthly Uptime SLA	Max Downtime/Month	Support Response SLA
Starter	99.5%	3.65 hours	24-hour email
Growth	99.9%	43.8 minutes	4-hour email, 8-hour chat
Scale	99.95%	21.9 minutes	1-hour chat, 4-hour email
Enterprise	99.99%	4.4 minutes	15-min phone/chat + dedicated CSM

How 99.99% Uptime Is Achieved

Multi-AZ deployment:

- LiveKit Server: deployed across 2 AWS Availability Zones with automatic failover (<30 seconds)
- Application load balancer: health checks every 5 seconds, automatic instance replacement
- RDS PostgreSQL: Multi-AZ with automatic failover (<60 seconds)
- ElastiCache Redis: Multi-AZ replication with automatic failover
- ECS agent workers: minimum 2 tasks running at all times, auto-scaling adds capacity in <90 seconds

Circuit breaker pattern:

```
class CircuitBreaker:
    states = {
        'closed': 'Normal operation – all requests pass through',
        'open': 'Failure threshold exceeded – requests fail immediately with fallback',
        'half_open': 'Testing recovery – limited requests pass through to test service health'
    }

    thresholds = {
        'failure_rate': 0.50,    # Open circuit if >50% of requests fail
        'slow_call_rate': 0.50,  # Open circuit if >50% of requests >2s
        'wait_duration_open': 30 # Try half-open after 30 seconds
    }
```

Failure mode graceful degradation:

1. LLM failure → fallback to cached response templates → human escalation
2. STT failure → fallback to DTMF input → human escalation
3. TTS failure → fallback to pre-recorded audio prompts

4. Memory failure → continue with session-only context (no long-term recall)
5. Complete platform failure → call forwarded to human agent queue via Exotel failover routing

Service Credits for SLA Breach

SLA Breach Severity	Credit Applied
Uptime 99.0-99.49%	10% of monthly invoice
Uptime 95.0-98.99%	25% of monthly invoice
Uptime below 95%	50% of monthly invoice
Incident response SLA missed	5% of monthly invoice per incident

2A.6 Enterprise Implementation Timeline Guarantee

Enterprise customers require a contractual implementation timeline — not just an estimate. This is a differentiator vs legacy platforms (Genesys: 3-6 months, NICE: 6-18 months) and a key selling point.

AgentCore Guaranteed Implementation Timeline

Deployment Scope	Timeline Guarantee	What's Included
Single vertical, API mode	4 weeks	Voice + Chat + WhatsApp on one industry vertical
Single vertical, white-label	6 weeks	All channels + branded portal + CRM lifecycle
Multi-vertical, white-label	10 weeks	2-3 verticals + all engines + compliance config
Full enterprise, self-hosted	12 weeks	All 24 engines + AWS deployment + compliance certification

Guarantee terms: If AgentCore is not live in production within the contractual timeline, tenant receives 1 month free service per week of delay.

Section 2B: Product Market Fit Analysis

This section answers three questions that determine whether AgentCore can succeed in the market:

1. Who are the buyers? (segmentation and ICP)
2. Are we positioned to win vs each tier? (competitive fit)
3. Is the enterprise grade sufficient? (readiness assessment)

2B.1 Market Segmentation & Ideal Customer Profile (ICP)

AgentCore serves three distinct market segments with different buying processes, pricing sensitivities, and feature requirements.

Segment 1: SMB (Small-Medium Business)

Definition: 1-50 seats, <\$5M revenue, typically a single-location or regional business

Industries: Local travel agencies, photography studios, independent clinics, coaching centers, boutique gyms, automotive service centers, restaurants, home service companies

Buying behavior:

- Self-serve signup or low-touch sales
- Decision made by owner/founder in <2 weeks
- Price-sensitive: budget ₹2,000-20,000/month
- Wants fast setup: "first call working in 1 hour"
- Judges value by: calls answered after hours, leads captured, time saved

AgentCore ICP (SMB):

- Travel agency with 2-5 staff, handling 50-200 calls/month
- Photography studio owner who misses 30% of booking calls
- Fitness studio with 200+ members, managing class booking by phone
- Independent clinic with appointment scheduling problem

Product-market fit indicators for SMB:

- ✓ Starter plan at ₹5,000/month with no setup fee
- ✓ First voice agent working in <30 minutes (template-based)
- ✓ Handles after-hours calls without hiring staff
- ✓ WhatsApp booking confirmation within 2 minutes of call
- ⚠ Gap: Self-serve onboarding not yet built — currently requires implementation support

Segment 2: Mid-Market

Definition: 50-500 seats, \$5M-\$100M revenue, multi-location or regional chain

Industries: Tour operator chains (50+ offices), hospital groups (5-20 locations), real estate developers (multiple projects), regional insurance companies, hotel chains, school/coaching chains, regional banks/NBFCs

Buying behavior:

- Inside sales + solution demo

- Decision involves 2-4 stakeholders (IT head, CX head, Finance head)
- Timeline: 3-8 weeks procurement to close
- Budget: ₹50,000-5,00,000/month
- Wants: proven ROI metrics, integration with existing systems (Zoho, HubSpot, HMS, LOS), compliance for regulated verticals

AgentCore ICP (Mid-Market):

- Travel operator with 5 branches, 20 agents, handling 1,000 calls/day
- Hospital group with 8 locations, complex appointment routing by doctor specialty
- Real estate developer with 3 active projects, 500+ inquiries/month
- Regional NBFC with 50 field agents and 10,000 customer service calls/month

Product-market fit indicators for Mid-Market:

- ✓ Growth plan at ₹50,000-2,00,000/month
- ✓ API integration with Zoho, HubSpot, custom HMS/LOS via webhook
- ✓ HIPAA/BFSI/RERA compliance for regulated vertical deals
- ✓ Multi-location tenant architecture with per-location analytics
- ✓ 4-week implementation guarantee
- ✓ Dedicated onboarding support
- ⚠ Gap: Pre-built CRM integrations (Zoho, HubSpot) not yet packaged as one-click connectors

Segment 3: Enterprise

Definition: 500+ seats, \$100M+ revenue, national or multi-national operation

Industries: Large private banks, insurance companies, hospital networks, national retail chains, telecom companies, large edtech platforms, FMCG companies with field sales force

Buying behavior:

- Enterprise sales + security review + legal review + procurement
- Decision committee of 5-10 stakeholders
- Timeline: 3-6 months from first demo to signed contract
- Budget: ₹5,00,000-50,00,000/month
- Requires: SOC 2 Type II, dedicated infrastructure, 99.99% SLA, on-site support, compliance audit support, custom contract terms

AgentCore ICP (Enterprise):

- Top-10 private sector bank in India with 2 million customer service calls/month
- Large general insurance company with 5,000 renewal calls/day

- Hospital network with 50 locations and 10,000 appointments/day
- National e-commerce company with 100,000 order support calls/month

Product-market fit indicators for Enterprise:

- ✓ Enterprise plan with dedicated infrastructure option
- ✓ Self-hosted AWS deployment for data sovereignty
- ✓ 9999% uptime SLA
- ✓ 12-week implementation guarantee
- ⚠ Gap: SOC 2 Type II certification not yet achieved (Month 13-18 roadmap)
- ⚠ Gap: HIPAA BAA formal process not yet documented
- ⚠ Gap: PCI-DSS certification not yet achieved

2B.2 Competitive Positioning Matrix

How AgentCore wins in each competitive scenario:

When Competing Against	Our Winning Argument	Key Differentiators
Retell / Vapi / Bland	"Why pay 3-5 separate platforms when AgentCore gives you voice + chat + WhatsApp + CRM lifecycle + memory + compliance in one platform at \$0.039-0.077/min?"	24 engines, memory, WhatsApp commerce, Indian languages, self-hosted
Genesys / NICE CXone	"Same capabilities at 1/5th the cost, deployed in 12 weeks vs 6 months, with full self-hosted option for data sovereignty"	Price, speed, Indian language, self-hosted, DPDP compliance
PolyAI / Cognigy	"All the enterprise-grade AI with Indian language depth, Indian pricing, Indian compliance, and the full CRM lifecycle — not just the conversation layer"	Indian focus, CRM ownership, WhatsApp commerce, DPDP, price
Salesforce Agentforce	"Works with any CRM, not just Salesforce. Proactive outbound, Indian languages, self-hosted, and the full CRM lifecycle at a fraction of the add-on cost"	CRM-agnostic, outbound, Indian languages, price
Zendesk / Freshdesk	"Full autonomous voice AI, not just tickets. Agent doesn't just route — it books, proposes, invoices, and resolves. Indian language voice support Zendesk cannot match."	Autonomous action, voice AI, Indian languages, WhatsApp commerce
AWS Connect DIY	"Same AWS infrastructure, pre-built and production-ready in 12 weeks instead of 12 months. All the AWS benefits with none of the assembly required."	Speed, pre-built platform, 24 engines, support SLA

2B.3 Go-To-Market Strategy (SMB vs Mid-Market vs Enterprise)

SMB GTM: Product-Led Growth

Strategy: Self-serve signup → free trial (14 days, 500 minutes) → convert to Starter plan → expand to Growth when volume increases

Channels:

1. SEO content: "AI voice agent for travel agents", "WhatsApp automation for gyms", "appointment booking AI for clinics"
2. Google Ads: industry-specific targeting (travel agency software, hospital appointment system)
3. Product directory listings: G2, Capterra, Product Hunt, SaaSwothy
4. WhatsApp Business API partner ecosystem: AiSensy, Interakt, Wati partner channels
5. Industry associations: TAAI (Travel Agents Association of India), IMA (Indian Medical Association), CREDAI (real estate)

Conversion triggers:

- 14-day free trial — no credit card required
- First AI call set up in <30 minutes with industry template
- After-hours call capture = immediate visible ROI

Mid-Market GTM: Inside Sales + Demo

Strategy: Inbound demo request from content → BDR follow-up → live demo → pilot (30 days, paid) → full deployment

Channels:

1. Content marketing: industry-specific ROI calculators ("how much revenue are you losing to missed calls?")
2. LinkedIn: target CX heads, IT heads, and founders in target industries
3. Referral program: 20% commission to referring partners (system integrators, CRM consultants)
4. Events: industry conferences (TAAI convention, ASSOCHAM insurance summit, CII healthcare summit)
5. System integrator partnerships: Zoho partner network, HubSpot agency partners

Enterprise GTM: Account-Based Sales

Strategy: Identify 100 target accounts → personalized outreach → discovery call → custom demo → RFP response → pilot → contract

Target account selection criteria:

- 500+ customer service calls per day
- Already using contact center software (switching, not greenfield)
- In a regulated vertical (BFSI, healthcare) where compliance differentiation wins
- India-HQ company (DPDP compliance is critical for them)

Enterprise sales enablement:

- Custom ROI model: "You currently spend ₹X on 50 agents handling Y calls. AgentCore handles 70% of those calls at \$0.050/min = ₹Z savings."
- Security questionnaire responses pre-drafted for SOC 2, HIPAA, GDPR
- Reference customers in same industry (case studies from HelpRevX's own deployments)
- Proof-of-concept (POC) program: 30-day POC with real customer calls, measured vs baseline

2B.4 Product Market Fit Evidence & Validation

Before external sales, AgentCore will generate PMF evidence from HelpRevX's own product deployments:

Product	Baseline Metric	AgentCore-Powered Target	Validation Method
ItinerarySuite	60% calls answered in business hours	100% calls answered 24/7	Call answer rate tracking
ItinerarySuite	Lead to proposal: 3 days avg	Lead to proposal: <2 hours (AI-assisted)	CRM time-to-proposal metric
PhotoAdda	40% booking inquiries convert	65% convert with 24/7 response + follow-up	Booking conversion rate
Aayiq	Manual expense categorization: 4 hours/week	Automated: <30 minutes	Time tracking metric
HelpRevX	Manual supplier quote requests: 2 hours	Automated via voice + WhatsApp: 15 minutes	Process time metric

These are the "founding customer" case studies that become the core of AgentCore sales decks, website case study pages, and G2 reviews.

2B.5 Competitive Gap Closure Roadmap

Based on the analysis above, these gaps must be closed to compete at enterprise tier:

Gap	Severity	Planned Closure	Timeline
SOC 2 Type II certification	HIGH — blocks enterprise deals	Vanta implementation + audit firm engagement	Month 13-18
HIPAA BAA formal process	HIGH — blocks healthcare enterprise	Legal review + sub-processor BAAs	Month 6
PCI-DSS SAQ-A compliance	MEDIUM — blocks BFSI/retail enterprise	Razorpay integration + DTMF masking	Month 8
Workforce scheduling module	MEDIUM — needed for 20+ agent deployments	WFM module in supervisor dashboard	Month 10
Conversation A/B testing	MEDIUM — needed for optimization	A/B testing engine	Month 9
Predictive routing ML model	MEDIUM — needed for enterprise routing	Training data from first 6 months of calls	Month 12
SOC 2 Type I (interim)	LOW — provides evidence of security posture	Achievable in 3-4 months	Month 9
GDPR data residency	LOW — EU customers, not immediate priority	Multi-region AWS with tenant-level residency	Month 10
Self-serve onboarding	MEDIUM — required for SMB GTM	Sign-up flow + template wizard	Month 6
Pre-built CRM connectors (Zoho, HubSpot)	MEDIUM — mid-market deals	OAuth connectors for top 3 CRMs	Month 7

Section 3: Product Definition -- All 24 Engines

3.1 The Platform Definition

AgentCore is a self-hosted voice AI infrastructure combined with a CRM intelligence layer. It is the bridge between raw AI capabilities (speech recognition, language models, speech synthesis) and real business outcomes (leads qualified, proposals sent, quotes accepted, bookings confirmed, cases resolved) -- across ANY industry.

AgentCore is NOT:

- A chatbot -- it is a full-stack voice and omnichannel intelligence platform
- A simple IVR replacement -- it understands intent, maintains context, takes actions
- A wrapper around OpenAI -- it orchestrates multiple AI providers, databases, and business systems
- A demo project -- it is production infrastructure designed for multi-tenant, high-concurrency deployment

- A travel-only product -- it is an industry-agnostic platform with configurable vertical deployments

AgentCore IS:

- An intelligence platform that makes every CRM product voice-native, regardless of industry
- A self-hosted infrastructure that gives customers complete data sovereignty
- A multi-agent orchestration system that handles complex business workflows in real-time
- A memory-persistent system that knows every customer across every interaction
- An omnichannel engine that unifies voice, chat, and WhatsApp into a single experience
- A horizontal platform serving 15+ industries through configurable tool registrations

3.2 Three Deployment Modes

Mode A: Embedded in Your Product

AgentCore runs as the voice/chat layer inside a vertical CRM product. Example: ItinerarySuite uses AgentCore to provide voice-based lead qualification, proposal generation, and booking support. The end user never sees "AgentCore" -- they see ItinerarySuite's voice assistant. Similarly, a healthcare CRM embeds AgentCore and patients interact with the clinic's virtual assistant.

Mode B: White-Label to Others

AgentCore is licensed to agencies or CRM builders who rebrand it as their own voice AI solution. Example: a digital marketing agency licenses AgentCore, adds their branding, and sells "VoiceBot Pro" to their clients across multiple industries. A BFSI technology company white-labels it as their voice banking solution.

Mode C: Standalone SaaS

AgentCore is sold directly as a platform product to enterprises, contact centers, or CRM builders who want to add voice AI to their existing systems. Example: a real estate CRM company subscribes to AgentCore's API, registers their custom tools, and immediately has voice AI capabilities. A hospital chain integrates AgentCore into their existing HMS.

3.3 Engine Groups Overview

AgentCore V2 consists of 24 engines organized into 5 groups:

```
GROUP A -- COMMUNICATION ENGINES (4):  Voice, Chat, WhatsApp, Email
GROUP B -- INTELLIGENCE ENGINES (3):   Memory, Multi-Agent, Proactive Outbound
GROUP C -- CRM LIFECYCLE ENGINES (8):  Lead, Proposal, Quote, Case, Approval,
                                         Booking, Invoice, Refund/Cancellation
GROUP D -- OPERATIONS ENGINES (6):     Document Intelligence, Analytics,
                                         Supplier Mgmt, Campaign, NPS/Feedback,
                                         Compliance, Tenant Management
GROUP E -- VERTICAL ENGINES (2):       Itinerary Builder, Expense Capture
```

Every engine in Groups A through D is industry-agnostic. They work identically whether the tenant is a travel agency, a hospital, a bank, or a gym. Only Group E contains vertical-specific engines, and even these serve as templates for building industry-specific extensions.

3.4 Group A -- Communication Engines (4)

Engine 1: VoiceAgent Engine

Technology: LiveKit Agents + Hybrid Cascaded/Speech-to-Speech Pipeline

The VoiceAgent Engine is the core real-time voice conversation service. It manages WebRTC connections, speech-to-text, language model inference, text-to-speech, and audio streaming -- all with sub-second latency. It serves every industry identically -- the same engine that handles a travel enquiry handles a medical appointment booking, an insurance claim query, or a legal consultation intake.

Key capabilities:

- Hybrid pipeline: intelligently routes between cascaded (STT -> LLM -> TTS) and speech-to-speech (OpenAI Realtime API) based on conversation requirements
- Sub-800ms first-token latency in cascaded mode
- Sub-300ms response latency in S2S mode for natural conversation
- Barge-in support: users can interrupt the AI mid-sentence
- Turn detection using LiveKit's MultilingualModel()
- Silence detection and graceful timeout handling
- Call recording with configurable retention policies
- DTMF tone detection for legacy system compatibility
- V2 UPDATE: Sarvam STT for Indian regional, ElevenLabs Flash API (not platform) for TTS

Cross-Industry Applications:

- Travel: "Tell me about Bali packages for a family of 4"
- Healthcare: "I need to book an appointment with Dr. Sharma for tomorrow"
- BFSI: "What is the status of my home loan application?"
- Legal: "I need to consult a lawyer about a property dispute"
- Automotive: "I want to schedule a test drive for the new Nexon this Saturday"

Engine 2: ChatAgent Engine [UPDATED -- ADVANCED ARCHITECTURE]

Technology: WebSocket + REST API + GPT-4.1-mini + Sarvam + ElevenLabs + Supabase Realtime

The ChatAgent Engine is a fully agentic, multilingual, omnichannel chat platform that handles the complete conversation lifecycle from initial contact through to resolution -- including intelligent escalation to live agents, virtual queueing, callback scheduling, AI co-pilot

assistance for agents, cobrowse, and real-time multilingual interpretation. It is not a simple chatbot. It is a complete customer engagement system.

AgentCore — Advanced Chat Agent Architecture

Comprehensive Design: Live Agent Escalation + Callback System

All Scenarios, All Edge Cases, Production-Ready

THE FOUR CORE SCENARIOS

SCENARIO 1: Customer requests live agent AND agent IS available

SCENARIO 2: Customer requests live agent AND NO agent is available

SCENARIO 3: Customer requests a callback

SCENARIO 4: AI decides proactively to escalate (without customer asking)

PLUS: Sub-scenarios for each (business hours, off-hours, VIP, priority, multi-language)

SCENARIO 1 — LIVE AGENT REQUEST: AGENT IS AVAILABLE

Detection Layer (How the AI Knows Customer Wants a Human)

EXPLICIT TRIGGERS (customer says exactly this):

- "Talk to a human" / "I want a human agent" / "Connect me to a person"
- "Give me a representative" / "Agent please" / "Can I speak to someone"
- "This is not helping" / "Your bot is useless" (frustration signal)
- "I need to speak to a manager" / "I want to escalate"
- "Real person please" / "Human support" / "Live chat"

IMPLICIT TRIGGERS (AI detects these patterns):

- Sentiment score drops below 35/100 for 2+ consecutive turns
- Customer uses profanity or all-caps (anger signal)
- Same question asked 3+ times (AI failed to resolve)
- Customer says "this is urgent" / "I need this NOW"
- Emotional keywords: "frustrated", "disappointed", "terrible", "disaster"
- Complaint mentions legal: "lawyer", "court", "file a complaint", "sue"
- Financial dispute: "charge back", "fraud", "dispute", "my money"
- Medical/health urgency detected
- Customer explicitly says "you don't understand"

CONFIDENCE-BASED TRIGGERS (AI self-assessment):

- AI confidence in response drops below 60%
- Query requires access to privileged data AI cannot reach
- Query involves account modification beyond AI permissions
- Query requires physical verification (ID check, in-person)
- Multi-turn conversation exceeds 15 exchanges without resolution

Agent Availability Check (Real-Time)

AVAILABILITY ENGINE:

1. Query agent_presence table in Redis (TTL: 30 seconds, refreshed by agent heartbeat)
2. Check: agents with status = "available" AND skill_match(customer_query)
3. Check: agent current load < max_concurrent_chats (default: 3)
4. Check: within business hours for tenant
5. Check: customer language -> agents with that language skill

AVAILABILITY RESPONSE:

```
{
  available: true,
  agents: [
    { id: "AGT-001", name: "Priya S.", skill_score: 94, current_load: 1, languages: ["en", "hi"] },
    { id: "AGT-002", name: "Rahul M.", skill_score: 87, current_load: 2, languages: ["en", "hi"] },
  ],
  estimated_wait_seconds: 0,    // immediate since agent available
  recommended_agent: "AGT-001"
}
```

The Warm Handoff Flow (Agent Available)

STEP 1: CUSTOMER SIGNALS ESCALATION REQUEST

Customer: "I need to speak with a real person please"

STEP 2: AI ACKNOWLEDGES AND SETS EXPECTATION (< 500ms response)

AI: "Of course! Let me connect you with [Agent Name] right now.
I will share everything we have discussed so you will not need
to repeat yourself. One moment..."

[If multilingual detected]:

AI responds in customer's language:

Hindi: "Bilkul! Main aapko abhi [Agent Name] se connect kar raha hoon.
Hamaari saari baat unhe bhejunga, aapko dobara bolna nahi padega."

STEP 3: CONTEXT PACKAGE ASSEMBLY (background, instant)

WHAT IS ASSEMBLED:

```
{
  "session_id": "CHT-20260328-8821",
  "customer_profile": {
    "name": "Rohit Sharma",
    "mobile": "9876543210",
    "tier": "gold",
    "customer_since": "2024-01",
    "lifetime_value": 285000,
    "past_interactions": 12,
    "sentiment_history": "generally_positive"
  },
  "current_session": {
    "channel": "website_chat",
    "duration_seconds": 187,
    "language_detected": "en",
    "message_count": 9,
    "ai_turns": 8,
  }
}
```

```

    "topic_summary": "Customer enquiring about Bali family trip, 4 pax, Oct 15-22",
    "intent_classification": "travel_enquiry",
    "current_sentiment": "frustrated",
    "sentiment_trend": "declining",
    "escalation_trigger": "sentiment_below_threshold",
    "ai_confidence_last_3": [82, 65, 48]
  },
  "extracted_entities": {
    "destination": "Bali",
    "travel_dates": "Oct 15-22",
    "pax_count": 4,
    "pax_type": "family_2_adults_2_children",
    "budget_mentioned": false,
    "special_requirements": "beach resort, kids activities"
  },
  "what_ai_already_did": [
    "Searched Bali family packages: 3 options found",
    "Sent package carousel to WhatsApp: not yet opened",
    "Checked visa requirements: not required for Indian passport",
    "Proposed date range: Oct 15-22 confirmed by customer"
  ],
  "what_ai_could_NOT_do": [
    "Customer wants custom package not in standard catalog",
    "Needs price matching against competitor quote"
  ],
  "recommended_next_steps": [
    "Show competitive pricing for 5-star Bali resorts",
    "Address the budget concern (customer seems price-sensitive)",
    "Offer anniversary upgrade (customer mentioned 10th anniversary)"
  ],
  "full_transcript": [
    { "role": "customer", "content": "...", "timestamp": "..." },
    ...
  ]
}

```

STEP 4: AGENT SCREEN-POP (< 1 second, Supabase Realtime push)

Agent's screen receives:

- Toast notification: "Incoming chat: Rohit Sharma (Gold) - Bali Family Trip"
- LEFT PANEL: Full customer 360 profile + interaction history
- CENTER: Live chat window (customer sees "Connecting you to an agent...")
- RIGHT SIDEBAR: AI Context Brief:

"Rohit wants: Bali family trip, 4 pax, Oct 15-22. Frustrated because AI could not find custom package. He mentioned 10th anniversary -- use this. Competitive pricing needed. He has NOT seen the WhatsApp quote yet.
SUGGESTED OPENER: 'Hi Rohit! I can see you are planning a special Bali trip for your anniversary. Let me find you something truly special.'"

STEP 5: AGENT JOINS CHAT (< 3 seconds total from escalation request)

Customer sees: "[Agent Name] has joined the conversation"

Customer does NOT see: "Connecting..." for more than 3 seconds

Agent sees: Full AI context, transcript, recommended opener

What NEVER happens: Customer is asked to repeat anything AI already collected.

STEP 6: AI CO-PILOT MODE (runs silently during human agent conversation)
AI continues working in background:

- Live transcription of entire conversation
- Real-time suggestions to agent (sidebar, not visible to customer):
"Customer used 'anniversary' -- suggest Honeymoon Suite upgrade (+INR 8,000)"
"Price objection detected -- you have 8% discount headroom"
"Customer comparing with MakeMyTrip -- here are 3 differentiators"
- Auto-complete: agent types partial response, AI suggests completion
- Knowledge base search: pulls relevant FAQs/policies in real-time
- Form fill: any new data customer shares auto-writes to lead record

STEP 7: POST-CONVERSATION

Full transcript (AI + agent portions combined) stored on Lead/Case record
AI generates unified summary of entire interaction (AI portion + agent portion)
Customer satisfaction prompt sent via WhatsApp 2 hours later

Agent Assignment Logic (Skills-Based Routing)

ROUTING PRIORITY ORDER:

1. DEDICATED: Does this customer have an assigned relationship manager?
-> Yes: route to that agent if available
-> No: continue to step 2
2. SKILL MATCH: Which agents have skills matching this query?
Query classified as: travel_package, complaint, billing, technical, general
Skills: travel_expert, billing_specialist, tier_1_support, tier_2_escalation
-> Filter to matching skilled agents only
3. LANGUAGE MATCH: Does customer need non-English support?
-> Filter to agents with customer's language skill
4. LOAD BALANCE: Among filtered agents, pick least loaded
(current_chat_count / max_chat_capacity)
5. TIER PRIORITY: Gold/Platinum customers jump to front of queue
-> If all matching agents are full, Gold waits max 2 min
-> Silver: 5 min, Bronze: standard queue
6. EXPERTISE SCORE: If tie, route to highest-rated agent (based on CSAT)

DATABASE: agent_skills, agent_presence, agent_queue tables
UPDATE: every 30 seconds via heartbeat from agent dashboard

SCENARIO 2 — LIVE AGENT REQUEST: NO AGENT AVAILABLE

The No-Agent State Detection

NO AGENT AVAILABLE means ANY of these:

- a) Outside business hours (all agents offline)
- b) All agents at maximum capacity (queue full)
- c) No agent with required skill/language available

- d) Agent capacity reached (queue limit set by tenant)
- e) High-priority incident: agents on emergency protocol

The Advanced No-Agent Response Flow

STEP 1: IMMEDIATE ACKNOWLEDGEMENT (never make customer wait > 500ms)

AI: "I completely understand you would like to speak with a person.
Our team is currently helping other customers right now.
I want to make sure you get the best support -- let me give you
your options:"

[Show interactive options, not just text:]

OPTION A: [Wait in Queue] -- "Join the queue (est. wait: X min)"

OPTION B: [Request Callback] -- "We call you back at a time you choose"

OPTION C: [WhatsApp Continuation] -- "Continue this on WhatsApp, agent replies there"

OPTION D: [Schedule for Later] -- "Book a specific time with an agent"

OPTION E: [Try to Resolve Now] -- "Let me try harder to help you right now"

STEP 2A: IF CUSTOMER CHOOSES "WAIT IN QUEUE"

- > Create queue entry with full context package
- > Show live queue position: "You are #3 in queue. Estimated wait: 4 minutes."
- > Update queue position in real-time as customers before them complete
- > Option to convert to callback while waiting: "Don't want to wait? Get a callback instead"
- > When agent becomes free: notify customer instantly, agent briefed with context
- > If customer closes window while in queue:
Send WhatsApp/email: "You were in our queue! Agent [Name] is now free.
Click here to continue: [resume link] (valid 15 min)"

STEP 2B: IF CUSTOMER CHOOSES "WHATSAPP CONTINUATION"

- > Full conversation transcript sent to WhatsApp immediately
- > "Hi Rohit! This is your chat transcript. Our agent will reply here within 2 hours. You can continue the conversation any time."
- > WhatsApp thread becomes the continuation channel
- > Agent sees both chat transcript AND any new WhatsApp messages
- > Unified timeline in CRM shows full journey

STEP 2C: IF CUSTOMER CHOOSES "SCHEDULE FOR LATER"

- > Show agent calendar with available slots (next 48 hours)
- > Customer picks slot: date + time
- > Booking confirmation: WhatsApp + email
- > 15-minute reminder before scheduled chat
- > Agent pre-briefed with full session context 5 minutes before

STEP 2D: IF CUSTOMER CHOOSES "LET AI TRY HARDER"

- > AI escalates internally: switches to GPT-4o (full quality, no cost limit)
- > AI accesses all knowledge bases, runs deeper searches
- > Tries 3 different resolution approaches
- > If still cannot resolve after 5 more exchanges:
"I have exhausted my ability to help with this specific request.
I have escalated this to our support team with Priority status.
They will reach you within [SLA: 2 hours] on WhatsApp."
-> Creates Priority Case, notifies supervisor

STEP 3: AFTER-HOURS SPECIAL HANDLING

[When entirely outside business hours]

AI does NOT just say "we are closed."

AI DOES:

- a) Solve as much as possible with full AI capability
- b) Create a complete case/lead/quote with all information
- c) Schedule a callback for next business day opening (or chosen time)
- d) Send WhatsApp summary: "Here is what we discussed. [Agent Name] will call you at 9:00 AM tomorrow to take this forward. Reference: #8821."
- e) Agent sees fully contextualized queue item when they log in next morning

[What makes this advanced vs basic]:

Basic: "We are closed. Please try during business hours."

Advanced: Full resolution attempt + guaranteed follow-up with context preservation

Queue Management System (Technical)

VIRTUAL QUEUE ARCHITECTURE:

QUEUE TABLE (Redis + PostgreSQL):

```
{
  queue_id: "Q-2026-8821",
  tenant_id: "T-042",
  customer_id: "C-9981",
  customer_name: "Rohit Sharma",
  customer_mobile: "9876543210",
  customer_tier: "gold",
  language: "en",
  required_skill: "travel_package",
  context_package: { ... full JSON from Step 3 above ... },
  queue_position: 3,
  entered_at: "2026-03-28T14:23:00Z",
  estimated_wait_seconds: 240,
  priority_score: 85,           // Gold tier boosts priority
  sentiment_score: 42,         // Low = high priority
  urgency_flag: false,
  channel_preference: "chat",
  whatsapp_fallback: "9876543210",
  status: "waiting"           // waiting | connecting | with_agent | abandoned | resolved
}
```

PRIORITY SCORING FORMULA:

```
base_score = 50
+ tier_bonus (Gold=+30, Silver=+15, Bronze=0)
+ urgency_bonus (legal/medical keyword=+20, frustration=+15, repeat=+10)
+ wait_time_bonus (each 60s waiting = +2, max +40)
+ value_bonus (LTV > 100k = +15)
- low_priority (FAQ query = -20)
= priority_score (0-100, higher = faster)
```

QUEUE DISPLAY TO CUSTOMER:

"You are #3 in line. Estimated wait: 4 minutes."

Live-updating counter (via WebSocket)

Option to convert to callback at any time

Progress bar showing queue movement

CAPACITY MANAGEMENT:

Max queue length: configurable per tenant (default 10)
When queue full: chat widget changes to "Request a Callback" only
Never show customer "Queue full" -- show "Our agents are fully engaged, would you like us to call you back?"

QUEUE ABANDONMENT HANDLING:

Customer closes window while in queue:
-> Mark as "queue_abandoned"
-> Send WhatsApp within 60 seconds: "We noticed you left our queue. You were #3 -- an agent is now free! Click to reconnect: [link]"
-> If not reconnected in 10 min: schedule automatic callback

SCENARIO 3 — CUSTOMER REQUESTS CALLBACK

Callback Request Flow (The Most Advanced Part)

CALLBACK TRIGGER PHRASES:

"Call me back" / "Can you call me?" / "I'll wait for your call"
"Give me a call" / "Ring me back" / "Call me at..."
"Better to talk on phone" / "I prefer a phone call"
"When can someone call?" / "Book a call with me"

CALLBACK TYPES (3 distinct types, each different flow):

TYPE 1: IMMEDIATE CALLBACK ("Call me right now")

-> AI checks: is any agent available to call right now?
-> YES:
"Our agent [Name] will call you within 2 minutes.
We will call: +91-XXXXXXXXXX. Is that correct?"
-> If yes: create callback job, push to available agent
-> Agent dashboard shows: "[Ring icon] CALLBACK REQUEST - Rohit Sharma - 2 min SLA"
-> Agent clicks [Call Now]: Exotel/Twilio initiates outbound call
-> Pre-call: agent sees full context package (same as chat escalation)
-> During call: same co-pilot assistance as any other call
-> AI handles call if agent not available (PathA fallback)

-> NO (no agent available):
"No agent is free right this moment. Here are your options:
[Callback in X min when next agent is free]
[Choose a specific time that works for you]"

TYPE 2: SCHEDULED CALLBACK ("Call me at 3pm" or "Call me tomorrow")

-> Customer specifies preferred time OR selects from available slots

INTELLIGENT SLOT SUGGESTION:

-> AI checks agent availability calendar (real-time)
-> AI checks: customer's timezone (from profile or IP detection)
-> AI suggests 3 slots:
"Based on your preference, here are available times:
1. Today 3:00 PM - 3:30 PM (Agent: Priya S.)
2. Today 5:00 PM - 5:30 PM (Agent: Rahul M.)

```

    3. Tomorrow 10:00 AM - 10:30 AM (Agent: Priya S.)
    Which works best?"
-> Customer selects
-> BOOKING CONFIRMATION:
    - Chat: "Confirmed! Priya will call you today at 3:00 PM."
    - WhatsApp: Full confirmation with calendar event (.ics attachment)
    - Email: Confirmation + ability to reschedule/cancel
    - Calendar invite sent to agent's Google/Outlook calendar

-> 15-minute reminder:
    WhatsApp to customer: "Reminder: Priya from [Company] is calling you at 3:00 PM.
    That's in 15 minutes. Make sure your line is free. [Reschedule | Cancel]"

-> 5-minute brief to agent:
    Dashboard notification: "SCHEDULED CALLBACK in 5 min: Rohit Sharma
    Context: [2-line summary of chat session]
    [View Full Context] [Start Call Now]"

-> At appointment time:
    Exotel/Twilio initiates outbound call to customer
    If customer does not answer: SMS + WhatsApp "We tried calling at 3 PM.
    [Reschedule | Call me now if you are free]"

```

TYPE 3: ANONYMOUS / NO MOBILE NUMBER

```

-> Customer has not shared mobile number
-> AI asks: "Of course! What number should we call you on?"
-> Validate: regex check, normalize format (+91, remove spaces)
-> Optional: "What is the best time for you?" (for slot booking)
-> Proceed to TYPE 1 or TYPE 2 flow above

```

CALLBACK DATA STORED:

```

model CallbackRequest {
    id                String
    tenantId          String
    customerId        String?
    customerMobile    String          // normalized
    customerName      String?
    customerLanguage  String          // for agent matching
    callbackType      String          // immediate|scheduled|queued
    scheduledFor      DateTime?
    assignedAgentId   String?
    agentName         String?
    status            String          // pending|confirmed|called|completed|no_answer|
    noAnswerCount     Int default 0
    contextPackage    Json           // full AI session context
    chatSessionId     String?        // link to original chat session
    reminderSentAt    DateTime?
    agentBriefedAt    DateTime?
    callInitiatedAt   DateTime?
    callCompletedAt   DateTime?
    callOutcome       String?        // resolved|needs_followup|no_answer|rescheduled
    noAnswerFollowup  String          // whatsapp|email|reschedule
    createdAt         DateTime
    updatedAt         DateTime

```

```

@@index([tenantId, status, scheduledFor])

```

```
    @@index([assignedAgentId, status])
}
```

Callback No-Answer Handling

CUSTOMER DID NOT ANSWER THE CALLBACK:

Attempt 1 (scheduled time):

- > Call placed, no answer after 6 rings
- > Immediately: WhatsApp "We tried calling you at 3:00 PM but couldn't reach you.
[Try again now] [Reschedule for later] [Switch to WhatsApp chat]"

Attempt 2 (auto-retry after 30 min):

- > If no response to WhatsApp above
- > Auto-retry call 30 minutes later
- > If no answer: "We tried twice. When is a good time?
[Choose another time] [Leave us a message]"

Attempt 3 (next business day):

- > Morning call attempt
- > If still no answer: case marked as "customer_unreachable"
- > Agent notes updated: "3 callback attempts, no answer"
- > ProactiveAgent schedules a 7-day re-engagement sequence

max_attempts: configurable per tenant (default: 3)

retry_intervals: [0 min, 30 min, next_business_day]

SCENARIO 4 — AI PROACTIVE ESCALATION (AI DECIDES, NOT CUSTOMER)

When AI Should Initiate Escalation Without Customer Asking

AI MONITORS THESE SIGNALS CONTINUOUSLY:

EMOTIONAL ESCALATION TRIGGERS:

- Sentiment drops below 30 (very frustrated)
- Customer uses words: "useless", "terrible", "disgusting", "pathetic"
- Customer threatens public action: "Twitter", "Facebook", "review", "media"
- Customer threatens legal action: "lawyer", "court", "sue", "police", "consumer court"
- Medical/safety emergency keywords: "emergency", "urgent", "dying", "hospital"

BUSINESS ESCALATION TRIGGERS:

- Same unresolved query after 5+ AI turns
- AI confidence below 50% for 2 consecutive responses
- Customer value > INR 5 lakhs AND any friction signal
- Financial dispute > configured threshold (default INR 10,000)
- Complaint about booking already in "confirmed" status

KILL-SWITCH TRIGGERS (immediate, no option given):

- Self-harm keywords -> crisis protocol
- Active fraud attempt detected -> security team
- Legal/regulatory mention -> compliance team

AI RESPONSE FOR PROACTIVE ESCALATION:

(Do NOT say "I am just a bot")

AI: "I want to make sure you get the best support for this.
Let me bring in a specialist who can help you directly.
I am connecting you now -- everything we have discussed
will be shared with them."

-> Escalates to Scenario 1 or 2 depending on availability

-> NO option for customer to say "no thanks" for kill-switch triggers

THE CHAT AGENT STATE MACHINE

COMPLETE STATE MACHINE:

States:

GREETING -> AI greets, qualifies intent
QUALIFICATION -> Collecting data (lead/case/quote)
RESOLUTION -> AI attempting to resolve
QUOTE_SHOWN -> AI showed product/package options
ESCALATION_REQUESTED -> Customer asked for human
CHECKING_AVAILABILITY -> Querying agent presence
QUEUEING -> Customer in virtual queue
CALLBACK_SCHEDULED -> Callback booked
TRANSFERRING -> Handoff in progress
AGENT_ACTIVE -> Human agent handling
AI_COPILOT -> Agent active, AI assisting
COMPLETED -> Session resolved
ABANDONED -> Customer left
AFTER_HOURS -> Outside business hours
ERROR -> Technical failure

Transitions:

GREETING -> QUALIFICATION (intent detected)
QUALIFICATION -> RESOLUTION (enough data to attempt resolution)
QUALIFICATION -> ESCALATION_REQUESTED (customer asks for agent)
RESOLUTION -> ESCALATION_REQUESTED (AI cannot resolve after threshold)
RESOLUTION -> QUOTE_SHOWN (package/service proposed)
QUOTE_SHOWN -> COMPLETED (customer accepts)
QUOTE_SHOWN -> RESOLUTION (customer wants changes)
ESCALATION_REQUESTED -> CHECKING_AVAILABILITY
CHECKING_AVAILABILITY -> TRANSFERRING (agent available)
CHECKING_AVAILABILITY -> QUEUEING (agents busy)
CHECKING_AVAILABILITY -> CALLBACK_SCHEDULED (no agents + customer chooses callback)
CHECKING_AVAILABILITY -> AFTER_HOURS (outside business hours)
QUEUEING -> TRANSFERRING (agent becomes available)
QUEUEING -> CALLBACK_SCHEDULED (customer converts queue to callback)
TRANSFERRING -> AGENT_ACTIVE + AI_COPILOT
AGENT_ACTIVE + AI_COPILOT -> COMPLETED
AFTER_HOURS -> CALLBACK_SCHEDULED (customer books for next day)
AFTER_HOURS -> COMPLETED (AI resolves without agent)
Any state -> ABANDONED (customer closes window)
ABANDONED -> CALLBACK_SCHEDULED (re-engagement sent)

COBROWSE AND SCREEN SHARING (ADVANCED)

WHEN AGENT IS ACTIVE IN CHAT:

- Agent can initiate cobrowse session:
- > Agent clicks [Cobrowse] in their dashboard
- > Customer receives: "Our agent would like to guide you on-screen. Click [Start Screen Share] to allow. You stay in control at all times."
- > Uses WebRTC (same LiveKit infrastructure)
- > Agent sees customer's browser (tab-only, not full screen)
- > Agent can highlight elements, draw attention arrows
- > NO access to passwords, payment forms (masked automatically)
- > Compliant with DPDP Act: consent logged with timestamp

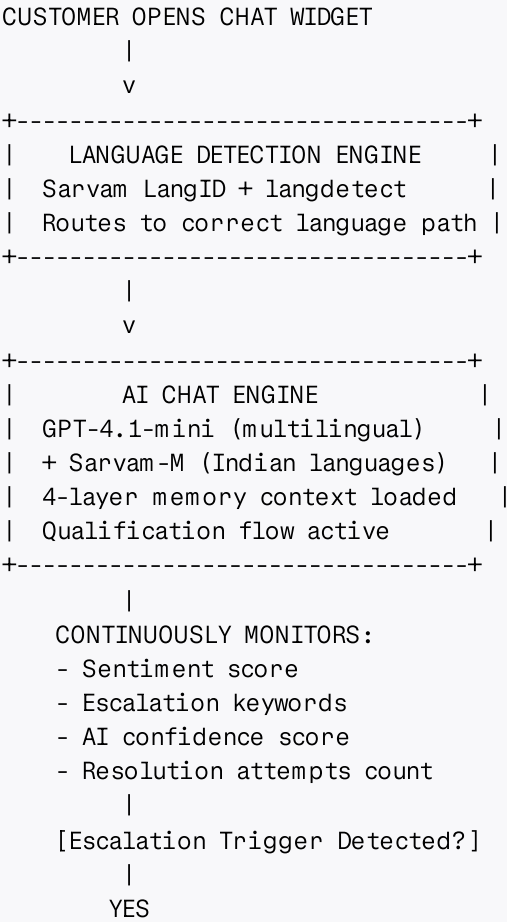
USE CASES:

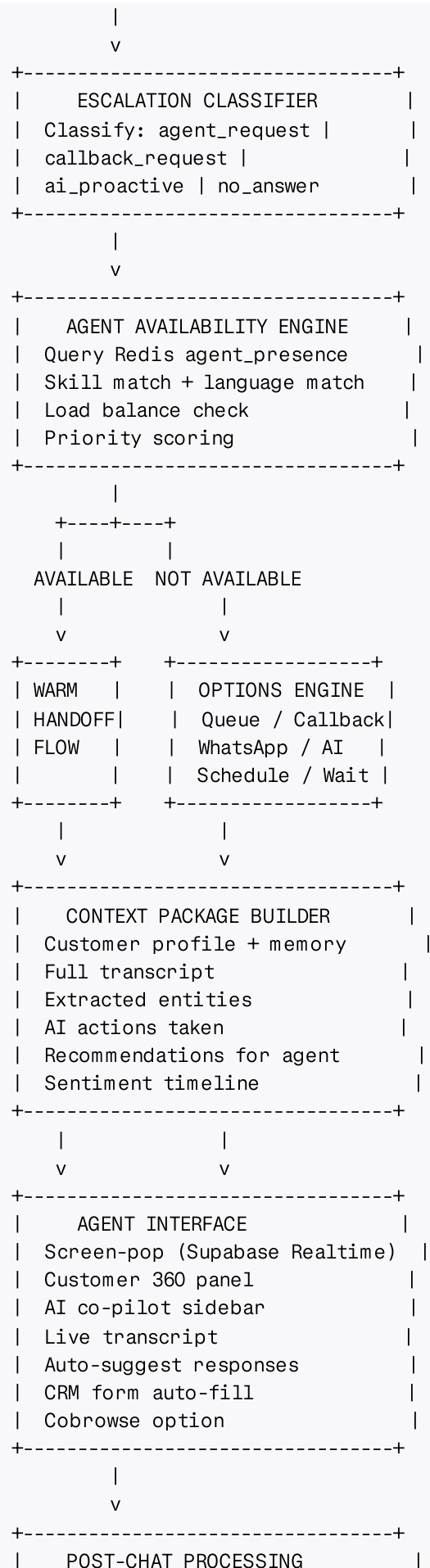
- Guiding customer through booking form
- Showing them where to upload documents
- Walking through itinerary builder together
- Explaining invoice/quote PDF

COBROWSE STATS (industry):

- 25% higher first-call resolution
- 30% boost in customer satisfaction
- 15% higher resolution rate vs without (Klarna case study)

COMPLETE CHAT AGENT ARCHITECTURE DIAGRAM





```
| Unified transcript storage |
| AI summary (AI + agent parts) |
| Memory write-back |
| CSAT request (2hr delay) |
| Proactive follow-up scheduled |
| Lead/Case/Quote updated |
+-----+
```

LIVE AGENT DASHBOARD — WHAT THE AGENT SEES

AGENT DASHBOARD PANELS:

LEFT PANEL: Queue & Status

- My current conversations (list)
- Incoming queue (waiting customers, with priority indicators)
- My availability toggle: [Available] [Busy] [Break]
- Skills assigned to me
- CSAT score (last 30 days)
- Today's stats: chats handled, avg resolution time

CENTER PANEL: Active Chat

- Customer name + tier badge
- Chat history (AI turns in gray, agent turns in blue)
- Message input with AI auto-complete
- [Cobrowse] [Transfer] [Add Note] [Close] buttons
- Customer is typing indicator
- Send button with channel selector (chat | WhatsApp | email)

RIGHT PANEL: AI Co-Pilot

- Customer 360 profile card
- AI-generated session brief (2-3 sentences)
- Extracted entities (destination, dates, budget, etc.)
- Suggested responses (agent can click to use)
- Objection handlers (auto-populated when objection detected)
- Upsell signals (highlighted in yellow)
- Knowledge base search (real-time, auto-searches on topic)
- Open leads/bookings for this customer
- "What AI already tried" list
- "Recommended next steps" list

TOP BAR:

- Notifications (incoming escalations, callbacks due)
- Team online status
- Supervisor escalation button
- Current load indicator

MULTILINGUAL CHAT ESCALATION

LANGUAGE PRESERVATION THROUGHOUT ESCALATION:

When customer is in Hindi and needs escalation:

1. AI escalation message in Hindi:
"Bilkul, main aapko Hindi mein baat karne waale agent se connect karta hoon. Ek minute..."
2. Agent matching: filters to agents with "hi" language skill
3. If no Hindi agent available:
"Hindi mein support ke liye abhi koi agent uplabdh nahi hai.
Lekin main aapko do options de sakta hoon:
[15 min mein callback chahiye?]
[Kal subah 10 baje slot book karein - Hindi mein]"
4. Agent receives context package with:
 - Language: "hi" (Hindi)
 - transcript in Hindi
 - English translation of transcript
 - Note: "Customer prefers Hindi communication"
5. If agent is available but not Hindi-speaking:
 - AI acts as real-time interpreter
 - Customer types in Hindi
 - AI translates to English for agent in real-time (in agent sidebar)
 - Agent responds in English
 - AI translates response back to Hindi for customer
 - Customer experience: seamless Hindi conversation
 - Agent experience: seamless English interaction
 - This is the "AI as interpreter" pattern for multi-language support

CALLBACK SCHEDULING WIDGET (UI SPEC)

CALLBACK WIDGET DESIGN:

Title: "Request a Call Back"

Subtitle: "Our agent will call you at a time that works for you"

FIELDS:

1. Name: [Pre-filled from chat if captured]
2. Mobile Number: [Pre-filled from profile if known]
 - Format hint: +91 XXXXX XXXXX
 - Validation: real-time E.164 check
3. Preferred Time:
TABS: [Call Now] [Choose a Time]

"Call Now" tab:

"An agent will call you within 5-10 minutes"

[Confirm Call Now] button

Estimated wait shown dynamically

```

"Choose a Time" tab:
Date picker (next 3 days, business hours only)
Time slot grid: 30-min slots, green=available, gray=booked
"Your timezone: Asia/Kolkata (IST)"

4. What is this about? (optional, pre-filled from AI summary)
Text area: "Briefly describe what you need help with"
(This briefs the agent before the call)

5. Preferred Language: [Auto-detected, editable]
Dropdown: English, Hindi, Tamil, Telugu, Bengali, Other

CONFIRMATION:
- Inline: "Booked! Priya will call you at 3:00 PM today."
- WhatsApp: Sent automatically
- Email: Sent automatically
- Add to Calendar: [Google] [Apple] [Outlook] (.ics)

RESCHEDULING:
Customer can reschedule via:
- WhatsApp reply: "Reschedule my 3pm call"
- Email link in confirmation
- Chat widget: "I need to reschedule my call"
AI handles rescheduling with same slot-picker flow

```

COMPLETE DATABASE SCHEMA (CHAT ESCALATION + CALLBACK)

```

-- Agent presence and availability
CREATE TABLE agent_presence (
  agent_id      TEXT PRIMARY KEY,
  tenant_id     TEXT NOT NULL,
  status        TEXT NOT NULL, -- available|busy|break|offline
  current_load  INT DEFAULT 0,  -- active chat count
  max_load      INT DEFAULT 3,  -- max concurrent chats
  skills        TEXT[],         -- ['travel', 'billing', 'tier1', 'tier2']
  languages     TEXT[],         -- ['en', 'hi', 'ta']
  last_heartbeat TIMESTAMPTZ DEFAULT NOW(),
  shift_ends_at TIMESTAMPTZ,
  updated_at    TIMESTAMPTZ DEFAULT NOW()
);
-- TTL-like: rows with last_heartbeat older than 60s considered offline

-- Virtual queue
CREATE TABLE chat_queue (
  id            UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id     TEXT NOT NULL,
  chat_session_id UUID,
  customer_id   TEXT,
  customer_mobile TEXT,
  customer_name TEXT,
  customer_tier TEXT DEFAULT 'bronze',
  language      TEXT DEFAULT 'en',
  required_skill TEXT,

```

```

context_package JSONB,          -- full AI session context
queue_position  INT,
priority_score  INT DEFAULT 50,
sentiment_score INT,
urgency_flag    BOOLEAN DEFAULT false,
entered_at      TIMESTAMPTZ DEFAULT NOW(),
estimated_wait_s INT,
status          TEXT DEFAULT 'waiting',
assigned_agent  TEXT,
assigned_at     TIMESTAMPTZ,
converted_to_callback BOOLEAN DEFAULT false,
abandoned_at   TIMESTAMPTZ,
resolved_at     TIMESTAMPTZ,
created_at      TIMESTAMPTZ DEFAULT NOW()
);
CREATE INDEX ON chat_queue(tenant_id, status, priority_score DESC);

-- Callback requests
CREATE TABLE callback_requests (
  id            UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id     TEXT NOT NULL,
  customer_id   TEXT,
  customer_mobile TEXT NOT NULL,
  customer_name TEXT,
  customer_language TEXT DEFAULT 'en',
  callback_type TEXT NOT NULL, -- immediate|scheduled|queued
  scheduled_for TIMESTAMPTZ,
  assigned_agent TEXT,
  agent_name    TEXT,
  status        TEXT DEFAULT 'pending',
  context_package JSONB,
  chat_session_id UUID,
  reminder_sent_at TIMESTAMPTZ,
  agent_briefed_at TIMESTAMPTZ,
  call_initiated_at TIMESTAMPTZ,
  call_completed_at TIMESTAMPTZ,
  call_duration_s INT,
  no_answer_count INT DEFAULT 0,
  call_outcome    TEXT,
  notes           TEXT,
  created_at      TIMESTAMPTZ DEFAULT NOW(),
  updated_at      TIMESTAMPTZ DEFAULT NOW()
);
CREATE INDEX ON callback_requests(tenant_id, status, scheduled_for);
CREATE INDEX ON callback_requests(assigned_agent, status);

-- Agent chat sessions (for load tracking)
CREATE TABLE agent_chat_assignments (
  id            UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  agent_id      TEXT NOT NULL,
  tenant_id     TEXT NOT NULL,
  chat_session_id UUID NOT NULL,
  customer_id   TEXT,
  assigned_at    TIMESTAMPTZ DEFAULT NOW(),
  first_response_at TIMESTAMPTZ,
  resolved_at    TIMESTAMPTZ,

```

```

    transferred_to TEXT,
    csat_score INT,
    resolution_type TEXT,
    created_at TIMESTAMPTZ DEFAULT NOW()
);

-- Cobrowse sessions
CREATE TABLE cobrowse_sessions (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id TEXT NOT NULL,
  chat_session_id UUID NOT NULL,
  agent_id TEXT NOT NULL,
  customer_id TEXT,
  started_at TIMESTAMPTZ DEFAULT NOW(),
  ended_at TIMESTAMPTZ,
  duration_s INT,
  consent_given_at TIMESTAMPTZ,
  consent_method TEXT,          -- chat_button|sms_link
  outcome TEXT,                -- resolved|abandoned
  created_at TIMESTAMPTZ DEFAULT NOW()
);

```

PROMPT FOR CLAUDE — PHASE 12: ADVANCED CHAT AGENT

CONTEXT: You are implementing the Advanced Chat Agent for AgentCore.
 Codebase: HelpRevX turborepo on AWS.
 Location: packages/agentcore-chat/ and apps/api/src/routes/chat/

WHAT ALREADY EXISTS:

- From Phase 4: packages/agentcore-whatsapp/ (use for WhatsApp delivery)
- From Phase 5: packages/agentcore-memory/ (use for customer context)
- From Phase 5: packages/agentcore-lead/ (use for lead creation)
- From Phase 3: apps/api/src/routes/telephony/ (use Exotel patterns)
- From Phase 8: packages/agentcore-document/ (reference for DB patterns)

IMPORTANT: The existing chat agent in packages/agentcore-chat/ only handles AI qualification and basic responses. This phase ADDS the escalation and callback systems. Do NOT rewrite what works. ADD the missing capabilities.

AUDIT FIRST:

```

cat packages/agentcore-chat/src/chat-service.ts
cat packages/agentcore-chat/src/index.ts
grep -r "escalat\|agent.*available\|handoff\|callback" packages/agentcore-chat/
Report: what exists, what is missing

```

TASK 1: DATABASE MIGRATIONS

Add to packages/db/prisma/schema.prisma:

```

model AgentPresence {full schema as defined in architecture doc}
model ChatQueue {full schema as defined}
model CallbackRequest {full schema as defined}
model AgentChatAssignment {full schema as defined}
model CobrowseSession {full schema as defined}

```

Also add raw SQL: packages/db/prisma/migrations/add_chat_escalation.sql

- Indexes for performance
- Function: update_queue_positions() trigger
(automatically reorders queue when item removed/priority changed)

TASK 2: AGENT PRESENCE SERVICE

Create: packages/agentcore-chat/src/agent-presence.ts

```
class AgentPresenceService {
  // Called by agent dashboard every 30 seconds (heartbeat)
  async updatePresence(agentId: string, tenantId: string, status: AgentStatus): Promise<
    - Upsert agent_presence table
    - Update Redis: key = `agent:${tenantId}:${agentId}`, TTL = 90s

  // Query for available agents matching criteria
  async getAvailableAgents(
    tenantId: string,
    requiredSkill: string,
    requiredLanguage: string,
    customerTier: string
  ): Promise<AvailableAgentResult>
    - Query Redis first (fast path, < 10ms)
    - Fallback: DB query if Redis miss
    - Filter: status=available AND load<maxLoad AND skill_match AND language_match
    - Sort by priority: tier_bonus + expertise_score - current_load
    - Return: { available: boolean, agents: Agent[], estimatedWait: number }

  // Assign agent to chat
  async assignAgent(agentId: string, chatSessionId: string): Promise<void>
    - Increment agent current_load in Redis and DB
    - Create AgentChatAssignment record
    - Push screen-pop event via Supabase Realtime

  // Release agent from chat
  async releaseAgent(agentId: string, chatSessionId: string, outcome: string): Promise<
    - Decrement agent current_load
    - Update AgentChatAssignment: resolved_at, resolution_type
    - Trigger queue processor: assign next waiting customer

  // Calculate priority score for queue
  calculatePriorityScore(customer: Customer, session: ChatSession): number
    - Formula: base(50) + tier_bonus + urgency + sentiment_inverted + wait_time + value
    - Return: 0-100

  // Get business hours for tenant
  async isWithinBusinessHours(tenantId: string): Promise<BusinessHoursResult>
    - Load tenant_voice_config.businessHoursJson
    - Check current time vs hours in tenant's timezone
    - Return: { isOpen: boolean, opensAt?: Date, timezone: string }
}
```

TASK 3: ESCALATION DETECTOR

Create: packages/agentcore-chat/src/escalation-detector.ts

```
class EscalationDetector {
  // Main detection function called on every customer message
  async detectEscalationNeed(
```

```

    message: string,
    sessionContext: ChatSessionContext
  ): Promise<EscalationSignal>
  Returns:
  {
    shouldEscalate: boolean,
    trigger: 'explicit_request'|'implicit_frustration'|'ai_failure'|'kill_switch'|'none',
    confidence: number,
    escalationType: 'agent'|'callback'|'supervisor'|'crisis',
    detectedKeywords: string[],
    sentimentScore: number,
    urgencyLevel: 'low'|'medium'|'high'|'critical'
  }

  Detection methods:
  1. Pattern matching for explicit triggers (list from architecture doc)
  2. Sentiment analysis (GPT-4.1-nano, fast+cheap)
  3. Session metrics: turn count, AI confidence history
  4. Kill-switch keywords (synchronous check, no API call)

  // Detect if customer is asking for callback specifically
  async isCallbackRequest(message: string): Promise<boolean>
    - Pattern match callback phrases
    - Return boolean

  // Classify what type of agent help is needed
  async classifyEscalationType(
    message: string,
    sessionContext: ChatSessionContext
  ): Promise<SkillRequired>
    - Options: general|travel_specialist|billing|complaints|technical|senior
    - Use LLM classification with cost-optimized model
}

```

TASK 4: WARM HANDOFF ENGINE

Create: packages/agentcore-chat/src/handoff-engine.ts

```

class HandoffEngine {
  // The core handoff function - called when escalation decided
  async initiateHandoff(
    chatSessionId: string,
    escalationSignal: EscalationSignal
  ): Promise<HandoffResult>

  STEPS (all must be production-complete with error handling):
  1. Acknowledge customer immediately with language-appropriate message
  2. Assemble full context package (function: buildContextPackage)
  3. Check agent availability (AgentPresenceService)
  4. If available: call executeWarmHandoff()
  5. If not available: call handleNoAgentScenario()
  6. Log to AgentAuditLog
  7. Update chat session state

  async buildContextPackage(chatSessionId: string): Promise<ContextPackage>
    MUST include (all fields, no placeholders):
    - Customer profile + memory context

```

- Full conversation transcript
- Extracted entities from session
- Sentiment timeline (array of scores per turn)
- Actions AI already took
- What AI could not resolve (reason)
- Recommended next steps for agent
- Upsell signals detected
- Relevant CRM records (open leads, bookings, cases)

```

async executeWarmHandoff(
  chatSessionId: string,
  assignedAgent: Agent,
  contextPackage: ContextPackage
): Promise<void>
  1. Push screen-pop to agent (Supabase Realtime channel: `agent:${agentId}`)
  2. AI says to customer: language-appropriate "connecting you" message
  3. Transfer chat session to agent in shared chat database
  4. Start AI co-pilot mode (background, assists agent)
  5. Update session state to AGENT_ACTIVE

async handleNoAgentScenario(
  chatSessionId: string,
  contextPackage: ContextPackage,
  businessHours: BusinessHoursResult
): Promise<NoAgentOptions>
  If within business hours:
    Return: queue | callback | whatsapp_continuation | try_again
  If outside business hours:
    Return: after_hours_callback | whatsapp_continuation | async_case

async activateAICopilot(chatSessionId: string, agentId: string): Promise<void>
  - Subscribe to agent's message events via WebSocket
  - For each agent message: analyze + generate sidebar suggestion
  - For each customer message: update sentiment score, flag objections
  - Auto-fill form fields from customer messages
  - All suggestions go to agent sidebar (never visible to customer)
}

```

TASK 5: VIRTUAL QUEUE MANAGER

Create: packages/agentcore-chat/src/queue-manager.ts

```

class VirtualQueueManager {
  async enqueue(chatSessionId: string, contextPackage: ContextPackage): Promise<QueueEntry>
    - Create ChatQueue record
    - Calculate priority score
    - Insert at correct position based on priority
    - Push position notification to customer (WebSocket)
    - Return: { queueId, position, estimatedWaitSeconds }

  async updateQueuePositions(tenantId: string): Promise<void>
    - Called whenever someone leaves queue
    - Recalculate all positions
    - Push updates to all waiting customers (WebSocket broadcast)

  async processNextInQueue(tenantId: string, agentId: string): Promise<void>
    - Called when agent becomes available

```

- Pop highest priority entry from queue
- Initiate warm handoff for that entry
- Notify agent: screen-pop with context

```

async convertToCallback(queueId: string, preferredTime?: Date): Promise<CallbackRequest> {
  - Remove from queue
  - Create CallbackRequest
  - Send confirmation to customer

async handleQueueAbandonment(queueId: string): Promise<void> {
  - Called when customer closes window
  - Mark as abandoned
  - Send WhatsApp re-engagement (within 60 seconds)
  - Schedule automatic callback attempt

// Push queue position to customer via WebSocket (Supabase Realtime)
async pushQueueUpdate(customerId: string, position: number, waitSeconds: number): Promise<void> {
  - Publish to Supabase channel: `chat:${chatSessionId}:queue`
  - Customer widget receives: "You are #X in line. Est. wait: Y min."

async getQueueStatus(tenantId: string): Promise<QueueStatus> {
  - total_waiting, average_wait_seconds, longest_wait_seconds
  - available_agents, total_agents_online
}

```

TASK 6: CALLBACK SERVICE

Create: packages/agentcore-chat/src/callback-service.ts

```

class CallbackService {
  // Create immediate callback request
  async requestImmediateCallback(
    chatSessionId: string,
    customerMobile: string,
    contextPackage: ContextPackage
  ): Promise<CallbackRequest> {
    - Create CallbackRequest with type=immediate
    - Check agent availability for outbound call
    - If agent available: push outbound call task to agent dashboard
    - If not: queue for next available agent
    - Send WhatsApp confirmation
    - Return callback record

  // Create scheduled callback with slot selection
  async requestScheduledCallback(
    chatSessionId: string,
    customerMobile: string,
    preferredTime: Date,
    contextPackage: ContextPackage,
    agentId?: string
  ): Promise<CallbackRequest> {
    - Validate preferredTime is within business hours
    - Check agent availability at requested time
    - If agent available at time: reserve slot
    - If not: suggest alternatives (3 nearest available slots)
    - Create CallbackRequest with type=scheduled
    - Create calendar event for agent

```

- Send confirmation to customer (WhatsApp + email)
- Schedule reminder job (BullMQ: 15 min before)

// Get available time slots for callback booking

async getAvailableSlots(

 tenantId: string,
 customerLanguage: string,
 requiredSkill: string,
 daysAhead: number = 3

): Promise<TimeSlot[]>

- Query agent schedules for next N days
- Filter by: language, skill, existing bookings
- Return 30-min slots grouped by day
- Each slot: { date, startTime, endTime, agentName, agentId }

// Initiate the actual outbound call (called by scheduler at appointment time)

async initiateOutboundCall(callbackId: string): Promise<void>

- Load callback record + context package
- Brief the assigned agent (dashboard notification + context push)
- Use Exotel/Twilio API to place outbound call to customer
- Update callback status: calling
- Start 6-ring timeout

// Handle no-answer

async handleNoAnswer(callbackId: string): Promise<void>

- Increment no_answer_count
- If count < max_attempts: schedule retry
- Send WhatsApp to customer immediately
- Log outcome

// Send confirmation messages

async sendCallbackConfirmation(callbackRequest: CallbackRequest): Promise<void>

- WhatsApp message: confirmation with reschedule/cancel buttons
- Email: confirmation with .ics calendar attachment
- Chat widget: inline confirmation message

// Handle reschedule request (from WhatsApp or chat)

async rescheduleCallback(callbackId: string, newTime: Date): Promise<CallbackRequest>

- Release old slot
- Book new slot
- Update agent calendar
- Send updated confirmation

TASK 7: AGENT CO-PILOT (Real-Time Assist)

Create: packages/agentcore-chat/src/agent-copilot.ts

class AgentCopilot {

 // Subscribe to a live chat session and assist the agent

 async startCopilotSession(chatSessionId: string, agentId: string): Promise<void>

- Subscribe to Supabase Realtime channel for this chat
- Initialize GPT-4.1-nano streaming session (cost-optimized)
- Load customer context into copilot context window

 // Called on every new customer message during agent session

 async processCustomerMessage(

 message: string,

```

    chatSessionId: string
  ): Promise<CopilotSuggestion>
  Returns:
  {
    suggestedResponse: string,          // one-click response for agent
    detectedObjection: string | null,   // price/timing/competitor objection
    objectionHandler: string | null,    // how to handle it
    upsellOpportunity: string | null,    // detected upsell signal
    upsellScript: string | null,        // suggested upsell pitch
    relevantKBArticle: KBArticle | null, // relevant KB entry
    extractedEntity: Record<string, string>, // new info from customer
    formFieldUpdate: Record<string, string>, // auto-update lead form
    sentimentUpdate: { score: number, trend: 'improving' | 'declining' | 'stable' }
  }

  // Called on every agent message (quality coaching)
  async processAgentMessage(
    message: string,
    chatSessionId: string
  ): Promise<AgentCoaching>
  Returns:
  {
    qualityScore: number,          // 0-100
    suggestions: string[],         // improvement suggestions
    complianceFlag: boolean,       // flagged any compliance issues?
    missedOpportunity: string | null // did agent miss an upsell/resolution?
  }

  // Real-time knowledge base search triggered by conversation topic
  async searchKnowledgeBase(topic: string, tenantId: string): Promise<KBArticle[]>
  - Vector search on tenant's knowledge base
  - Return top 3 most relevant articles
  - Displayed in agent sidebar as clickable cards

```

TASK 8: API ROUTES

Create: apps/api/src/routes/chat/escalation.ts

```

POST /api/chat/escalate/:sessionId    -- trigger escalation
GET  /api/chat/agent-availability     -- check availability
POST /api/chat/queue/join/:sessionId  -- join queue
GET  /api/chat/queue/position/:queueId -- get queue position (polling)
POST /api/chat/queue/abandon/:queueId -- leave queue

```

Create: apps/api/src/routes/chat/callback.ts

```

POST /api/chat/callback/immediate     -- request immediate callback
POST /api/chat/callback/schedule      -- book scheduled callback
GET  /api/chat/callback/slots         -- get available time slots
PUT  /api/chat/callback/:id/reschedule -- reschedule callback
PUT  /api/chat/callback/:id/cancel     -- cancel callback
POST /api/chat/callback/:id/initiate  -- internal: trigger outbound call
POST /api/chat/callback/:id/no-answer -- internal: handle no answer

```

Create: apps/api/src/routes/chat/agent.ts

```

POST /api/agent/presence/heartbeat    -- agent sends heartbeat
PUT  /api/agent/presence/status       -- agent changes status
GET  /api/agent/queue/:tenantId       -- get agent's queue
POST /api/agent/chat/accept/:queueId  -- accept chat from queue

```

```
POST /api/agent/chat/transfer/:sessionId -- transfer to another agent
POST /api/agent/cobrowse/start/:sessionId -- initiate cobrowse
```

Create: apps/voice-dashboard/src/api/copilot.ts

```
GET /api/copilot/suggestion/:sessionId -- real-time suggestions
POST /api/copilot/form-fill/:sessionId -- auto-fill form field
```

All routes: complete TypeScript, Zod validation, auth middleware, error handling.

TASK 9: WIDGET FRONTEND

Create: apps/voice-dashboard/src/components/EscalationModal.tsx

SCENARIO A: Agent Available

- "Connecting you to [Agent Name]..." with animated indicator
- Agent photo + name + "Travel Specialist" badge
- "I am sharing our full conversation so you will not need to repeat anything"

SCENARIO B: No Agent - Options

- Four option cards: Wait | Callback | WhatsApp | Schedule
- Each with icon, title, subtitle, estimated time
- Queue wait: live counter updating in real-time via WebSocket

SCENARIO C: Callback Booking

- Name + phone pre-filled
- TAB: "Now" / "Choose Time"
- Calendar date picker (grayed-out = fully booked)
- Time slot grid (30-min slots)
- Language preference selector

SCENARIO D: After-Hours

- Business hours shown: "We're available Mon-Sat 9AM-7PM IST"
- "Schedule callback for tomorrow morning" (1-click)
- "Continue on WhatsApp" option
- "Leave a message" (creates case)

Create: apps/voice-dashboard/src/components/QueueWidget.tsx

- Live position counter: "You are #X in line"
- Progress bar showing movement
- Estimated wait time (updates every 30 seconds)
- "Convert to callback" button (always visible while queuing)
- "Leave queue" button with confirmation

TASK 10: MULTILINGUAL INTERPRETATION LAYER

If no agent with customer's language is available, activate AI interpreter:

Create: packages/agentcore-chat/src/interpreter.ts

```
class RealTimeInterpreter {
  async translateForAgent(
    customerMessage: string,
    fromLanguage: string
  ): Promise<string>
    // Uses Sarvam Translate for Indian languages, GPT for global

  async translateForCustomer(
    agentMessage: string,
    toLanguage: string
```

```

): Promise<string>
  // Translates agent English response to customer's language

// Called when agent message received, before delivery to customer
async interceptAndTranslate(
  message: string,
  chatSessionId: string,
  direction: 'to_agent' | 'to_customer'
): Promise<string>
  - Check if interpretation needed (customer_language != agent_language)
  - If yes: translate
  - Log translation in transcript (both original and translated)
  - Mark translated messages with [translated] tag in audit log
}

VALIDATION CHECKLIST:
[ ] npx tsc --noEmit (zero TypeScript errors across all packages)
[ ] DB migrations run successfully
[ ] Agent heartbeat endpoint working (POST /api/agent/presence/heartbeat)
[ ] Escalation trigger detected when customer says "agent"
[ ] Warm handoff: agent receives screen-pop within 3 seconds
[ ] Queue position updates in real-time via WebSocket
[ ] Callback confirmation sent via WhatsApp
[ ] Scheduled callback initiates outbound call at correct time
[ ] No-answer: WhatsApp retry sent within 60 seconds
[ ] AI co-pilot suggestions appear in agent sidebar
[ ] Cobrowse session starts successfully (WebRTC)
[ ] Multilingual interpretation works (Hindi message -> English for agent -> Hindi back)
[ ] All routes return correct HTTP status codes
[ ] Error handling: graceful degradation if agent service unavailable

```

SUMMARY: COMPLETE FEATURE MATRIX

FEATURE	IMPLEMENTED
Explicit escalation trigger detection	[YES]
Implicit escalation (sentiment/keywords)	[YES]
AI proactive escalation	[YES]
Kill-switch triggers (legal/crisis)	[YES]
Skills-based agent routing	[YES]
Language-based agent routing	[YES]
Tier-based priority (Gold/Silver/Bronze)	[YES]
Agent availability real-time check	[YES]
Warm handoff (< 3 seconds)	[YES]
Full context package to agent	[YES]
Agent screen-pop via realtime push	[YES]
AI co-pilot for agent (live suggestions)	[YES]
Auto-form-fill during agent chat	[YES]
Virtual queue with position display	[YES]
Queue position real-time updates	[YES]
Convert queue to callback	[YES]
Queue abandonment re-engagement	[YES]
Immediate callback	[YES]
Scheduled callback with slot picker	[YES]

Calendar invite (agent + customer)	[YES]
Callback reminder (15 min before)	[YES]
Callback confirmation (WhatsApp + email)	[YES]
No-answer retry (3 attempts)	[YES]
No-answer WhatsApp notification	[YES]
After-hours intelligent handling	[YES]
After-hours callback booking	[YES]
WhatsApp conversation continuation	[YES]
Cobrowse / screen share	[YES]
Multilingual chat (30+ languages)	[YES]
AI real-time interpreter (no native agent)	[YES]
Unified transcript (AI + agent)	[YES]
Post-chat CSAT request	[YES]

Engine 3: WhatsApp Commerce Engine

Technology: WhatsApp Cloud API + WhatsApp Flows + WhatsApp Pay

WhatsApp is India's dominant communication channel with 500M+ users. The WhatsApp Commerce Engine goes beyond simple messaging to provide a full commerce experience within WhatsApp. It serves every industry -- appointment booking for hospitals, loan application for banks, property enquiry for real estate, trial booking for gyms.

Key capabilities:

- WhatsApp Flows: structured multi-step forms (lead capture, booking, feedback) rendered natively in WhatsApp
- WhatsApp Pay integration: in-chat payment collection
- Product carousels: browseable product/package/service displays within chat
- Document collection: request and receive documents (Aadhaar, PAN, prescriptions, property papers) within the WhatsApp conversation
- Template message management: pre-approved message templates for proactive outreach
- Catalog integration: link WhatsApp product catalogs to CRM inventory

Cross-Industry WhatsApp Flows:

- Travel: Destination selection -> Dates -> Travelers -> Budget -> Proposal link
- Healthcare: Symptom -> Doctor selection -> Available slots -> Confirm appointment
- BFSI: Loan type -> Amount -> Employment details -> Application status
- Real Estate: Property type -> Budget -> Location -> Schedule site visit
- Fitness: Goal -> Preferred time -> Trial class booking

Engine 4: Email Intelligence Engine

Technology: Email parsing + GPT-4.1-nano classification + Thread tracking

The Email Intelligence Engine handles inbound and outbound email automation with AI-powered parsing, routing, and draft generation.

Key capabilities:

- Inbound email parsing and intent classification (enquiry, complaint, booking change, document submission, application, renewal)
- Auto-routing to correct engine based on intent (Lead, Case, Booking, etc.)
- Thread tracking across replies and forwards
- Auto-draft responses for agent review
- Attachment processing (forward documents to Document Intelligence Engine)
- Multi-language email support (detect and respond in customer's language)

3.5 Group B -- Intelligence Engines (3)

Engine 5: 4-Layer Memory Service

Technology: Redis + PostgreSQL + pgvector + Custom Procedural Engine

Memory is what separates AgentCore from every competitor. While other platforms treat each conversation as isolated, AgentCore maintains a comprehensive, multi-layered memory system that makes every interaction smarter -- regardless of industry.

- **Layer 1 -- Short-Term Memory (Redis):** Current session context with TTL-based expiry. Stores the active conversation state, recent tool calls, current customer intent, and in-progress workflow data. Sub-millisecond read/write. TTL of 30 minutes after last interaction.
- **Layer 2 -- Episodic Memory (PostgreSQL):** Every interaction ever, structured and queryable. Full conversation transcripts, tool invocations, outcomes, sentiment scores, escalation events, and resolution data. Industry examples:
 - Travel: "Last time you called about your Bali trip, we discussed the visa requirements."
 - Healthcare: "Your last visit was on March 15 for a blood pressure follow-up. Dr. Sharma prescribed Amlodipine."
 - BFSI: "You applied for a home loan of INR 50 lakhs on February 20. The status was pending verification."
 - Legal: "Your property dispute case was last heard on March 10. The next hearing is April 5."
 - Fitness: "You attended 12 out of 20 sessions this month. Your trainer recommended increasing to 4x/week."

- **Layer 3 -- Semantic Memory (pgvector):** Embeddings for RAG retrieval. Customer preferences, product knowledge, company policies, FAQ answers, and historical interaction summaries are embedded as vectors and retrieved using similarity search. This enables context-aware responses without stuffing the LLM context window with irrelevant data.
- **Layer 4 -- Procedural Memory (PostgreSQL):** Agent workflows, tenant-specific business rules, escalation procedures, and compliance requirements. This layer tells the AI how to behave differently per tenant and per industry:
 - For a bank: "Always verify customer identity before sharing account details"
 - For a clinic: "Never provide diagnosis. Only share doctor-confirmed results."
 - For a law firm: "Do not discuss fees until consultation is scheduled"
 - For a gym: "Always offer trial class to new enquiries"

Engine 6: Multi-Agent Orchestrator

Technology: LangGraph Supervisor/Worker Pattern

The orchestrator manages up to 7 parallel worker agents per call, each specialized for a specific task. This is not sequential processing -- agents work simultaneously, dramatically reducing response time for complex queries.



The Supervisor Agent is the conversational controller. It:

1. Receives the user's intent from the voice pipeline
2. Determines which worker agents are needed
3. Dispatches tasks to workers in parallel
4. Aggregates worker responses
5. Synthesizes a coherent response for the user
6. Maintains conversation flow and context

Worker Agent 1: Qualification Agent

Purpose: Qualify leads by collecting structured information through natural conversation.

Capabilities: Dynamic question selection, adaptive questioning style, real-time lead scoring, buying signal detection, objection handling, writes to Lead Engine.

Industry-specific qualification:

- Travel: destination, dates, travelers, budget, occasion
- Healthcare: symptoms, urgency, preferred doctor, insurance
- BFSI: product type, amount, employment, existing relationship

- Real Estate: property type, budget, location, timeline
- Legal: case type, parties, urgency, existing documentation

Worker Agent 2: Customer Memory Agent (RAG Retrieval)

Purpose: Retrieve and contextualize customer history from the 4-layer memory system.

Capabilities: Pre-call context loading in sub-200ms, semantic search via pgvector, episodic memory retrieval, sentiment trend identification, communication style insights.

Worker Agent 3: Quote Generation Agent

Purpose: Generate accurate, real-time quotes during conversation.

Capabilities: Real-time CRM inventory queries, multi-component quote calculation, PDF generation with tenant branding, WhatsApp delivery, quote status tracking.

Industry-specific quotes:

- Travel: trip package quote with flights, hotels, activities
- Photography: event photography quote with hours, deliverables, editing
- Legal: legal fee estimate based on case complexity and jurisdiction
- Healthcare: treatment cost estimate with insurance coverage
- Real Estate: property valuation or rental quote
- Fitness: membership package quote with add-ons

Worker Agent 4: Document Intelligence Agent

Purpose: Process, verify, and manage documents within conversations.

Capabilities: Passport OCR (MRZ extraction), ID verification, document completeness checking, photo quality assessment, secure document handling.

Industry-specific documents:

- Travel: passport, visa, travel insurance
- Healthcare: prescriptions, lab reports, insurance cards, ABDM health records
- BFSI: PAN card, Aadhaar, income proof, bank statements
- Real Estate: property papers, sale deed, encumbrance certificate
- Legal: court orders, contracts, affidavits, power of attorney

Worker Agent 5: Disruption Monitor Agent

Purpose: Monitor for and respond to disruptions affecting customers.

Capabilities: Service disruption alerts, schedule change detection, automatic rebooking suggestions.

Industry-specific disruptions:

- Travel: flight delays, hotel overbooking, weather events
- Healthcare: doctor schedule changes, lab equipment downtime
- Logistics: delivery delays, route disruptions, warehouse issues
- Automotive: service part unavailability, recall notices

Worker Agent 6: Handoff/Escalation Agent

Purpose: Manage seamless transitions between AI and human agents.

Capabilities: Escalation trigger detection (frustration, complex issues, explicit request), sentiment-based escalation, context packaging, warm transfer, cool-down handling, post-handoff tracking.

Worker Agent 7: Compliance & Governance Agent

Purpose: Ensure every interaction complies with regulatory requirements.

Capabilities: Real-time compliance monitoring, consent tracking, PII detection and masking, prohibited topic detection, audit log generation, regulatory rule updates.

Industry-specific compliance:

- BFSI: RBI/SEBI/IRDAI guidelines, financial advice disclaimers
- Healthcare: HIPAA (international), ABDM data sharing norms, no-diagnosis rule
- Legal: attorney-client privilege protections, Bar Council rules
- Real Estate: RERA disclosures, anti-money laundering checks

Engine 7: Proactive Outbound Engine

Technology: Scheduler + Event Engine + Multi-Channel Delivery

AgentCore does not just wait for customers to call -- it proactively reaches out when there is value to deliver.

Key capabilities:

- Price drop alerts: notify customers when packages/products they viewed drop in price
- Re-engagement campaigns: reach out to leads that went cold with personalized offers
- NPS/feedback collection: automated post-service satisfaction surveys
- Disruption notifications: alert customers about schedule changes, service disruptions
- Reminder sequences: payment reminders, document submission reminders, appointment reminders, renewal reminders
- Multi-channel delivery: voice call, WhatsApp message, or SMS based on customer preference and time of day
- Throttling and compliance: respect DND registries, time-of-day restrictions, and frequency caps

Industry-specific proactive outreach:

- Travel: pre-trip checklists, visa reminders, price drops on watched destinations
- Healthcare: appointment reminders, medication refill alerts, follow-up scheduling
- BFSI: policy renewal reminders, SIP date alerts, EMI due reminders
- Fitness: inactive member re-engagement, renewal reminders, class schedule changes
- Automotive: service due reminders, insurance renewal, recall notifications
- Education: fee payment reminders, exam date alerts, result notifications

3.6 Group C -- CRM Lifecycle Engines (8)

Engine 8: Lead Engine

Technology: REST API + PostgreSQL + Deduplication Service

Universal lead management that works across every channel, product, and industry.

Key capabilities:

- Universal lead creation from any channel (voice, chat, WhatsApp, web form)
- Lead scoring based on conversation quality, engagement signals, and qualification criteria
- Deduplication: automatic detection of duplicate leads across channels using phone, email, and fuzzy name matching
- Lead assignment: rule-based and round-robin assignment to sales teams
- Lead lifecycle tracking: new -> contacted -> qualified -> converted -> lost
- Integration with CRM lead objects in each vertical product

What "Lead" Means Across Industries:

- Travel: enquiry for a trip package
- Photography: request for event photography quote
- Real Estate: property enquiry or site visit request
- Healthcare: patient appointment request or consultation enquiry
- Legal: client intake or consultation request
- BFSI: loan application, insurance enquiry, investment query
- Fitness: trial class request or membership enquiry
- Education: admission enquiry or demo class request
- Automotive: test drive request or vehicle enquiry
- Hospitality: reservation request or event booking enquiry

Engine 9: Proposal Engine [NEW -- The Most Important V2 Addition]

Technology: GPT-4.1 + PDF Generator + Web Renderer + Tracking Service + WhatsApp Delivery

The Proposal Engine is the critical missing link between lead qualification and quote generation. It generates beautiful, personalized proposal documents that tell a story, show relevant images/details, build excitement, and convert a qualified lead into a paying customer.

What a Proposal Is vs a Quote:

QUOTE:

"4 nights Bali at Hotel X. Flights DEL-DPS. Total: INR 52,000/person."
= Pricing only. Numbers. No story. No persuasion.
= Cold, transactional.

PROPOSAL:

- "Your Dream Bali Getaway -- Curated by [Agency Name]"
- = Beautiful visual document with destination imagery
- = Day-by-day narrative itinerary
- = Why this trip suits THIS customer specifically
- = Multiple options presented (Standard vs Deluxe vs Ultimate)
- = Pricing with value justification
- = Social proof (reviews, testimonials)
- = Agent's personal note
- = Call-to-action with expiry
- = Shareable link (customer can view on phone)
- = Customer can approve/request changes directly

A PROPOSAL is what converts a lead into a booking/sale/engagement.

A QUOTE is just one component of the proposal.

What "Proposal" Means Across Industries:

- Travel: curated trip package with narrative itinerary and imagery
- Photography: event photography proposal with portfolio samples and packages
- Legal: legal service proposal with scope, timeline, and fee estimate
- Healthcare: treatment plan with procedure details and cost breakdown
- Real Estate: property presentation with floor plans, amenities, and EMI calculation
- Fitness: membership proposal with package comparison and trainer profiles
- Education: admission offer with course details, fees, and scholarship options
- Automotive: vehicle configuration proposal with finance options
- Hospitality: event or wedding proposal with venue details and packages

Proposal Engine Architecture -- Complete 8-Step Flow:

TRIGGER: Lead qualifies (key information collected per industry)

STEP 1: AI PROPOSAL GENERATION

Input: lead_data, customer_preferences, tenant_catalog, memory_context

AI-Generated sections:

- a) Cover page: hero image + personalized headline
- b) Personal greeting from agent
- c) Context story: why this solution fits this customer
- d) Detailed offering: day-by-day, item-by-item, or service-by-service
- e) Package options: 2-3 tiers with visual comparison
- f) Inclusions/exclusions table
- g) Testimonials: pulled from past similar customers
- h) Agent's personal recommendation note
- i) Pricing section (linked to Quote Engine)
- j) Next steps + expiry date
- k) Business branding + contact

STEP 2: PROPOSAL RENDERING

Format options:

- Interactive web page (shareable link, works on phone)
- PDF (downloadable, printable)
- WhatsApp version (carousel + text summary)

STEP 3: PROPOSAL DELIVERY

Primary: WhatsApp shareable link

Secondary: Email with PDF attachment

In-chat: Interactive cards if chat agent

STEP 4: PROPOSAL TRACKING

Track: opened, time spent, pages viewed, sections read

If not opened in 24h: proactive follow-up

If opened but not responded: "I see you were looking at the proposal..."

STEP 5: CUSTOMER INTERACTION WITH PROPOSAL

Customer taps [Modify] -> sends modification to agent queue

Customer taps [Select Package] -> Quote Engine triggered

Customer taps [Approve & Proceed] -> Approval Engine triggered

Customer adds comments inline -> agent notified with context

STEP 6: VERSION MANAGEMENT

Every modification creates new version

Customer can compare V1 vs V2

Agent can see exactly what changed and why

STEP 7: APPROVAL FLOW

Customer approves via tap, e-signature, or verbal confirmation

If high-value: internal approval required before proceeding

Approval Engine handles multi-step workflows

STEP 8: CONVERSION TO QUOTE

Approved proposal -> formal quote generated automatically

Quote linked to proposal version

Quote includes all pricing from selected package option

Proposal Engine Database Schema:

```
model Proposal {
  id          String    @id @default(cuid())
  tenantId    String
  leadId      String
  customerId  String?
  status      String    -- draft|sent|viewed|modified|approved|expired|rejected
  version     Int       @default(1)
  coverTitle  String?
  personalGreeting String? @db.Text
  destinationStory String? @db.Text
  itineraryNarrative Json? -- day-by-day or item-by-item narrative array
  packageOptions Json?   -- 2-3 tier options
  selectedPackage String?
  pricingData  Json?     -- linked from Quote Engine
  mediaUrls    Json?     -- images used
  testimonials Json?     -- selected testimonials
  agentNote    String?   @db.Text
```

```

expiresAt      DateTime?
shareableLink  String?   @unique
pdfUrl        String?

-- Tracking
firstOpenedAt  DateTime?
lastViewedAt   DateTime?
totalViewTimeMs Int?
sectionsViewed Json?

-- Customer interactions
modifications  Json?      -- customer-requested changes
customerComments String?  @db.Text
approvedAt     DateTime?
approvedByMethod String?  -- whatsapp|email|esign|verbal

-- AI generation metadata
aiGeneratedBy  String?    -- model used
aiPromptVersion String?
generationTimeMs Int?

createdAt      DateTime @default(now())
updatedAt      DateTime @updatedAt

@@index([tenantId, leadId])
@@index([shareableLink])
@@index([status, tenantId])
}

```

Proposal Engine API Endpoints:

```

POST /api/proposals/generate    -- AI generates proposal from lead
GET  /api/proposals/:id        -- Get proposal details
PUT  /api/proposals/:id        -- Update proposal
POST /api/proposals/:id/send    -- Send via WhatsApp + Email
GET  /api/proposals/:id/preview -- Web preview (shareable link)
GET  /api/proposals/:id/pdf     -- Download PDF
POST /api/proposals/:id/version -- Create new version
GET  /api/proposals/:id/tracking -- View tracking data
POST /api/proposals/:id/approve -- Customer approves (from link)
POST /api/proposals/:id/modify  -- Customer requests modification
POST /api/proposals/:id/reject  -- Customer rejects
GET  /api/proposals/:id/timeline -- Full audit trail

```

Engine 10: Quote Engine

Technology: REST API + Pricing Service + PDF Generator + WhatsApp Delivery

Real-time quote generation during voice and chat conversations. V2 UPDATE: Now linked to Proposal Engine -- quotes are generated from approved proposals.

Key capabilities:

- Live pricing calculation based on CRM inventory and pricing rules
- Dynamic quote generation: multi-line quotes with configurable components
- PDF quote generation with tenant branding
- WhatsApp delivery: send formatted quotes directly to customer's WhatsApp
- Quote tracking: opened, viewed, accepted, expired, revised
- Quote-to-booking conversion with one-click acceptance
- Multi-currency support with real-time exchange rates
- [NEW] Link to approved proposal version
- [NEW] GST breakdown included

Engine 11: Case Engine

Technology: REST API + PostgreSQL + SLA Timer Service

Customer support case management with intelligent routing and SLA enforcement. Works identically across all industries -- a complaint is a complaint whether it is about a delayed flight, a wrong prescription, a defective product, or a missed service appointment.

Key capabilities:

- Case creation from any channel with automatic categorization
- SLA timer management: configurable per-tenant, per-category SLA targets
- Intelligent routing: route cases to the right team based on category, priority, language, and agent availability
- Escalation workflows: automatic escalation when SLA is approaching breach
- Case merging: detect and merge duplicate cases from the same customer
- Resolution tracking and customer satisfaction scoring
- Integration with CRM case objects in each vertical product

Engine 12: Approval/Workflow Engine [NEW]

Technology: Workflow Engine + Notification Service + SLA Timer

The Approval Engine handles multi-step approval workflows across all business objects and industries.

Needed for:

1. Proposal approval (customer approves proposal before full quote)
2. Quote approval (high-value quotes need manager approval before sending)
3. Booking approval (for corporate accounts with travel/purchase policy)
4. Refund approval (refunds above threshold need manager sign-off)
5. Expense approval (Aayiq: multi-level approval workflow)

6. Supplier/vendor contract approval (new partner onboarding)
7. Loan approval (BFSI: multi-level credit approval)
8. Treatment plan approval (Healthcare: specialist review before procedure)

Engine design:

```
class ApprovalEngine {
    // Create a multi-step approval workflow
    createWorkflow(config: WorkflowConfig): Workflow

    // Submit item for approval
    submitForApproval(itemType, itemId, workflowId, submittedBy): ApprovalRequest

    // Get pending approvals for a user
    getPendingApprovals(approverId): ApprovalRequest[]

    // Approve with optional comment
    approve(requestId, approverId, comment?): ApprovalResult

    // Reject with mandatory reason
    reject(requestId, approverId, reason): ApprovalResult

    // Escalate (if SLA breached)
    escalate(requestId): void

    // Get full approval history
    getHistory(itemType, itemId): ApprovalHistory
}
```

Workflow config example (Quote approval > INR 5L):

```
{
  "trigger": "quote.created AND quote.total > 500000",
  "steps": [
    { "approver": "senior_agent", "timeout_hours": 4, "on_timeout": "escalate" },
    { "approver": "manager", "timeout_hours": 8, "on_timeout": "auto_approve" }
  ],
  "notifications": {
    "on_submit": "whatsapp_to_approver",
    "on_approve": "email_to_submitter",
    "on_reject": "whatsapp_to_submitter_with_reason",
    "on_timeout": "escalation_alert"
  }
}
```

Engine 13: Booking Engine

Technology: REST API + Payment Gateway Integration + Voucher Generator

End-to-end booking management from payment to confirmation. "Booking" is the universal term for a confirmed transaction -- whether it is a travel booking, a medical appointment, a legal consultation, a gym membership, a test drive, or a hotel reservation.

Key capabilities:

- Payment collection via integrated payment gateways (Razorpay, Stripe)
- Booking confirmation with automated voucher/receipt generation
- Calendar integration for availability management
- Cancellation and modification workflows
- Refund processing with configurable policies
- Post-booking communication: reminders, updates, feedback requests
- Integration with CRM booking objects in each vertical product
- [NEW] Partial payment and installment support
- [NEW] Links to Invoice Engine for billing

Engine 14: Invoice Engine [NEW]

Technology: REST API + GST Calculation Service + PDF Generator + Payment Tracking

Separate from Quote. Invoice is the billing document generated after booking/service confirmation.

Features:

- GST-compliant invoices (CGST/SGST/IGST based on Place of Supply)
- Multi-currency invoicing with exchange rate lock
- Proforma invoice (before full payment)
- Tax invoice (after payment)
- Credit note (for cancellations/refunds)
- Debit note (for additional charges)
- E-invoice (GST portal integration for B2B)
- Automatic numbering series per tenant
- WhatsApp delivery of invoice PDF
- Payment link embedded in invoice
- Payment reminders (3/7/14 days overdue)
- Partial payment tracking
- Receipt generation on each payment

Invoice schema key fields:

```
invoiceNumber:  tenant-specific series (e.g., INV-ABC-2025-001)
invoiceType:    proforma | tax_invoice | credit_note | debit_note
bookingId, customerId, tenantId
lineItems:      Json (description, qty, rate, discount, tax_rate, amount)
subtotal, discountAmount, taxableAmount
```

```
cgst, sgst, igst, cess (GST fields)
totalAmount, amountPaid, balanceDue
currency, exchangeRate, exchangeRateDate
placeOfSupply: for GST determination
invoiceDate, dueDate
paymentStatus: unpaid | partial | paid | overdue | cancelled
payments:      relation to Payment[]
receipts:      relation to Receipt[]
```

Engine 15: Refund/Cancellation Engine [NEW]

Technology: REST API + Policy Engine + Payment Gateway Refund API

Complete lifecycle management after a booking/transaction is made. Works universally across industries.

Customer Cancellation Flow:

```
Customer requests cancellation (WhatsApp/phone/chat)
-> AI checks cancellation policy (loaded from booking terms)
-> AI calculates refund amount based on: days to service, policy terms
-> If within policy: auto-process refund
-> If borderline: escalate to agent with recommendation
-> Generate cancellation note (debit note or credit note)
-> Initiate refund via Razorpay/Stripe
-> Update booking status
-> Notify service provider to release inventory/slot
-> WhatsApp confirmation to customer with refund timeline
```

Refund Processing:

```
-> Razorpay Refund API or Stripe Refund API
-> Timeline: UPI 1-2 days, card 5-7 days, bank transfer 3-5 days
-> WhatsApp notification with refund reference number
-> Update financial records
```

3.7 Group D -- Operations Engines (6)

Engine 16: Document Intelligence Agent

Technology: OCR Engine + Validation Service + Compliance Checker

Automated document processing and verification within conversations. Industry-agnostic-- processes any document type.

Key capabilities:

- Passport OCR: extract name, nationality, expiry, passport number from passport images

- ID verification: extract and validate government ID documents (Aadhaar, PAN, Voter ID, Driving License)
- Document compliance checking: verify documents meet submission requirements
- Secure document storage with encryption at rest and configurable retention
- Integration with WhatsApp for document collection and ChatAgent for document upload
- [NEW] E-signature integration layer for proposal and contract signing

Industry-Specific Document Types:

- Travel: passport, visa, travel insurance, vaccination certificates
- Healthcare: prescriptions, lab reports, insurance cards, ABDM records, consent forms
- BFSI: PAN, Aadhaar, salary slips, bank statements, ITR, Form 16
- Real Estate: sale deed, encumbrance certificate, property tax receipts, RERA certificates
- Legal: court orders, FIR copies, agreements, power of attorney, affidavits
- Education: marksheets, certificates, transfer certificates, migration certificates

Engine 17: Analytics & Observability Service

Technology: PostgreSQL + Time-Series Metrics + Dashboard API

Comprehensive analytics for platform operators and tenants.

Key capabilities:

- Per-tenant dashboards: call volume, resolution rate, average handle time, cost per call
- Agent scoring: AI agent performance metrics (resolution rate, escalation rate, customer satisfaction)
- Conversation analytics: intent distribution, common failure points, topic clustering
- Cost tracking: per-call and per-tenant cost breakdown across all AI providers
- Feedback loops: customer satisfaction data feeds back into agent improvement
- Real-time monitoring: active calls, queue depth, error rates, latency percentiles
- Alerting: configurable alerts for SLA breaches, error spikes, cost anomalies

Engine 18: Supplier Management Engine [NEW]

Technology: REST API + Supplier Catalog + Rate Management + Integration Layer

Tenant's supplier/vendor catalog. In travel, these are hotels, airlines, activities. In healthcare, these are labs, pharmacies, specialists. In real estate, these are builders, agents, service providers. The engine is generic -- it manages any type of supplier relationship.

Features:

- **Supplier profiles:** Configurable fields per industry (star rating for hotels, specialization for doctors, capacity for venues)

- **Rate management:** Seasonal rates, contracted vs published rates, markup rules per supplier per tenant tier
- **Availability:** Blackout dates, capacity limits, real-time check via supplier API if available
- **Contract management:** Contract upload (PDF), cancellation policy per supplier, commission rate, payment terms
- **Performance tracking:** Customer ratings, complaint frequency, response time, completion rate
- **Integration:** Pull from industry-specific systems (GDS for travel, ABDM for healthcare, MLS for real estate)

Engine 19: Campaign Engine [NEW]

Technology: Segment Builder + Template Engine + Multi-Channel Delivery + Analytics

Marketing automation for tenants to reach their customer base. Not a generic email marketing tool -- configurable per industry with AI-powered personalization.

Features:

- **Segment builder:** By interest, service date, spending tier, engagement recency, product type, custom attributes
- **Campaign types:** WhatsApp broadcast (Meta-approved templates), email newsletter, SMS (time-sensitive), automated drip sequences (7-touch lead nurture)
- **AI-powered content:** AI writes campaign copy in customer's preferred language, AI selects best send time per segment, A/B test two versions automatically, personalise each message with customer context
- **Analytics:** Open rate, click rate, conversion rate per campaign, revenue attributed, unsubscribe rate, segment performance comparison
- **GDPR/DPDP compliance:** Consent required before sending, one-click unsubscribe, opt-in tracking with timestamp, blacklist management

Engine 20: NPS/Feedback Engine [NEW]

Technology: Rating Collection + Sentiment Analysis + Review Routing + Insights Dashboard

Post-service feedback collection that was embedded in Proactive in V1 but is now a standalone engine.

Features:

- Multi-channel NPS collection (WhatsApp, voice, email, SMS)
- 5-star rating with follow-up logic: 4-5 stars -> Google Review prompt; 1-3 stars -> Case created, retention agent assigned
- Open-ended feedback analysis via GPT-4.1-mini (sentiment, topics, actionable items)
- Per-supplier/provider feedback tracking

- Per-agent feedback (track AI agent quality)
- Trend dashboards: NPS over time, by product, by segment
- Automated re-engagement for detractors

Engine 21: Compliance Service

Technology: Policy Engine + Audit Logger + Consent Manager

Regulatory compliance built into the platform, not bolted on. Enforces industry-specific rules per tenant.

Key capabilities:

- DPDP Act 2023 compliance: consent collection, purpose limitation, data erasure rights
- TRAI compliance: call recording disclosure, DND registry checking
- GDPR readiness: data portability, right to erasure, consent management
- Industry-specific compliance rules (HIPAA for healthcare, SEBI for financial services, RERA for real estate, Bar Council for legal)
- Audit trail: immutable log of every AI decision, tool call, and data access
- Kill-switch: emergency shutdown capability for individual tenants or the entire platform
- Data retention policies: configurable per-tenant retention with automatic purging
- Consent management: granular consent tracking for voice recording, data processing, marketing outreach

Engine 22: Tenant Management Service

Technology: REST API + PostgreSQL + Billing Metering

Multi-tenant infrastructure management for the platform.

Key capabilities:

- Tenant provisioning: automated setup of new tenants with configurable defaults
- Industry template selection: pre-configured tool sets, compliance rules, and workflows per industry
- Configuration management: per-tenant settings for voice, LLM, language, branding, and business rules
- Billing metering: real-time usage tracking for per-minute, per-call, and per-feature billing
- White-label support: custom branding, custom domains, custom voice personas per tenant
- Resource isolation: configurable isolation levels from shared to dedicated infrastructure
- Usage quotas and rate limiting: prevent individual tenants from consuming excessive resources
- API key management: per-tenant API keys with scope-based permissions

3.8 Group E -- Vertical Extension Engines (2)

Engine 23: Itinerary Builder Engine [NEW -- Travel Specific]

Technology: Day-by-Day Planner + Template Engine + AI Narrative + Cost Calculator

The hero feature of ItinerarySuite. Day-by-day trip planner that feeds into Proposal Engine and Quote Engine.

Features:

- **Builder interface:** Day-by-day drag-and-drop planner, service types (flight, hotel, transfer, activity, meal, free time, custom), each service with time/supplier/description/cost, group FIT and GIT support, multi-destination itineraries
- **Templates:** Pre-built destination templates, agency's custom templates, clone from past bookings
- **AI assistance:** "Build a 7-night Bali itinerary for a family with 2 kids, budget INR 2L/person" -- AI suggests day-by-day services from supplier catalog + GDS
- **Costing:** Auto-calculate total cost as services are added, per-person and group cost, gross vs net, margin visibility, GST calculation per service type
- **Versions:** Draft, Shared with Customer, Approved, Booked -- track what customer changed
- **Output:** Beautiful itinerary PDF with images, WhatsApp-shareable version
- **Integration:** Feeds into Proposal Engine, Quote Engine, Booking Engine, Supplier Management

This engine serves as a template for building similar vertical extensions in other industries (treatment plan builder for healthcare, event plan builder for event management, etc.).

Engine 24: Expense Capture Engine [NEW -- Aaiyiq/SpendPilot Specific]

Technology: Voice Capture + Receipt OCR + Policy Engine + Approval Workflow

Voice-first expense capture designed for Aaiyiq and SpendPilot products.

Features:

- Voice expense capture: "I spent fifteen hundred rupees on dinner at Taj with 3 clients"
- Receipt OCR via camera or WhatsApp photo
- Automatic categorization (travel, meals, accommodation, transport, office)
- Policy compliance check in real-time (amount limits, category rules, required approvals)
- Multi-level approval routing based on amount and category
- GST input credit extraction from receipts
- Mileage calculation for travel expenses
- Per diem calculation for multi-day trips

- Integration with Approval Engine for workflow
- Integration with Analytics for spend reporting

3.9 Feature Comparison Matrix

Feature	AgentCore	Retell AI	Vapi	ElevenLabs Conv. AI	Bland AI
Self-hosted deployment	[YES]	[NO]	[NO]	[NO]	[NO]
Hybrid cascaded + S2S pipeline	[YES]	Cascaded only	Cascaded only	N/A	Cascaded only
Multi-agent orchestration (7 parallel)	[YES]	[NO]	[NO]	[NO]	[NO]
4-layer persistent memory	[YES]	[NO]	[NO]	[NO]	[NO]
WhatsApp Commerce (Flows, Pay)	[YES]	[NO]	[NO]	[NO]	[NO]
Proactive outbound intelligence	[YES]	Basic outbound	Basic outbound	[NO]	Outbound only
Document Intelligence (OCR, ID)	[YES]	[NO]	[NO]	[NO]	[NO]
22 Indian languages + Hinglish	[YES]	Limited	Limited	TTS only	[NO]
Multi-industry deployment templates	[YES] 15+ industries	[NO]	[NO]	[NO]	[NO]
Vertical CRM integration	Deep	None	None	None	None
Lead scoring + deduplication	[YES]	[NO]	[NO]	[NO]	[NO]
Proposal generation + tracking	[YES]	[NO]	[NO]	[NO]	[NO]
Quote generation + PDF + delivery	[YES]	[NO]	[NO]	[NO]	[NO]
Booking + payment integration	[YES]	[NO]	[NO]	[NO]	[NO]
Invoice + GST compliance	[YES]	[NO]	[NO]	[NO]	[NO]
Case routing + SLA management	[YES]	[NO]	[NO]	[NO]	[NO]
Multi-tenant white-label	[YES]	[NO]	Limited	[NO]	[NO]

Feature	AgentCore	Retell AI	Vapi	ElevenLabs Conv. AI	Bland AI
Per-tenant billing metering	[YES]	[NO]	[NO]	[NO]	[NO]
DPDP Act 2023 compliance	[YES]	[NO]	[NO]	[NO]	[NO]
TRAI compliance (DND, disclosure)	[YES]	[NO]	[NO]	[NO]	[NO]
HIPAA / BFSI / RERA compliance	[YES]	[NO]	Partial	[NO]	[NO]
Kill-switch framework	[YES]	[NO]	[NO]	[NO]	[NO]
Omnichannel unified memory	[YES]	[NO]	[NO]	[NO]	[NO]
Cost per minute (self-hosted)	\$0.039-0.077	\$0.10-0.15	\$0.13-0.31	\$0.08-0.10	\$0.09+

3.10 All 24 Engines Manifest Table

#	Engine	Group	Priority	New/Updated	Cost/min Impact
1	VoiceAgent Engine	Communication	P0	Updated (Sarvam+ElevenLabs API)	\$0.039-0.059
2	ChatAgent Engine	Communication	P0	Updated (full multilingual)	Near zero
3	WhatsApp Commerce Engine	Communication	P0	Existing	~\$0.011/msg
4	Email Intelligence Engine	Communication	P1	Existing	Near zero
5	4-Layer Memory Service	Intelligence	P0	Existing	Near zero
6	Multi-Agent Orchestrator	Intelligence	P0	Existing	\$0.002/min
7	Proactive Outbound Engine	Intelligence	P1	Existing	\$0.011/msg
8	Lead Engine	CRM Lifecycle	P0	Existing	Near zero
9	Proposal Engine	CRM Lifecycle	P0	[NEW]	Near zero (AI gen)
10	Quote Engine	CRM Lifecycle	P0	Existing (updated flow)	Near zero
11	Case Engine	CRM Lifecycle	P0	Existing	Near zero
12	Approval/Workflow Engine	CRM Lifecycle	P1	[NEW]	Near zero
13	Booking Engine	CRM Lifecycle	P0	Existing (enhanced)	Near zero

#	Engine	Group	Priority	New/Updated	Cost/min Impact
14	Invoice Engine	CRM Lifecycle	P1	[NEW]	Near zero
15	Refund/Cancellation Engine	CRM Lifecycle	P1	[NEW]	Near zero
16	Document Intelligence	Operations	P1	Existing (enhanced)	\$0.005/call
17	Analytics & Observability	Operations	P2	Existing	Near zero
18	Supplier Management Engine	Operations	P1	[NEW]	Near zero
19	Campaign Engine	Operations	P2	[NEW]	\$0.011/msg
20	NPS/Feedback Engine	Operations	P2	[NEW]	Near zero
21	Compliance Service	Operations	P0	Existing	Near zero
22	Tenant Management Service	Operations	P0	Existing	Near zero
23	Itinerary Builder Engine	Vertical-Travel	P0	[NEW]	Near zero
24	Expense Capture Engine	Vertical-Expense	P0	[NEW]	Near zero

Section 4: Complete Customer Journey -- Universal

4.1 The Universal Journey

The customer lifecycle is fundamentally the same across all industries. Only the terminology changes. AgentCore's 24 engines map to every step of this universal journey:

UNIVERSAL JOURNEY:

ENQUIRY/LEAD

- > QUALIFICATION
- > PROPOSAL / ESTIMATE / PLAN
- > APPROVAL
- > QUOTE / PRICING
- > BOOKING / APPOINTMENT / ENROLLMENT / PURCHASE
- > INVOICE
- > PAYMENT
- > SERVICE DELIVERY
- > POST-SERVICE NPS / FEEDBACK
- > RETENTION / RE-ENGAGEMENT

PARALLEL TRACKS (at any stage):

- > CASE / COMPLAINT / SUPPORT
- > DISRUPTION / CHANGE MANAGEMENT

-> DOCUMENT COLLECTION
-> CAMPAIGN / MARKETING

4.2 Journey Mapped Across 5 Industries

Journey Stage	Travel	Healthcare	Real Estate	Legal	Fitness
Lead	Trip enquiry	Appointment request	Property enquiry	Consultation request	Trial class request
Qualification	Destination, dates, pax, budget	Symptoms, urgency, doctor preference	Budget, location, type, timeline	Case type, urgency, parties	Goals, schedule, preferences
Proposal	Curated trip package	Treatment plan	Property presentation	Legal service proposal	Membership package
Approval	Customer selects package	Patient agrees to plan	Buyer shortlists property	Client accepts engagement	Member selects plan
Quote	Package pricing	Treatment cost estimate	Property price + EMI	Legal fee estimate	Membership pricing
Booking	Trip booked, deposit paid	Appointment confirmed	Site visit / Agreement	Consultation scheduled	Membership activated
Invoice	Travel invoice + GST	Medical bill + insurance	Brokerage / Agreement fee	Legal fee invoice	Monthly membership
Payment	Razorpay / WhatsApp Pay	Online / Insurance	Bank transfer / EMI	Retainer / milestone	Auto-debit / UPI
Service	Trip executed	Treatment delivered	Property shown / Deal closed	Legal representation	Training sessions
NPS	Post-trip feedback	Post-visit satisfaction	Post-deal review	Case outcome satisfaction	Monthly check-in
Retention	Next trip suggestion	Annual health check	Investment property	Ongoing advisory	Renewal + upsell

4.3 Detailed Flow with Engine Mapping

```
CHANNEL (Phone/Chat/WhatsApp/Email/Web)
  |
  v
LEAD CAPTURED <-- [Lead Engine]
- Source tagged (which channel, which campaign)
- Duplicate check (mobile/email)
- Initial AI score
- Memory: returning customer context loaded [Memory Service]
  |
  v
QUALIFICATION <-- [VoiceAgent / ChatAgent + Qualification Worker Agent]
```

```
- AI voice/chat collects industry-specific information
- Lead score updated in real-time
- If score < 40: nurture sequence starts [Campaign Engine]
- If score >= 40: move to Proposal
  |
  v
PROPOSAL GENERATED <-- [Proposal Engine]
- AI generates industry-appropriate proposal document
- Includes: offering narrative, package options, pricing, testimonials
- Sent via WhatsApp shareable link + email PDF [WhatsApp Commerce Engine]
- Tracked: opened, time spent, modifications
- Customer can comment/modify directly
  |
  v
PROPOSAL APPROVED <-- [Approval Engine]
- Customer taps [Accept] or signs via e-signature
- Internal sign-off if high value
- Modification loop: customer requests changes -> new version -> re-sent
  |
  v
QUOTE GENERATED <-- [Quote Engine]
- Linked to approved proposal version
- Includes: itemized pricing, payment schedule, T&Cs
- GST breakdown
- Valid for X days (tenant configurable)
- Sent via WhatsApp + email
  |
  v
BOOKING CREATED <-- [Booking Engine]
- Customer pays deposit (via WhatsApp Pay or payment link)
- Booking record created in DB
- Service provider/supplier bookings initiated [Supplier Mgmt Engine]
- Proforma invoice sent [Invoice Engine]
- Document collection starts [Document Intelligence]
  |
  v
INVOICE GENERATED <-- [Invoice Engine]
- Tax invoice on full payment
- GST-compliant
- Sent via WhatsApp + email
- Stored in booking documents
  |
  v
SERVICE EXECUTED
- Real-time updates if disruptions [Disruption Monitor Agent]
- Reminders and alerts [Proactive Outbound Engine]
- Emergency contact access
  |
  v
POST-SERVICE <-- [NPS/Feedback Engine]
- Rating request sent
- If 4-5 stars: Google Review prompt
- If 1-3 stars: Case created [Case Engine], retention agent assigned
- Upsell: suggest next service based on history [Memory Service]
- Re-engagement sequence starts [Campaign Engine]
```

4.4 Parallel Tracks

At any stage during the customer journey, regardless of industry:

- **Case Engine:** Customer can raise a complaint or issue via any channel
- **Document Intelligence:** ID verification, document collection, compliance checking
- **Disruption Monitor:** Monitors for schedule changes, service disruptions, external events
- **Proactive Outbound:** Reminders, alerts, follow-ups, re-engagement
- **Campaign Engine:** Nurture sequences, seasonal offers, cross-sell campaigns

4.5 Cancellation Branch

CANCELLATION BRANCH (if applicable, any industry):

- > Refund/Cancellation Engine calculates refund per policy
- > Credit note generated [Invoice Engine]
- > Refund initiated via payment gateway
- > Invoice updated
- > Service provider notified to release slot/inventory [Supplier Mgmt Engine]
- > Customer notified with refund timeline [WhatsApp Commerce Engine]

Section 5: Technical Architecture

NOTE: The technical architecture described in this section is entirely industry-agnostic. The same hybrid pipeline, multi-agent orchestration, 4-layer memory system, and AWS infrastructure serve a travel agency, a hospital, a bank, a gym, or a law firm identically. Industry-specific behavior is achieved through configurable tool registrations and tenant-level procedural memory -- not through architectural changes.

5.1 The Hybrid Pipeline (Cascaded + Speech-to-Speech)

AgentCore employs a unique hybrid voice pipeline that intelligently routes between two processing modes based on conversation requirements. This dual-pipeline approach is a key architectural differentiator -- no competitor offers this capability.

Pipeline A: Cascaded Pipeline

The cascaded pipeline processes speech through discrete stages, each optimized independently:

```
User Speech
|
v
[Silero VAD] -- Voice Activity Detection (runs locally, ~10ms)
|
v
[Deepgram Nova-3 / Sarvam Saaras V3] -- Speech-to-Text (~200-400ms, streaming)
|
v
```

```
[Sentence Buffer] -- Accumulates tokens until sentence boundary
|
v
[GPT-4.1-mini / Sarvam-M] -- Language Model inference (~200-400ms, streaming)
|
v
[ElevenLabs Flash v2.5 API / Sarvam Bulbul v3] -- Text-to-Speech (~150-250ms)
|
v
[LiveKit Server] -- WebRTC audio delivery (~10-20ms)
|
v
User Hears Response
```

Total end-to-end latency: 600-1000ms (first audio token)

Component breakdown:

- **Silero VAD:** Local voice activity detection running on the agent worker. Detects speech onset and offset with ~10ms latency. Critical for barge-in handling.
- **Deepgram Nova-3:** Streaming STT for English, Hinglish, and global languages. Best accuracy for Indian English accents.
- **Sarvam Saaras V3:** STT for pure Indian regional languages (Hindi, Tamil, Telugu, Bengali, etc.). 22 languages, best accuracy for regional.
- **Sentence Buffer:** Accumulates STT tokens until natural sentence boundary. Prevents LLM from receiving fragments.
- **GPT-4.1-mini:** Primary LLM for most conversations. Streams response tokens for minimal latency.
- **Sarvam-M (105B):** FREE LLM for Indian regional context queries. Used as first-pass for simple Indian language interactions.
- **ElevenLabs Flash v2.5 API:** Ultra-low-latency TTS via direct API (not platform). 74 languages. Used for English and global.
- **Sarvam Bulbul v3:** TTS for Indian regional languages. 11 native Indian voices.
- **LiveKit Server:** Self-hosted WebRTC media server. Adaptive bitrate, echo cancellation, noise suppression.

When to use cascaded:

- When function calling / tool use is required (book a flight, generate a quote, look up a customer)
- When structured data extraction is needed (collect name, email, phone)
- When compliance logging requires full text transcripts at each stage
- When using Sarvam AI for regional Indian language STT (not available in S2S)

Pipeline B: Speech-to-Speech (S2S)

```
User Speech
  |
  v
[LiveKit Agent] -- Captures audio frames
  |
  v
[OpenAI Realtime API (gpt-4o-realtime-mini)] -- Speech in -> Speech out
  |
  v
[LiveKit Server] -- WebRTC audio delivery
  |
  v
User Hears Response
```

Total end-to-end latency: 200-400ms (first audio token)

The S2S pipeline skips discrete STT and TTS. The model receives raw audio and produces raw audio, preserving prosody, emotion, and conversational rhythm.

When to use S2S:

- Natural conversation flow where low latency is critical (greeting, small talk)
- Emotional or nuanced interactions where tone matters
- Simple queries that do not require tool calling
- Conversations in English where S2S quality is highest

When NOT to use S2S:

- Function calling is required
- Non-English Indian languages (quality drops significantly)
- Compliance requires discrete text transcripts
- Complex multi-step workflows

The Intelligent Router

```
class PipelineRouter:
    def select_pipeline(self, context):
        # S2S for natural conversation in English
        if context.language == "en" and not context.requires_tools:
            return Pipeline.SPEECH_TO_SPEECH

        # Cascaded for function calling
        if context.requires_tools:
            return Pipeline.CASCADED

        # Cascaded for Indian regional languages
        if context.language in INDIAN_REGIONAL_LANGUAGES:
            return Pipeline.CASCADED
```

```

# Cascaded for compliance-heavy tenants
if context.tenant.requires_full_transcript:
    return Pipeline.CASCADED

# Default to cascaded for safety
return Pipeline.CASCADED

```

Mid-call switching: The router can switch pipelines during a single call. A conversation might start in S2S mode for the greeting, switch to cascaded when the customer asks to book a package (requiring tool calls), and switch back to S2S for closing.

5.2 The Multi-Agent Architecture (LangGraph)

Detailed in Section 3.5 (Engine 6). Architecture uses LangGraph Supervisor/Worker pattern with up to 7 parallel workers per call.

5.3 The 4-Layer Memory System (Detailed Architecture)

Layer 1: Short-Term Memory (Redis)

```

Key Structure: session:{tenant_id}:{session_id}
TTL: 1800 seconds (30 minutes from last activity)

Stored Data:
- Current conversation state (greeting, qualification, quoting, closing)
- Active intent stack (what the customer is trying to accomplish)
- In-progress tool calls and their status
- Temporary data buffers (partially collected form data)
- Pipeline state (which pipeline is active, switch history)
- Current worker agent assignments

Performance:
- Read latency: <1ms (same VPC)
- Write latency: <1ms
- Memory per session: ~2-5KB typical

```

Layer 2: Episodic Memory (PostgreSQL)

```

CREATE TABLE customer_memory_episodic (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    tenant_id         UUID NOT NULL REFERENCES tenants(id),
    customer_id       UUID NOT NULL,
    session_id        UUID REFERENCES ai_call_sessions(id),
    channel           VARCHAR(20) NOT NULL,
    event_type        VARCHAR(50) NOT NULL,
    event_data        JSONB NOT NULL,
    sentiment_score    FLOAT,
    intent            VARCHAR(100),
    created_at        TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    metadata          JSONB DEFAULT '{}'

```

```
);

CREATE INDEX idx_episodic_customer_time ON customer_memory_episodic(
    tenant_id, customer_id, created_at DESC
);
CREATE INDEX idx_episodic_session ON customer_memory_episodic(session_id);
CREATE INDEX idx_episodic_intent ON customer_memory_episodic(
    tenant_id, intent
);
```

Layer 3: Semantic Memory (pgvector)

```
CREATE EXTENSION IF NOT EXISTS vector;

CREATE TABLE customer_memory_semantic (
    id            UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    tenant_id     UUID NOT NULL REFERENCES tenants(id),
    customer_id   UUID,
    content_type  VARCHAR(50) NOT NULL,
    content_text  TEXT NOT NULL,
    embedding     vector(1536) NOT NULL,
    source_id     UUID,
    created_at    TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    metadata     JSONB DEFAULT '{}'
);

CREATE INDEX idx_semantic_embedding ON customer_memory_semantic
    USING ivfflat (embedding vector_cosine_ops)
    WITH (lists = 100);
CREATE INDEX idx_semantic_tenant_type ON customer_memory_semantic(
    tenant_id, content_type
);
CREATE INDEX idx_semantic_customer ON customer_memory_semantic(
    tenant_id, customer_id
);
```

Layer 4: Procedural Memory (PostgreSQL)

```
CREATE TABLE agent_procedures (
    id            UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    tenant_id     UUID NOT NULL REFERENCES tenants(id),
    procedure_type VARCHAR(50) NOT NULL,
    procedure_name VARCHAR(200) NOT NULL,
    conditions     JSONB NOT NULL,
    actions        JSONB NOT NULL,
    priority       INT DEFAULT 0,
    is_active      BOOLEAN DEFAULT true,
    created_at     TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    updated_at     TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

CREATE INDEX idx_procedures_tenant ON agent_procedures(
```

```
tenant_id, procedure_type, is_active
);
```

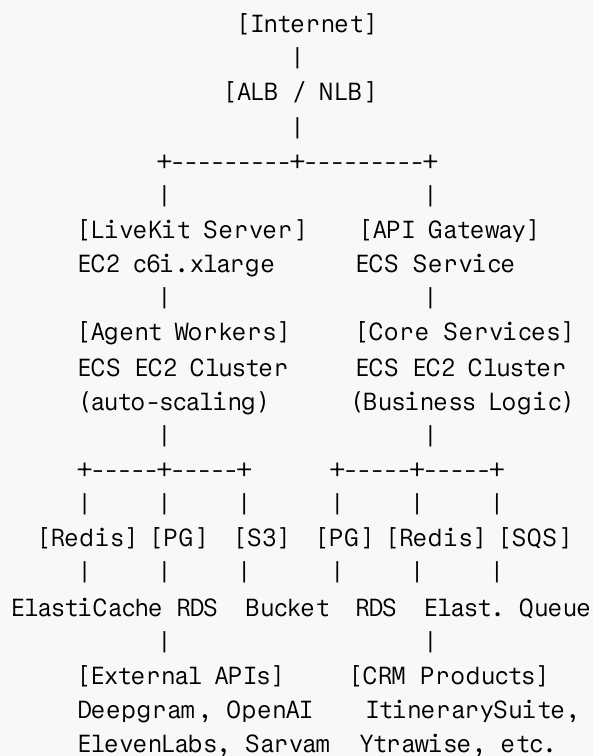
RAG Retrieval Architecture

Pre-call context loading in sub-200ms:

1. Incoming call detected (caller ID identified)
 - |
 - v
2. Parallel lookups (all execute simultaneously):
 - a. Redis: Check for active session (sub-1ms)
 - b. PostgreSQL: Load recent episodic memories, last 5 interactions (10-20ms)
 - c. pgvector: Semantic search for relevant context (50-100ms)
 - d. PostgreSQL: Load active procedures for this tenant (5-10ms)
 - |
 - v
3. Context assembly (10-20ms):
 - Merge episodic memories into chronological summary
 - Rank semantic results by relevance
 - Apply procedural rules to determine greeting and flow
 - |
 - v
4. Context injected into LLM system prompt (before first user utterance)

Total: 80-150ms (well within the ringing-to-answer window)

5.4 AWS Infrastructure (Self-Hosted)



Component Specifications

LiveKit Server: EC2 c6i.xlarge

- vCPUs: 4, Memory: 8 GB
- Cost: ~\$49/month (1-year reserved instance, US East)
- Purpose: WebRTC media server handling real-time audio routing
- Network: Enhanced networking enabled, placement group for lowest latency

Agent Workers: ECS EC2 Cluster (c6i.xlarge instances)

- Auto-scaling group: min 1, max 10 instances
- Cost: ~\$49/month per instance (1-year reserved)
- Purpose: Run LiveKit Agent processes
- Why NOT ECS Fargate: Voice calls are stateful. Fargate's aggressive task draining (30-second default) would terminate active calls. EC2-backed ECS allows custom drain periods (10+ minutes).
- Auto-scaling: Scale UP aggressively (1-minute cooldown), scale DOWN slowly (10-minute grace period)

Security Groups -- Critical Configuration:

LiveKit Server Security Group:

- Inbound TCP 443 (HTTPS/WSS from ALB)
- Inbound TCP 7880 (LiveKit API from Agent Workers)
- Inbound TCP 7881 (LiveKit RTC from Agent Workers)
- Inbound UDP 50000-60000 (WebRTC media -- MUST be open)
- Outbound: All traffic

Agent Worker Security Group:

- Inbound TCP 8080 (health check from ALB)
- Outbound TCP 443 (to Deepgram, OpenAI, ElevenLabs, Sarvam APIs)
- Outbound TCP 5432 (to RDS PostgreSQL)
- Outbound TCP 6379 (to ElastiCache Redis)
- Outbound TCP 7880-7881 (to LiveKit Server)

Important: UDP ports 50000-60000 MUST be open for WebRTC media traffic. Without these ports, voice calls will connect (signaling works over TCP) but no audio will flow. This is the most common deployment issue.

Additional AWS Services

Service	Purpose	Specification	Monthly Cost
RDS PostgreSQL	Episodic + procedural memory	db.r6g.large, Multi-AZ	~\$180
RDS PostgreSQL (pgvector)	Semantic memory	db.r6g.large, pgvector extension	~\$180

Service	Purpose	Specification	Monthly Cost
ElastiCache Redis	Short-term memory	cache.r6g.large, cluster mode	~\$130
S3	Call recordings, documents, quotes	Standard tier	~\$10-50
SQS	Async job queuing	Standard queues	~\$5
CloudWatch	Monitoring, logging, alerting	Standard metrics + custom	~\$30
ALB	Load balancing, SSL termination	Application Load Balancer	~\$25
Route 53	DNS management	Hosted zone + queries	~\$5

Total AWS infrastructure cost: ~\$660-\$750/month at base scale

5.5 Language Architecture (Updated with Sarvam + ElevenLabs API Routing)

STT (Speech-to-Text) Routing

Language	Primary STT	Fallback STT	Notes
English	Deepgram Nova-3	OpenAI Whisper	Best-in-class English accuracy
Hinglish (Hindi-English mix)	Deepgram Nova-3	Sarvam Saaras V3	Nova-3 handles code-switching
Hindi	Sarvam Saaras V3	Deepgram Nova-3	Sarvam best for pure Hindi
Tamil	Sarvam Saaras V3	Deepgram Nova-3	Sarvam trained on Indian languages
Telugu	Sarvam Saaras V3	Deepgram Nova-3	Sarvam trained on Indian languages
Kannada	Sarvam Saaras V3	Deepgram Nova-3	Sarvam trained on Indian languages
Malayalam	Sarvam Saaras V3	Deepgram Nova-3	Sarvam trained on Indian languages
Bengali	Sarvam Saaras V3	Deepgram Nova-3	Sarvam trained on Indian languages
Marathi	Sarvam Saaras V3	Deepgram Nova-3	Sarvam trained on Indian languages
Gujarati	Sarvam Saaras V3	Deepgram Nova-3	Sarvam trained on Indian languages
Punjabi	Sarvam Saaras V3	Deepgram Nova-3	Sarvam trained on Indian languages
Odia	Sarvam Saaras V3	Deepgram Nova-3	Sarvam trained on Indian languages
Assamese	Sarvam Saaras V3	Deepgram Nova-3	Sarvam trained on Indian languages

Language	Primary STT	Fallback STT	Notes
Urdu	Sarvam Saaras V3	Deepgram Nova-3	Sarvam trained on Indian languages
Global (Arabic, Chinese, etc.)	Deepgram Nova-3 multi	ElevenLabs Scribe v2	Best global coverage

TTS Routing (Updated V2)

Language	Primary TTS	Cost/min	Notes
English	ElevenLabs Flash v2.5 API	\$0.045	74 languages, 75ms latency
Hindi	ElevenLabs Flash or Sarvam Bulbul v3	\$0.045 or \$0.027	Choice based on quality vs cost
Tamil	Sarvam Bulbul v3	\$0.027	Native Tamil voice
Telugu	Sarvam Bulbul v3	\$0.027	Native Telugu voice
Bengali	Sarvam Bulbul v3	\$0.027	Native Bengali voice
Marathi	Sarvam Bulbul v3	\$0.027	Native Marathi voice
Kannada	Sarvam Bulbul v3	\$0.027	Native Kannada voice
Malayalam	Sarvam Bulbul v3	\$0.027	Native voice
Global	ElevenLabs Flash v2.5 API	\$0.045	74 languages

Turn Detection

LiveKit's `MultilingualModel()` provides intelligent turn detection:

- Detects end-of-utterance based on prosodic cues, not just silence
- Handles code-switching (Hindi and English mid-sentence)
- Adapts to conversational pace
- Prevents false triggers from background noise

5.6 The Database Schema (Full SQL)

Tenants and Configuration

```
CREATE TABLE tenants (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  name        VARCHAR(200) NOT NULL,
  slug        VARCHAR(100) UNIQUE NOT NULL,
  plan_type   VARCHAR(50) NOT NULL DEFAULT 'starter',
  status      VARCHAR(20) NOT NULL DEFAULT 'active',
  owner_email VARCHAR(255) NOT NULL,
  company_name VARCHAR(200),
  industry    VARCHAR(100),
  max_concurrent_calls INT DEFAULT 5,
  monthly_minute_limit INT DEFAULT 5000,
```

```

        billing_email      VARCHAR(255),
        metadata           JSONB DEFAULT '{}',
        created_at         TIMESTAMPTZ NOT NULL DEFAULT NOW(),
        updated_at         TIMESTAMPTZ NOT NULL DEFAULT NOW()
    );

CREATE TABLE tenant_voice_config (
    id                     UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    tenant_id              UUID NOT NULL REFERENCES tenants(id) ON DELETE CASCADE,
    default_pipeline       VARCHAR(20) DEFAULT 'cascaded',
    allow_s2s              BOOLEAN DEFAULT false,
    primary_stt            VARCHAR(50) DEFAULT 'deepgram_nova3',
    fallback_stt           VARCHAR(50) DEFAULT 'sarvam_saaras_v3',
    primary_llm            VARCHAR(50) DEFAULT 'gpt-4.1-mini',
    llm_temperature        FLOAT DEFAULT 0.7,
    llm_max_tokens         INT DEFAULT 500,
    system_prompt          TEXT,
    tts_provider           VARCHAR(50) DEFAULT 'elevenlabs_flash',
    tts_voice_id           VARCHAR(100),
    tts_speed              FLOAT DEFAULT 1.0,
    primary_language       VARCHAR(10) DEFAULT 'en',
    supported_languages    VARCHAR(10)[] DEFAULT ARRAY['en', 'hi'],
    auto_detect_language   BOOLEAN DEFAULT true,
    max_call_duration      INT DEFAULT 600,
    silence_timeout        INT DEFAULT 30,
    escalation_threshold   FLOAT DEFAULT 0.3,
    record_calls           BOOLEAN DEFAULT true,
    recording_retention_days INT DEFAULT 90,
    created_at            TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    updated_at            TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    UNIQUE(tenant_id)
);

```

AI Call Sessions

```

CREATE TABLE ai_call_sessions (
    id                     UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    tenant_id              UUID NOT NULL REFERENCES tenants(id),
    customer_id            UUID,
    session_token          VARCHAR(100) UNIQUE NOT NULL,
    livekit_room_id        VARCHAR(100),
    direction              VARCHAR(10) NOT NULL,
    channel                VARCHAR(20) NOT NULL,
    caller_phone           VARCHAR(20),
    caller_name            VARCHAR(200),
    pipeline_used          VARCHAR(20),
    pipeline_switches      INT DEFAULT 0,
    started_at            TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    answered_at           TIMESTAMPTZ,
    ended_at              TIMESTAMPTZ,
    duration_seconds       INT,
    first_response_ms      INT,
    avg_response_ms        INT,
    total_llm_tokens       INT DEFAULT 0,
    total_stt_seconds      FLOAT DEFAULT 0,

```

```

total_tts_chars      INT DEFAULT 0,
disposition          VARCHAR(50),
sentiment_score      FLOAT,
resolution_status    VARCHAR(50),
cost_stt             DECIMAL(10,6) DEFAULT 0,
cost_llm             DECIMAL(10,6) DEFAULT 0,
cost_tts             DECIMAL(10,6) DEFAULT 0,
cost_infra           DECIMAL(10,6) DEFAULT 0,
cost_total           DECIMAL(10,6) DEFAULT 0,
recording_url        VARCHAR(500),
transcript_url       VARCHAR(500),
tools_invoked        VARCHAR(100)[] DEFAULT '{}',
agents_used          VARCHAR(100)[] DEFAULT '{}',
escalation_reason    TEXT,
metadata             JSONB DEFAULT '{}',
created_at           TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

CREATE INDEX idx_sessions_tenant ON ai_call_sessions(tenant_id, started_at DESC);
CREATE INDEX idx_sessions_customer ON ai_call_sessions(tenant_id, customer_id, started_at DESC);
CREATE INDEX idx_sessions_disposition ON ai_call_sessions(tenant_id, disposition);

```

Audit and Compliance

```

CREATE TABLE agent_audit_log (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    tenant_id         UUID NOT NULL REFERENCES tenants(id),
    session_id        UUID REFERENCES ai_call_sessions(id),
    agent_name        VARCHAR(100) NOT NULL,
    action_type       VARCHAR(50) NOT NULL,
    action_detail     JSONB NOT NULL,
    input_data        JSONB,
    output_data       JSONB,
    latency_ms        INT,
    success           BOOLEAN NOT NULL,
    error_message     TEXT,
    created_at        TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

-- Append-only enforcement
CREATE OR REPLACE FUNCTION prevent_audit_modification()
RETURNS TRIGGER AS $$
BEGIN
    RAISE EXCEPTION 'Audit log records cannot be modified or deleted';
    RETURN NULL;
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER audit_log_immutable
BEFORE UPDATE OR DELETE ON agent_audit_log
FOR EACH ROW EXECUTE FUNCTION prevent_audit_modification();

CREATE INDEX idx_audit_tenant_time ON agent_audit_log(tenant_id, created_at DESC);
CREATE INDEX idx_audit_session ON agent_audit_log(session_id);

```

Proactive Messages

```
CREATE TABLE proactive_messages (  
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  tenant_id   UUID NOT NULL REFERENCES tenants(id),  
  customer_id UUID NOT NULL,  
  message_type VARCHAR(50) NOT NULL,  
  channel     VARCHAR(20) NOT NULL,  
  priority    INT DEFAULT 5,  
  subject     VARCHAR(200),  
  content_template VARCHAR(100),  
  content_data JSONB NOT NULL,  
  scheduled_at TIMESTAMPTZ NOT NULL,  
  timezone    VARCHAR(50) DEFAULT 'Asia/Kolkata',  
  status      VARCHAR(20) DEFAULT 'pending',  
  sent_at     TIMESTAMPTZ,  
  delivered_at TIMESTAMPTZ,  
  response_data JSONB,  
  consent_verified BOOLEAN DEFAULT false,  
  dnd_checked  BOOLEAN DEFAULT false,  
  retry_count  INT DEFAULT 0,  
  max_retries  INT DEFAULT 3,  
  created_at   TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
  updated_at   TIMESTAMPTZ NOT NULL DEFAULT NOW()  
);  
  
CREATE INDEX idx_proactive_schedule ON proactive_messages(  
  status, scheduled_at  
) WHERE status IN ('pending', 'queued');
```

Booking Documents

```
CREATE TABLE booking_documents (  
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  tenant_id   UUID NOT NULL REFERENCES tenants(id),  
  customer_id UUID NOT NULL,  
  booking_id  UUID,  
  document_type VARCHAR(50) NOT NULL,  
  file_name   VARCHAR(255) NOT NULL,  
  file_size   INT NOT NULL,  
  mime_type   VARCHAR(100) NOT NULL,  
  s3_key      VARCHAR(500) NOT NULL,  
  ocr_status  VARCHAR(20) DEFAULT 'pending',  
  ocr_data    JSONB,  
  validation_status VARCHAR(20) DEFAULT 'pending',  
  validation_notes TEXT,  
  encrypted   BOOLEAN DEFAULT true,  
  retention_until TIMESTAMPTZ,  
  uploaded_via VARCHAR(20),  
  created_at   TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
  updated_at   TIMESTAMPTZ NOT NULL DEFAULT NOW()  
);
```

Master Interaction Timeline

```
CREATE TABLE interactions (  
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  tenant_id   UUID NOT NULL REFERENCES tenants(id),  
  customer_id UUID NOT NULL,  
  channel     VARCHAR(20) NOT NULL,  
  direction   VARCHAR(10) NOT NULL,  
  session_id  UUID,  
  interaction_type VARCHAR(50) NOT NULL,  
  related_object_type VARCHAR(50),  
  related_object_id  UUID,  
  summary       TEXT,  
  detail        JSONB,  
  intent        VARCHAR(100),  
  sentiment_score FLOAT,  
  outcome       VARCHAR(50),  
  handled_by    VARCHAR(20) NOT NULL,  
  agent_id      UUID,  
  started_at    TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
  ended_at      TIMESTAMPTZ,  
  duration_seconds INT,  
  metadata      JSONB DEFAULT '{}',  
  created_at    TIMESTAMPTZ NOT NULL DEFAULT NOW()  
);
```

```
CREATE INDEX idx_interactions_customer ON interactions(tenant_id, customer_id, created_at);  
CREATE INDEX idx_interactions_object ON interactions(tenant_id, related_object_type, related_object_id);
```

Consent Records

```
CREATE TABLE consent_records (  
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  tenant_id   UUID NOT NULL REFERENCES tenants(id),  
  customer_id UUID NOT NULL,  
  consent_type VARCHAR(50) NOT NULL,  
  consent_given BOOLEAN NOT NULL,  
  consent_method VARCHAR(50) NOT NULL,  
  consent_text TEXT NOT NULL,  
  ip_address   VARCHAR(45),  
  collected_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
  expires_at   TIMESTAMPTZ,  
  withdrawn_at TIMESTAMPTZ,  
  metadata     JSONB DEFAULT '{}'  
);
```

Section 6: STT/TTS/LLM Stack (V2 Updated)

NOTE: The STT/TTS/LLM stack described in this section serves ALL industries with the same language routing regardless of vertical. A Hindi-speaking insurance customer gets the same Sarvam STT/TTS quality as a Hindi-speaking travel customer. Language selection is based on the customer's language, not the industry.

6.1 The Decision: ElevenLabs API + Sarvam API (Not Platforms)

V1 recommended ElevenLabs Conversational AI Platform (\$0.08-\$0.10/min). V2 switches to direct API access for both ElevenLabs TTS and Sarvam STT/TTS/LLM.

Why direct API beats managed platforms:

- More control over each component (can swap providers per-language)
- Lower cost (no platform markup)
- Best Indian language accuracy (Sarvam is purpose-built for India)
- Sarvam-M LLM is FREE for Indian language processing
- Can mix and match: Sarvam for Indian, ElevenLabs for global, Deepgram for English STT

6.2 Complete Component Selection Table

COMPONENT	PROVIDER	REASON	COST
Media Server	LiveKit (self-hosted)	Open source, AWS EC2	\$0.002/min
VAD	Silero (local)	Free, no API call	\$0.000
STT English	Deepgram Nova-3	Best accents, latency	\$0.0077/min
STT Indian	Sarvam Saaras V3	22 Indian langs, cheapest	\$0.006/min
STT Global	Deepgram Nova-3 multi	Best global coverage	\$0.0092/min
STT Realtime	ElevenLabs Scribe v2	For global + high quality	\$0.008/min
LLM Primary	GPT-4.1-mini	Best cost/quality	\$0.006/min
LLM Indian	Sarvam-M (105B)	FREE for Indian context	\$0.000/min
LLM S2S	OpenAI Realtime mini	Sub-300ms conversation	\$0.070/min
TTS English/Global	ElevenLabs Flash API	74 languages, 75ms	\$0.045/min
TTS Indian	Sarvam Bulbul v3	22 Indian, native voices	\$0.027/min
Translation	Sarvam Translate	INR-priced, accurate	INR 20/10K chars
Language ID	Sarvam LangID	Real-time detection	INR 3.5/10K chars
Telephony India	Exotel	India's largest	\$0.005/min
Telephony Global	Twilio	Standard	\$0.0085/min
Chat LLM	GPT-4.1-mini	Multilingual natively	\$0.001/exchange
WhatsApp	Meta Cloud API	Native, via AiSensy	\$0.011/msg
OCR Documents	AZAPI.ai	99.94% accuracy	By request
Visa Check	VisaHQ/Sherpa	Accurate, global	By request
Flight Monitor	Aviation Stack/FA	Real-time disruption	By request
Payment India	Razorpay	UPI + Cards + Netbanking	2% per txn
Payment Global	Stripe	International cards	2.9% + \$0.30
E-Signature	DocuSign/Zoho Sign	ITAC 2000 compliant	Per doc
Hosting	AWS EC2 + ECS EC2	Same VPC, cost-effective	\$147/mo fixed

6.3 Intelligent Language Routing

CALL LANGUAGE DETECTED -> ROUTE TO:

ENGLISH / HINGLISH:

STT: Deepgram Nova-3 Multilingual (\$0.0077/min)

LLM: GPT-4.1-mini (\$0.006/min)

TTS: ElevenLabs Flash API (\$0.045/min via API)

TOTAL: \$0.059/min

PURE HINDI / HINGLISH HEAVY:

STT: Sarvam Saaras V3 (\$0.006/min) -- better accuracy

LLM: Sarvam-M (FREE) for Indian context queries

GPT-4.1-mini for complex/tool calls

TTS: Sarvam Bulbul v3 (\$0.027/min) -- native Hindi voice

OR ElevenLabs Flash (\$0.045/min) for premium quality

TOTAL with Sarvam TTS: \$0.039/min (INR 3.2/min) -- LOWEST COST

TOTAL with ElevenLabs TTS: \$0.057/min -- better voice quality

PURE REGIONAL (Tamil/Telugu/Bengali/Marathi/Kannada/Malayalam):

STT: Sarvam Saaras V3 (\$0.006/min) -- best accuracy

LLM: Sarvam-M (FREE) for regional context

TTS: Sarvam Bulbul v3 (\$0.027/min) -- only option with native voices

TOTAL: \$0.039/min (INR 3.2/min)

GLOBAL (Arabic/Chinese/Japanese/Thai/etc.):

STT: Deepgram Nova-3 Multilingual (\$0.0077/min)

LLM: GPT-4.1-mini (\$0.006/min)

TTS: ElevenLabs Flash API (\$0.045/min) -- 74 languages

TOTAL: \$0.059/min

SPEECH-TO-SPEECH (fast conversational moments):

OpenAI Realtime mini: \$0.070/min (all-in-one)

LiveKit: \$0.002/min

Exotel: \$0.005/min

TOTAL: \$0.077/min

6.4 Sarvam AI Complete Reference

Sarvam AI (India's Sovereign AI -- YC F24, IIT Delhi spin-off)

Backed by: Government of India + VC funding

Website: sarvam.ai

KEY PRODUCTS FOR AGENTCORE:

1. Saaras V3 (Speech-to-Text)

- 22 Indian languages
- REST API (files < 30 sec) + Batch API + Streaming WebSocket
- Accuracy: Best for Indian accents (beats Deepgram for regional)
- Latency: ~400-600ms TTFB (vs Deepgram's ~300ms)
- Price: INR 30/hour = \$0.006/min (CHEAPEST for Indian STT)
- With Diarization: INR 45/hour = \$0.009/min
- STT + Translate (to English): INR 30/hour = \$0.006/min (SAME price!)

2. Bulbul v3 (Text-to-Speech)

- 11 Indian languages with native voices
- REST streaming API
- Price: INR 30/10K characters (v3) = ~\$0.027/min
- Quality: Very good for regional, slightly below ElevenLabs for English

3. Sarvam-M / 105B (LLM)

- FREE per token (open beta as of March 2026)
- 22 Indian language understanding + response
- Good for Indian context queries, cultural understanding
- Use as: first-pass for Indian language intent, GPT-4.1-mini for complex

4. Sarvam Translate

- INR 20/10K chars bidirectional
- All 22 Indian languages <-> English
- Use for: translating chat messages before LLM processing

5. Language Identification

- INR 3.5/10K chars
- Identifies which Indian language from text/audio
- Near-zero cost for routing decisions

GETTING STARTED:

sarvam.ai/api-pricing

docs.sarvam.ai

Free INR 1,000 credit on signup

Pro plan: INR 10,000 + INR 1,000 bonus = INR 11,000 credits

Business: INR 50,000 + INR 7,500 bonus = INR 57,500 credits

6.5 ElevenLabs API Direct Pricing

ElevenLabs TTS API (Flash/Turbo model -- for voice):

Tier	Characters Included	Price per 1K chars

Entry	20,000/month	\$0.08/1K (add'l)
Starter	60,000/month	\$0.11/1K (add'l)
Growth	200,000/month	\$0.10/1K (add'l) <-- best for scale
Scale	1,000,000/month	\$0.08/1K (add'l)
Business	4,000,000/month	\$0.06/1K (add'l)

COST PER VOICE MINUTE (TTS API only):

~750 chars/min of speech

At Growth tier: $750 \times \$0.00010 = \$0.075/1K = \$0.000075$ per char

Per minute: $750 \times \$0.000075 = \$0.056/\text{min}$ (TTS only)

Flash/Turbo model at volume: \$0.06/1K chars

Per minute: $750 \times \$0.00006 = \$0.045/\text{min}$ -- cheaper than platform

ElevenLabs Scribe v2 Realtime (STT API):

Price: \$0.48/hour = \$0.008/min realtime transcription

90+ languages including all Indian languages

150ms latency, word-level timestamps

VERDICT: Use Sarvam for Indian, Deepgram for English.

ElevenLabs Scribe for global languages where Deepgram lacks coverage.

6.6 Multilingual Chat Agent (Updated)

Language Detection Flow

CUSTOMER MESSAGE -> Language Detection -> Route

Detection methods (in priority order):

1. Sarvam Language ID API (INR 3.5/10K chars) -- best for Indian scripts
2. langdetect Python library (FREE) -- good for Latin/CJK scripts
3. GPT-4.1-mini inline detection -- fallback for ambiguous cases

Full TypeScript Implementation

```
// packages/agentcore-chat/src/multilingual.ts

class MultilingualChatEngine {

  async detectLanguage(text: string): Promise<LanguageCode> {
    // Method 1: Sarvam Language ID (most accurate for Indian languages)
    if (this.looksIndian(text)) {
      return await sarvam.identifyLanguage(text);
    }
    // Method 2: Langdetect for global languages (free)
    return langdetect.detect(text);
  }

  async processMessage(
    message: string,
    sessionId: string,
    tenantConfig: TenantConfig
  ): Promise<ChatResponse> {
    const lang = await this.detectLanguage(message);

    const systemPrompt = buildMultilingualPrompt(tenantConfig, lang, {
      instruction: `Always respond in ${LANGUAGE_NAMES[lang]}.`,
      Current conversation language: ${lang}.
      Customer context: ${await memory.recall(sessionId)}`,
    });

    let processMessage = message;
    let responseTranslation = false;

    if (PURE_REGIONAL_LANGS.includes(lang) && tenantConfig.useTranslation) {
      processMessage = await sarvam.translate(message, lang, 'en');
      responseTranslation = true;
    }

    const response = await openai.chat({
      model: 'gpt-4.1-mini',
      messages: buildConversation(systemPrompt, processMessage, sessionId),
    });

    let responseText = response.choices[0].message.content;
```

```

    if (responseTranslation) {
      responseText = await sarvam.translate(responseText, 'en', lang);
    }

    if (tenantConfig.voiceChatEnabled) {
      const audio = await this.generateVoiceResponse(responseText, lang);
      return { text: responseText, audio, language: lang };
    }

    return { text: responseText, language: lang };
  }

  async generateVoiceResponse(text: string, lang: LanguageCode): Promise<Buffer> {
    if (INDIAN_LANGUAGES.includes(lang)) {
      return sarvam.tts(text, lang); // Bulbul v3 native Indian voice
    }
    return elevenlabs.tts(text, { model: 'eleven_flash_v2_5' }); // 74 languages
  }
}

```

Supported Chat Languages Table

CODE	LANGUAGE	STT PROVIDER	TTS PROVIDER	LLM
----	-----	-----	-----	---
hi	Hindi	Sarvam	Sarvam or ElevenLabs	GPT-4.1-mini + Sarvam-M
ta	Tamil	Sarvam	Sarvam	GPT-4.1-mini
te	Telugu	Sarvam	Sarvam	GPT-4.1-mini
bn	Bengali	Sarvam	Sarvam	GPT-4.1-mini
mr	Marathi	Sarvam	Sarvam	GPT-4.1-mini
gu	Gujarati	Sarvam	Sarvam	GPT-4.1-mini
kn	Kannada	Sarvam	Sarvam	GPT-4.1-mini
ml	Malayalam	Sarvam	Sarvam	GPT-4.1-mini
pa	Punjabi	Sarvam	Sarvam	GPT-4.1-mini
or	Odia	Sarvam	Sarvam	GPT-4.1-mini
ur	Urdu	Sarvam	Sarvam	GPT-4.1-mini
en	English	Deepgram	ElevenLabs	GPT-4.1-mini
ar	Arabic	Deepgram	ElevenLabs	GPT-4.1-mini
zh	Chinese	Deepgram	ElevenLabs	GPT-4.1-mini
ja	Japanese	Deepgram	ElevenLabs	GPT-4.1-mini
ko	Korean	Deepgram	ElevenLabs	GPT-4.1-mini
th	Thai	Deepgram	ElevenLabs	GPT-4.1-mini
id	Bahasa Indonesia	Deepgram	ElevenLabs	GPT-4.1-mini
fr	French	Deepgram	ElevenLabs	GPT-4.1-mini
de	German	Deepgram	ElevenLabs	GPT-4.1-mini
es	Spanish	Deepgram	ElevenLabs	GPT-4.1-mini
pt	Portuguese	Deepgram	ElevenLabs	GPT-4.1-mini
it	Italian	Deepgram	ElevenLabs	GPT-4.1-mini
ru	Russian	Deepgram	ElevenLabs	GPT-4.1-mini
tr	Turkish	Deepgram	ElevenLabs	GPT-4.1-mini
vi	Vietnamese	Deepgram	ElevenLabs	GPT-4.1-mini
pl	Polish	Deepgram	ElevenLabs	GPT-4.1-mini
nl	Dutch	Deepgram	ElevenLabs	GPT-4.1-mini

Section 7: Vertical Deployments -- All Industries

This section describes deployment configurations for all HelpRevX products (7.1-7.6) and 10 additional industry verticals (7.7-7.16) that demonstrate AgentCore's horizontal platform capability.

For each industry deployment, we describe:

- a) Specific pain points this solves
- b) Custom AgentCore tools registered
- c) Sample qualification questions the AI asks
- d) WhatsApp flow design
- e) Sample call script (how the AI sounds)
- f) Expected call volume and cost estimate
- g) Key metrics (what success looks like)

7.1 ItinerarySuite -- Travel CRM

Pain Points Solved

- Travel agents spend 60% of their time on repetitive queries (visa requirements, package pricing, availability)
- Lead response time averages 4-6 hours -- by which time the customer has contacted 3 other agencies
- Quote generation takes 30-60 minutes manually; customers expect instant responses
- Document collection for visa applications is a multi-week email chain
- [NEW] Proposals are created manually in Word/Canva -- no tracking, no interaction

Custom Tools Registered

```
@tool("search_travel_packages")
def search_travel_packages(
    destination: str, travel_dates: str, num_travelers: int,
    budget_range: str, travel_type: str
) -> list[Package]:
    """Search GDS and internal inventory for matching travel packages."""

@tool("check_visa_requirement")
def check_visa_requirement(
    nationality: str, destination_country: str, passport_expiry: str
) -> VisaRequirement:
    """Check visa requirements including type, processing time, documents, fees."""

@tool("check_flight_availability")
def check_flight_availability(
```

```
origin: str, destination: str, departure_date: str,  
return_date: str, passengers: int, cabin_class: str  
) -> list[FlightOption]:  
    """Search GDS for available flights with real-time pricing."""
```

Custom Qualification Questions

1. "Where are you planning to travel?" (destination)
2. "When are you looking to travel?" (dates)
3. "How many travelers including children?" (group size)
4. "What's your approximate budget per person?" (budget)
5. "Is this for a special occasion?" (travel type / upsell opportunity)

WhatsApp Flow Design

```
[Welcome Message with Carousel of Popular Destinations]  
|  
v  
[Package Selection -> Quick Reply Buttons]  
|  
v  
[WhatsApp Flow: Travel Details Form]  
- Destination, Dates, Travelers, Budget  
|  
v  
[AI generates proposal -> Shareable link via WhatsApp]  
|  
v  
[Customer reviews proposal -> Approve/Modify]  
|  
v  
[Quote generated from approved proposal -> PDF in WhatsApp]  
|  
v  
[Quick Reply: "Book Now" / "Modify" / "Call Me"]  
|  
v  
[WhatsApp Pay for deposit / Razorpay link for full payment]  
|  
v  
[Booking confirmation + Document collection flow]
```

Expected Volume and Cost

- Monthly call volume: 5,000-15,000 minutes
- Average call duration: 4-6 minutes
- Cost per call: INR 8-12 (\$0.10-0.15)
- Lead conversion rate target: 15-25% (vs. industry 5-8%)

7.2 Ytrawise -- Travel Public Portal

Pain Points Solved

- Public travel portals have high bounce rates (70%+)
- No personalized assistance -- static, one-size-fits-all
- Customer intent is lost when they leave
- Booking abandonment at checkout due to unanswered questions

Custom Tools Registered

```
@tool("search_portal_packages")
def search_portal_packages(
    destination: str, theme: str, budget_per_person: int, duration_days: int
) -> list[PortalPackage]:
    """Search public portal inventory for matching packages."""

@tool("get_destination_content")
def get_destination_content(
    destination: str, content_type: str
) -> DestinationContent:
    """Retrieve rich destination content for conversational delivery."""

@tool("book_package_from_chat")
def book_package_from_chat(
    package_id: str, customer_name: str, customer_email: str,
    customer_phone: str, travel_date: str, num_travelers: int
) -> BookingInitiation:
    """Initiate booking from chat/voice conversation."""
```

Expected Volume and Cost

- Monthly call volume: 3,000-10,000 minutes
- Average interaction: 3-4 minutes
- Cost per interaction: INR 6-10 (\$0.07-0.12)
- Booking conversion from voice/chat: 8-15%

7.3 Aayiq -- Expense Management

Pain Points Solved

- Expense reporting is tedious: employees delay submissions
- Policy compliance is checked manually
- Receipt capture requires opening an app -- friction kills compliance
- Approval workflows are slow: managers forget, emails get buried

Custom Tools Registered

```
@tool("capture_expense")
def capture_expense(
    amount: float, currency: str, category: str,
    description: str, date: str, receipt_available: bool
) -> ExpenseRecord:
    """Create expense entry from voice/chat input."""

@tool("check_policy_compliance")
def check_policy_compliance(
    expense_category: str, amount: float,
    employee_level: str, trip_type: str
) -> PolicyCheck:
    """Validate expense against company policy limits."""

@tool("initiate_approval_workflow")
def initiate_approval_workflow(
    expense_report_id: str, total_amount: float,
    num_items: int, requires_special_approval: bool
) -> ApprovalStatus:
    """Submit expense report for approval."""
```

Expected Volume and Cost

- Monthly voice minutes: 2,000-5,000
- Average interaction: 2-3 minutes
- Cost per expense capture: INR 3-5 (\$0.04-0.06)

7.4 SpendPilot -- Spend Analytics

Pain Points Solved

- Finance teams spend hours generating reports
- Budget anomalies detected too late
- Executives want real-time visibility but will not log into dashboards

Custom Tools Registered

```
@tool("get_spend_summary")
def get_spend_summary(
    department: str, time_period: str, category: str, comparison: str
) -> SpendSummary:
    """Retrieve spend summary with comparisons and trend data."""

@tool("flag_anomaly")
def flag_anomaly(
    anomaly_type: str, department: str, details: str, severity: str
) -> AnomalyFlag:
    """Flag a spending anomaly for investigation."""
```

```
@tool("generate_budget_alert")
def generate_budget_alert(
    department: str, budget_category: str,
    current_spend: float, budget_limit: float, projected_end_of_month: float
) -> BudgetAlert:
    """Generate budget alert when spend approaches limits."""
```

Expected Volume and Cost

- Monthly voice minutes: 1,000-3,000
- Average interaction: 3-5 minutes
- Cost per query session: INR 5-8 (\$0.06-0.10)

7.5 PhotoAdda -- Photography CRM

Pain Points Solved

- Studios miss inquiries because they are on shoots
- Quoting is inconsistent across staff
- Trial session booking requires multiple back-and-forth messages
- Studio availability checked manually, leading to double bookings

Custom Tools Registered

```
@tool("check_studio_availability")
def check_studio_availability(
    studio_id: str, date: str, time_slot: str, session_type: str
) -> StudioAvailability:
    """Check real-time studio availability."""

@tool("get_package_quote")
def get_package_quote(
    session_type: str, duration_hours: int, location: str, deliverables: list
) -> PackageQuote:
    """Generate photography package quote with itemized pricing."""

@tool("book_trial_session")
def book_trial_session(
    customer_name: str, phone: str, session_type: str,
    preferred_date: str, preferred_time: str, special_requirements: str
) -> TrialBooking:
    """Book a trial/mini photography session."""
```

Expected Volume and Cost

- Monthly voice minutes:1,000-3,000
- Average call:3-4 minutes
- Cost per interaction:INR 5-7 (\$0.06-0.08)
- Trial session conversion rate target:40-60%

7.6 HelpRevX -- Internal Platform Operations

Pain Points Solved

- Tenant support requests pile up
- Supplier issues require manual status checking
- Platform metrics scattered across tools
- Escalation routing is ad-hoc

Custom Tools Registered

```
@tool("check_supplier_status")
def check_supplier_status(supplier_id: str, check_type: str) -> SupplierStatus:
    """Check supplier integration health and business status."""

@tool("escalate_tenant_issue")
def escalate_tenant_issue(
    tenant_id: str, issue_type: str, severity: str,
    description: str, affected_users: int
) -> EscalationTicket:
    """Create and route tenant escalation."""

@tool("get_platform_metrics")
def get_platform_metrics(metric_type: str, time_range: str) -> PlatformMetrics:
    """Retrieve platform operational metrics."""
```

Expected Volume and Cost

- Monthly voice minutes:500-1,000
- Average interaction:2-3 minutes
- Cost per interaction:INR 3-5 (\$0.04-0.06)

7.7 Financial Services (BFSI)

Sub-Verticals

- Insurance companies (life, general, health)
- Mutual fund distributors and AMCs
- NBFCs and lending platforms
- Wealth management and advisory firms
- Banks (retail banking customer service)

Specific Pain Points This Solves

- KYC verification calls automated end-to-end
- Policy renewal reminders made proactively (currently done by less than 20% of agents)
- Loan application status queries (30-40% of all inbound) handled instantly by AI
- Insurance claim status updates without agent lookup in 3 systems
- Mutual fund SIP queries answered with accurate, compliant product knowledge
- Compliance risk eliminated: AI never gives unauthorized financial advice
- After-hours card blocking and fraud reporting
- EMI payment reminders with automated payment link delivery

Custom AgentCore Tools Registered

```
@tool("check_policy_status")
def check_policy_status(
    policy_number: str, customer_id: str
) -> PolicyStatus:
    """Check insurance policy status, premium due, coverage details."""

@tool("get_premium_due")
def get_premium_due(
    policy_number: str
) -> PremiumDetails:
    """Get premium amount, due date, payment modes, grace period."""

@tool("initiate_claim")
def initiate_claim(
    policy_number: str, claim_type: str, incident_date: str,
    description: str, estimated_amount: float
) -> ClaimInitiation:
    """Initiate an insurance claim with basic details."""

@tool("check_loan_status")
def check_loan_status(
    application_id: str, customer_phone: str
) -> LoanStatus:
    """Check loan application status, pending documents, next steps."""

@tool("get_portfolio_summary")
def get_portfolio_summary(
```

```
client_id: str, as_of_date: str
) -> PortfolioSummary:
    """Get investment portfolio summary with returns and allocation."""
```

Sample Qualification Questions

1. "What product are you interested in -- insurance, loan, or investment?" (product type)
2. "Can you share your policy number or application ID?" (identification)
3. "What specific help do you need today?" (intent classification)
4. "Are you an existing customer or new to [Company Name]?" (customer type)
5. "What is your preferred language for this conversation?" (language routing)

WhatsApp Flow Design

```
[Welcome: "Hi! I'm [Company]'s virtual assistant. How can I help?"]
|
v
[Quick Replies: "Policy Status" / "Claim" / "Renewal" / "New Product" / "Other"]
|
v
[Policy Status Flow]:
-> Enter policy number
-> AI verifies identity (last 4 digits of registered phone)
-> Display policy details, premium due, next steps
-> Quick Reply: "Pay Now" / "Download Policy" / "Speak to Agent"
|
[Claim Flow]:
-> Claim type selection
-> WhatsApp Flow: incident details form
-> Document upload (photos, medical bills, FIR copy)
-> Claim reference number generated
-> Status tracking link shared
|
[Renewal Flow]:
-> AI shows renewal amount, due date, coverage summary
-> "Pay Now" -> Razorpay payment link
-> Confirmation + updated policy document via WhatsApp
```

Sample Call Script (Insurance Renewal)

AI: "Hello! I'm calling from [Insurance Company]. Am I speaking with Mr. Aditya Padhy?"

CUSTOMER: "Yes, speaking."

AI: "Thank you, Mr. Padhy. I'm calling about your health insurance policy ending in 4587. Your renewal is due in 15 days, on April 12th. Would you like me to walk you through your renewal options?"

CUSTOMER: "Yes, please. What's the premium this year?"

AI: "Your current premium is INR 18,500 for the family floater plan covering you, your spouse, and two children. This year, the renewal premium is INR 19,200 -- a 3.8% increase due to age adjustment. However, I can also share two upgraded options if you're interested. The first is an enhanced plan with INR 10 lakh coverage for INR 22,400, and the second is a super top-up of INR 25 lakh for just INR 3,200 additional. Would you like details on either?"

CUSTOMER: "Tell me about the top-up."

AI: "The super top-up works above your base policy's deductible. If your medical bill exceeds INR 5 lakh, the top-up kicks in and covers up to INR 25 lakh additionally. At INR 3,200 per year, it's excellent value for catastrophic coverage. Shall I add this to your renewal?"

CUSTOMER: "Yes, add it."

AI: "I've prepared your renewal with the base plan at INR 19,200 plus the super top-up at INR 3,200, totaling INR 22,400. I'm sending a payment link to your WhatsApp right now. You can pay via UPI, card, or net banking. Would you also like me to update your nominee details while we're at it?"

Expected Call Volume and Cost

- Monthly call volume per insurance company: 10,000-50,000 minutes
- Average call duration: 4-5 minutes
- Cost per call: INR 8-10 (\$0.10-0.12)
- Renewal conversion target: 70-85% (vs. industry 50-60% with human agents)

Key Metrics

- Policy renewal rate improvement: +15-25%
- Claim initiation time: from 15 minutes to 3 minutes
- After-hours query resolution: from 0% to 90%
- Compliance violation rate: target zero (AI never deviates from approved scripts)
- Customer satisfaction (CSAT): target 4.2+/5.0

7.8 Healthcare

Sub-Verticals

- Multi-specialty clinics and polyclinics
- Hospitals (outpatient departments)
- Diagnostic laboratories and imaging centers
- Pharmacies and medical stores
- Telemedicine platforms
- Dental clinics
- Veterinary clinics

Specific Pain Points This Solves

- Appointment booking automated 24/7 (currently requires human for every slot)
- Prescription refill calls handled instantly
- Lab result delivery via AI with appropriate medical context
- Patient follow-up after procedures automated (currently rarely done)
- Multilingual patient communication (Tamil, Telugu, Hindi, Bengali patients)
- After-hours emergency triage and on-call doctor connection
- ABDM health record integration for seamless data sharing

Custom AgentCore Tools Registered

```
@tool("check_doctor_availability")
def check_doctor_availability(
    doctor_id: str, specialization: str, date: str, time_preference: str
) -> DoctorAvailability:
    """Check doctor's available slots for appointment booking."""

@tool("book_appointment")
def book_appointment(
    patient_name: str, patient_phone: str, doctor_id: str,
    date: str, time_slot: str, reason: str, insurance_id: str
) -> AppointmentConfirmation:
    """Book a patient appointment with confirmation."""

@tool("get_lab_result_status")
def get_lab_result_status(
    patient_id: str, test_id: str
) -> LabResultStatus:
    """Check if lab results are ready and retrieve summary."""

@tool("send_prescription_reminder")
def send_prescription_reminder(
    patient_id: str, medication_name: str, dosage: str, pharmacy_id: str
) -> ReminderStatus:
    """Send medication refill reminder and connect to pharmacy."""
```

```
@tool("check_insurance_coverage")
def check_insurance_coverage(
    insurance_provider: str, policy_number: str, procedure_code: str
) -> CoverageDetails:
    """Verify insurance coverage for a specific procedure."""
```

Sample Qualification Questions

1. "Are you an existing patient or visiting us for the first time?" (patient type)
2. "Which doctor or department would you like to consult?" (routing)
3. "Can you briefly describe your concern?" (urgency assessment)
4. "Do you have a preferred date and time?" (scheduling)
5. "Do you have health insurance you'd like to use?" (billing preparation)

WhatsApp Flow Design

```
[Welcome: "Hello! Welcome to [Clinic/Hospital Name]. How can I help?"]
|
v
[Quick Replies: "Book Appointment" / "Lab Results" / "Prescription Refill" / "Other"]
|
v
[Appointment Booking Flow]:
-> Department selection (carousel of specializations)
-> Doctor selection (with photo, qualification, available dates)
-> WhatsApp Flow: date and time slot selection
-> Patient details (name, age, phone, insurance)
-> Confirmation with appointment card
-> Reminder: 24h before and 1h before
|
[Lab Results Flow]:
-> Patient ID or phone verification
-> AI retrieves and summarizes results
-> "Your CBC report is ready. Key findings: Hemoglobin 13.2 (normal),
    WBC 8,500 (normal), Platelets 2.1 lakh (normal).
    All values are within normal range."
-> PDF report sent via WhatsApp
-> "Would you like to book a follow-up with your doctor?"
```

Sample Call Script (Clinic Appointment)

AI: "Good morning! I'm the virtual assistant for [Clinic Name].
How can I help you today?"

CUSTOMER: "I need to see Dr. Sharma. I visited last week."

AI: "Of course! I can see from our records that you visited Dr. Sharma
on March 21st for a blood pressure follow-up. Dr. Sharma had
recommended a follow-up within 10 days. Shall I book a slot
for you? She has availability tomorrow at 10:30 AM and Thursday

at 3:00 PM."

CUSTOMER: "Tomorrow 10:30 works."

AI: "I've booked your appointment with Dr. Sharma for tomorrow, March 29th at 10:30 AM. You'll receive a confirmation on WhatsApp shortly, along with a reminder tomorrow morning. Is there anything specific you'd like Dr. Sharma to review? I can add a note to your appointment."

CUSTOMER: "Yes, my BP readings have been higher this week."

AI: "I've noted that your blood pressure readings have been elevated this week. Dr. Sharma will see this note before your appointment. If you have your recent BP readings, you can share them via WhatsApp and I'll add them to your patient record. Is there anything else I can help with?"

HIPAA / ABDM Compliance Notes

- All patient data encrypted at rest (AES-256) and in transit (TLS1.3)
- AI never provides diagnosis -- only shares doctor-confirmed results
- Patient identity verified before sharing any health information
- Consent collected for data processing and AI interaction recording
- ABDM integration for digital health record sharing with patient consent
- Audit trail for every patient data access
- Dedicated infrastructure option for HIPAA-compliant deployments

Expected Call Volume and Cost

- Monthly call volume per clinic: 2,000-8,000 minutes
- Monthly call volume per hospital: 15,000-50,000 minutes
- Average call duration: 2-3 minutes (appointment booking is fast)
- Cost per call: INR 4-6 (\$0.05-0.07)
- Appointment no-show reduction target: 30-40% (via AI reminders)

Key Metrics

- Appointment booking rate: 95%+ (vs. 60-70% with human receptionist)
- Average time to book: 90 seconds (vs. 5-8 minutes with human)
- After-hours appointment capture: from 0% to 100%
- Patient follow-up completion: from 15% to 85%
- Lab result delivery time: from 24-48 hours to instant

79 Real Estate

Sub-Verticals

- Residential sales (new projects, secondary market)
- Commercial leasing
- Rental management and PG/hostel operators
- Property management firms

Specific Pain Points This Solves

- 80% of property enquiries come outside business hours -- all captured by AI
- Site visit scheduling automated with agent calendar integration
- Property details served accurately from inventory database (carpet area, floor plan, amenities)
- Tenant renewal reminders automated
- Document collection for agreements compressed from weeks to days
- RERA-compliant disclosures delivered consistently

Custom AgentCore Tools Registered

```
@tool("search_properties")
def search_properties(
    property_type: str, budget_min: int, budget_max: int,
    location: str, bedrooms: int, amenities: list
) -> list[Property]:
    """Search property inventory matching customer criteria."""

@tool("check_unit_availability")
def check_unit_availability(
    project_id: str, tower: str, floor: int, unit_number: str
) -> UnitDetails:
    """Check specific unit availability with carpet area, price, floor plan."""

@tool("schedule_site_visit")
def schedule_site_visit(
    property_id: str, customer_name: str, customer_phone: str,
    preferred_date: str, preferred_time: str
) -> SiteVisitConfirmation:
    """Schedule a property site visit with agent assignment."""

@tool("calculate_emi")
def calculate_emi(
    property_price: int, down_payment_percent: float,
    loan_tenure_years: int, interest_rate: float
) -> EMICalculation:
    """Calculate home loan EMI with amortization schedule."""

@tool("check_application_status")
```

```
def check_application_status(
    application_id: str, customer_phone: str
) -> ApplicationStatus:
    """Check property purchase or rental application status."""
```

Sample Call Script (Property Enquiry)

AI: "Hello! Thank you for calling [Developer Name]. I'm your property assistant. Are you looking to buy or rent?"

CUSTOMER: "I'm looking to buy a 3BHK in Whitefield, Bangalore."

AI: "We have several 3BHK options in our Whitefield projects. Can you share your approximate budget?"

CUSTOMER: "Around 1.2 crore."

AI: "Within that budget, I have 3 excellent options for you:
First, [Project A] on Whitefield Main Road -- 3BHK, 1,450 sq ft carpet area, starting at INR 1.15 crore. RERA registered, possession by December 2027.
Second, [Project B] near ITPL -- 3BHK, 1,380 sq ft, starting at INR 1.08 crore, ready to move in.
Third, [Project C] near Varthur -- 3BHK, 1,520 sq ft, at INR 1.22 crore, possession March 2028.
Would you like to schedule a site visit for any of these?"

CUSTOMER: "The ready-to-move one sounds good. What's the EMI?"

AI: "For Project B at INR 1.08 crore, with 20% down payment of INR 21.6 lakhs and a home loan of INR 86.4 lakhs at 8.5% for 20 years, your EMI would be approximately INR 74,800 per month. Shall I schedule a site visit this weekend?
We have slots available Saturday at 11 AM and Sunday at 2 PM."

Expected Call Volume and Cost

- Monthly call volume per developer: 3,000-10,000 minutes
- Average call duration: 5-7 minutes
- Cost per call: INR 10-14 (\$0.12-0.17)
- Site visit booking rate target: 40-50% of qualified leads

Key Metrics

- After-hours lead capture: from 0% to 100%
- Average response time: from 4-6 hours to instant
- Site visit scheduling rate: +60% improvement
- Lead-to-visit conversion: 35-50% (vs. industry 15-25%)

7.10 Legal Services

Sub-Verticals

- Law firms (civil, criminal, corporate)
- Corporate legal departments
- Independent advocates and notaries
- Legal-tech platforms

Specific Pain Points This Solves

- First consultation intake automated -- collect case facts before meeting
- Case status queries answered instantly from case management system
- Document checklist sent and tracked via WhatsApp
- Fee estimate provided based on case parameters
- Court date reminders sent proactively
- Client confidentiality maintained through compliance engine

Custom AgentCore Tools Registered

```
@tool("check_case_status")
def check_case_status(
    case_id: str, client_id: str
) -> CaseStatus:
    """Check case status, next hearing date, pending actions."""

@tool("schedule_consultation")
def schedule_consultation(
    lawyer_id: str, client_name: str, client_phone: str,
    case_type: str, preferred_date: str, urgency: str
) -> ConsultationBooking:
    """Schedule a legal consultation with intake details."""

@tool("get_document_checklist")
def get_document_checklist(
    case_type: str, jurisdiction: str
) -> DocumentChecklist:
    """Generate document checklist for a specific case type."""

@tool("calculate_fee_estimate")
def calculate_fee_estimate(
    case_type: str, complexity: str, jurisdiction: str,
    estimated_hearings: int
) -> FeeEstimate:
    """Calculate preliminary legal fee estimate."""

@tool("send_court_date_reminder")
def send_court_date_reminder(
    case_id: str, hearing_date: str, court_name: str,
```

```
client_phone: str
) -> ReminderStatus:
    """Send court date reminder to client via WhatsApp."""
```

Sample Call Script (Legal Consultation Intake)

AI: "Good afternoon! Thank you for calling [Law Firm Name].
I'm the virtual legal assistant. How can I help you today?"

CUSTOMER: "I need a lawyer for a property dispute."

AI: "I understand. I can help you schedule a consultation with one
of our property law specialists. To prepare for the meeting,
may I ask a few questions about your case?"

CUSTOMER: "Sure."

AI: "First, is this dispute about residential or commercial property?"

CUSTOMER: "Residential. A flat in Pune."

AI: "And who are the parties involved in the dispute?"

CUSTOMER: "The builder has not given possession despite full payment."

AI: "I see -- this is a builder delay case. We handle many such cases.
Can you tell me when the possession was originally promised?"

CUSTOMER: "June 2024. It's been over a year now."

AI: "Based on what you've described, this would likely involve filing
a complaint with RERA Maharashtra and potentially the consumer
court. Our property law specialist, Advocate Priya Nair, handles
these cases regularly. She has availability this Thursday at
4:30 PM or Friday at 11 AM. Shall I book a consultation?
The initial consultation fee is INR 2,500."

CUSTOMER: "Thursday 4:30 works."

AI: "I've scheduled your consultation with Advocate Priya Nair for
Thursday at 4:30 PM. I'm sending a document checklist to your
WhatsApp -- please bring your sale agreement, payment receipts,
builder correspondence, and the possession letter. The more
documentation you have, the stronger your case will be."

Bar Council Compliance Notes

- AI clearly identifies as virtual assistant, not a lawyer
- No legal advice provided -- only scheduling, information, and intake
- Attorney-client privilege protections in place from first interaction
- All communications encrypted and stored with restricted access

- Conflict of interest check before accepting new client enquiries

Expected Call Volume and Cost

- Monthly call volume per firm: 500-3,000 minutes
- Average call duration: 6-8 minutes
- Cost per call: INR 12-16 (\$0.15-0.20)
- Consultation booking rate target: 50-65%

Key Metrics

- Client intake time: from 30 minutes to 8 minutes
- After-hours enquiry capture: from 0% to 100%
- Document collection completion: from 3 weeks to 5 days
- Client satisfaction (CSAT): target 4.5+/5.0

7.11 Education & Coaching

Sub-Verticals

- IIT-JEE, NEET, UPSC, CAT coaching centers
- K-12 schools
- Colleges and universities
- Online tutoring platforms and EdTech companies

Specific Pain Points This Solves

- Admission enquiry surge handled without additional staff
- Fee payment reminders automated for hundreds of students
- Course guidance personalized based on student performance
- Parent queries (results, attendance, fees) answered instantly
- Demo class scheduling streamlined
- Scholarship and loan guidance provided accurately

Custom AgentCore Tools Registered

```
@tool("check_batch_availability")
def check_batch_availability(
    course_id: str, batch_type: str, preferred_timing: str
) -> BatchAvailability:
    """Check available batches, seats, and schedule."""

@tool("schedule_demo_class")
```

```

def schedule_demo_class(
    student_name: str, phone: str, course: str,
    preferred_date: str, current_level: str
) -> DemoClassBooking:
    """Schedule a free demo class for prospective students."""

@tool("get_fee_structure")
def get_fee_structure(
    course_id: str, duration: str, scholarship_eligible: bool
) -> FeeStructure:
    """Get complete fee structure with payment options and scholarships."""

@tool("check_admission_status")
def check_admission_status(
    application_id: str, student_phone: str
) -> AdmissionStatus:
    """Check admission application status and pending requirements."""

@tool("get_exam_results")
def get_exam_results(
    student_id: str, exam_name: str
) -> ExamResults:
    """Retrieve student exam results with percentile and ranking."""

```

Sample Call Script (Coaching Admission)

AI: "Hello! Welcome to [Coaching Center]. I'm here to help with admissions, course information, or any queries. How can I assist?"

CUSTOMER: "My son is in Class 11. I want to know about your IIT-JEE program."

AI: "Our IIT-JEE program for Class 11 students is our most comprehensive offering. We have two formats available:
 The Classroom Program with daily 3-hour sessions, Monday to Saturday, and the Hybrid Program with 3 days classroom plus online practice.
 The Classroom Program fee is INR 1,85,000 for the two-year course, and the Hybrid is INR 1,45,000. We also have scholarship tests that can reduce fees by up to 90%. Would you like to schedule a free demo class for your son?"

CUSTOMER: "What about the scholarship test?"

AI: "The next scholarship test is on April 15th. It's a 2-hour test covering Maths and Science at the Class 10 level. Based on the score, scholarships range from 10% to 90% fee waiver:
 Above 90%: 90% scholarship. Above 80%: 50% scholarship.
 Above 70%: 25% scholarship. Above 60%: 10% scholarship.
 Registration is free. Shall I register your son for the April 15th test?"

CUSTOMER: "Yes please."

AI: "I've registered your son. You'll receive the test details, sample paper, and venue information on WhatsApp shortly.
 The test is at our Koramangala center at 10 AM. Would you

also like to book a demo class before the test so he can experience our teaching style?"

Expected Call Volume and Cost

- Monthly call volume per coaching center: 3,000-15,000 minutes (seasonal spikes 3-5x)
- Average call duration: 4-5 minutes
- Cost per call: INR 8-10 (\$0.10-0.12)
- Demo class booking rate target: 60-70%

Key Metrics

- Enquiry-to-enrollment conversion: 25-35% (vs. industry 10-15%)
- Fee collection reminder effectiveness: 85%+ payment within 7 days
- After-hours enquiry capture: from 0% to 100%
- Parent satisfaction score: target 4.3+/5.0

7.12 Fitness & Wellness

Sub-Verticals

- Gyms and fitness centers
- Yoga studios
- Spas and salons
- Personal trainers
- Swimming academies and sports facilities

Specific Pain Points This Solves

- Trial class booking available 24/7 without front desk
- Renewal reminders made proactively before expiry
- Inactive member detection and re-engagement
- Package upsell pitched at the right time
- Feedback collected after sessions
- Class schedule queries answered instantly

Custom AgentCore Tools Registered

```
@tool("check_trainer_availability")
def check_trainer_availability(
    trainer_id: str, date: str, time_preference: str, session_type: str
) -> TrainerAvailability:
    """Check personal trainer availability for booking."""

@tool("book_trial_session")
def book_trial_session(
    customer_name: str, phone: str, fitness_goal: str,
    preferred_date: str, preferred_time: str, experience_level: str
) -> TrialBooking:
    """Book a free trial session at the gym/studio."""

@tool("get_membership_options")
def get_membership_options(
    facility_id: str, membership_type: str, duration_months: int
) -> MembershipOptions:
    """Get membership packages with pricing and benefits."""

@tool("check_member_attendance")
def check_member_attendance(
    member_id: str, period: str
) -> AttendanceReport:
    """Check member attendance pattern for re-engagement triggers."""

@tool("send_renewal_reminder")
def send_renewal_reminder(
    member_id: str, expiry_date: str, renewal_offer: str
) -> ReminderStatus:
    """Send membership renewal reminder with special offer."""
```

Sample Call Script (Trial Class Booking)

AI: "Hi! Welcome to [Gym Name]. Are you looking to join us or are you an existing member?"

CUSTOMER: "I'm new. I want to try your gym."

AI: "We'd love to have you! We offer a free trial session so you can experience our facilities and meet our trainers. What's your primary fitness goal -- weight loss, muscle building, general fitness, or something specific?"

CUSTOMER: "Weight loss mostly."

AI: "For weight loss, I'd recommend trying our combination session -- 30 minutes of cardio plus 30 minutes of strength training with a certified trainer. We have trial slots available tomorrow morning at 7 AM and evening at 6 PM, or Saturday at 9 AM. Which works best for you?"

CUSTOMER: "Tomorrow evening."

AI: "I've booked your trial session for tomorrow at 6 PM. You'll be working with Trainer Vikram who specializes in weight management. Please bring comfortable workout clothes and a water bottle. I'll send you the gym location and parking details on WhatsApp. After your session, Vikram will discuss a personalized plan. Our monthly membership starts at INR 2,500. See you tomorrow!"

Expected Call Volume and Cost

- Monthly call volume per gym: 500-2,000 minutes
- Average call duration: 2-3 minutes
- Cost per call: INR 4-6 (\$0.05-0.07)
- Trial-to-member conversion target: 45-60%

Key Metrics

- Trial class booking rate: 70%+ of enquiries
- Member renewal rate: 75%+ (vs. industry 55-65%)
- Inactive member re-engagement: 30% return rate
- Monthly retention improvement: +10-15%

7.13 Automotive

Sub-Verticals

- New car dealerships
- Used car platforms
- Authorized service centers
- Fleet management companies
- EV charging networks

Specific Pain Points This Solves

- Test drive scheduling available 24/7
- Service reminders automated based on km/time intervals
- Vehicle availability answered accurately from dealer inventory
- Insurance and finance queries handled during purchase conversations
- Post-purchase follow-up for service package upsell
- Recall notification and compliance management

Custom AgentCore Tools Registered

```
@tool("check_vehicle_availability")
def check_vehicle_availability(
    make: str, model: str, variant: str, color: str, dealer_id: str
) -> VehicleStock:
    """Check real-time vehicle availability at dealer."""

@tool("schedule_test_drive")
def schedule_test_drive(
    customer_name: str, phone: str, vehicle_model: str,
    preferred_date: str, preferred_time: str, location_preference: str
) -> TestDriveBooking:
    """Schedule a test drive with vehicle preparation."""

@tool("get_service_quote")
def get_service_quote(
    vehicle_number: str, service_type: str, mileage: int
) -> ServiceQuote:
    """Generate service quote based on vehicle and service type."""

@tool("check_service_status")
def check_service_status(
    job_card_number: str, vehicle_number: str
) -> ServiceStatus:
    """Check vehicle service status and estimated completion."""

@tool("calculate_loan_emi")
def calculate_loan_emi(
    vehicle_price: int, down_payment: int,
    tenure_months: int, interest_rate: float
) -> AutoLoanEMI:
    """Calculate auto loan EMI with total cost breakdown."""
```

Sample Call Script (Service Reminder)

AI: "Hello, Mr. Sharma! I'm calling from [Dealer Name] service center. Our records show your Tata Nexon, registration MH-02-XX-1234, is due for its 20,000 km service. Would you like to book a service appointment?"

CUSTOMER: "Yes, but what will it cost?"

AI: "The 20,000 km service for your Nexon includes oil change, air filter replacement, brake inspection, and multi-point check. The estimated cost is INR 5,800 including parts and labor. If you'd like, I can also check if your extended warranty covers any of these items. Shall I book a slot?"

CUSTOMER: "Book for Saturday."

AI: "I've booked your service for Saturday, March 29th at 9 AM. Drop-off is at our Andheri service center. Estimated completion is same-day by 4 PM. I'll send you the confirmation with the

quote breakdown on WhatsApp. We also offer complimentary pickup and drop within 10 km -- shall I arrange that?"

Expected Call Volume and Cost

- Monthly call volume per dealership: 2,000-8,000 minutes
- Average call duration: 3-4 minutes
- Cost per call: INR 6-8 (\$0.07-0.10)
- Test drive booking rate target: 55-70%

Key Metrics

- Service reminder response rate: 65%+ (vs. 30% with SMS)
- Test drive booking rate: +40% improvement
- Service revenue per customer: +20% through upsell
- Customer retention rate: +15%

7.14 Hospitality & Food

Sub-Verticals

- Hotels, resorts, and boutique properties
- Restaurants and cafes
- Homestays and vacation rentals
- Catering services and cloud kitchens
- Banquet halls and event venues

Specific Pain Points This Solves

- Reservation queries handled at all hours for global travelers
- Room upgrade upsell pitched consistently
- Local experience recommendations delivered with local knowledge
- Repetitive queries (check-in time, parking, breakfast, WiFi) automated
- Post-stay feedback collected proactively
- Restaurant table booking and menu queries managed efficiently

Custom AgentCore Tools Registered

```
@tool("check_room_availability")
def check_room_availability(
    property_id: str, check_in: str, check_out: str,
    room_type: str, guests: int
) -> RoomAvailability:
    """Check room availability with rates and upgrade options."""

@tool("make_reservation")
def make_reservation(
    property_id: str, guest_name: str, guest_phone: str,
    check_in: str, check_out: str, room_type: str, special_requests: str
) -> ReservationConfirmation:
    """Create hotel reservation or restaurant booking."""

@tool("get_menu_info")
def get_menu_info(
    restaurant_id: str, cuisine_type: str, dietary_restrictions: list
) -> MenuInfo:
    """Get menu information with dietary accommodation."""

@tool("track_order")
def track_order(
    order_id: str
) -> OrderStatus:
    """Track food delivery or room service order status."""

@tool("handle_feedback")
def handle_feedback(
    reservation_id: str, rating: int, feedback_text: str
) -> FeedbackResponse:
    """Process guest feedback with routing for low ratings."""
```

Expected Call Volume and Cost

- Monthly call volume per hotel: 2,000-10,000 minutes
- Monthly call volume per restaurant: 500-3,000 minutes
- Average call duration: 3-4 minutes
- Cost per call: INR 6-8 (\$0.07-0.10)
- Direct booking conversion target: 30-45%

Key Metrics

- After-hours booking capture: from 0% to 100%
- Room upgrade conversion: 15-25%
- Guest satisfaction improvement: +0.5 points
- Direct booking ratio: +20% (reducing OTA dependency)

7.15 Retail & E-Commerce

Sub-Verticals

- D2C (Direct-to-Consumer) brands
- Physical retail stores
- Online marketplaces
- Subscription box companies

Specific Pain Points This Solves

- Order status queries (40-60% of all calls) automated completely
- Return initiation process streamlined to 2 minutes
- Product recommendations delivered with catalog knowledge
- Abandoned cart follow-up calls timed perfectly
- Loyalty program queries answered instantly

Custom AgentCore Tools Registered

```
@tool("track_order")
def track_order(
    order_id: str, customer_phone: str
) -> OrderTrackingDetails:
    """Track order status with carrier details and ETA."""

@tool("initiate_return")
def initiate_return(
    order_id: str, product_id: str, reason: str, preferred_resolution: str
) -> ReturnInitiation:
    """Initiate product return with pickup scheduling."""

@tool("check_product_availability")
def check_product_availability(
    product_id: str, size: str, color: str, pincode: str
) -> ProductAvailability:
    """Check product availability and delivery timeline."""

@tool("get_loyalty_points")
def get_loyalty_points(
    customer_id: str
) -> LoyaltyBalance:
    """Get loyalty points balance, expiry, and redemption options."""

@tool("process_complaint")
def process_complaint(
    order_id: str, complaint_type: str, description: str, urgency: str
) -> ComplaintTicket:
    """Process customer complaint with SLA assignment."""
```

Expected Call Volume and Cost

- Monthly call volume per D2C brand: 5,000-30,000 minutes
- Average call duration: 2-3 minutes
- Cost per call: INR 4-6 (\$0.05-0.07)
- Order query automation rate: 85-95%

Key Metrics

- First-call resolution rate: 80%+
- Average handle time: from 5 minutes to 2 minutes
- Customer satisfaction: target 4.0+/5.0
- Abandoned cart recovery: 15-20%

7.16 Home Services

Sub-Verticals

- HVAC installation and maintenance
- Plumbing services
- Electrical services
- Home cleaning and pest control
- Renovation and painting

Specific Pain Points This Solves

- Service booking available 24/7 (emergencies happen at night)
- Technician scheduling optimized with availability checking
- Service quotes provided instantly based on job description
- Technician tracking shared with customer in real-time
- Post-service feedback and warranty management
- Complaint handling with SLA-based escalation

Custom AgentCore Tools Registered

```
@tool("check_technician_availability")
def check_technician_availability(
    service_type: str, area_pincode: str, date: str, urgency: str
) -> TechnicianAvailability:
    """Check available technicians for a service type and area."""

@tool("book_service")
def book_service(
```

```

        service_type: str, customer_name: str, customer_phone: str,
        address: str, date: str, time_slot: str, description: str
    ) -> ServiceBooking:
        """Book a home service with technician assignment."""

@tool("get_service_quote")
def get_service_quote(
    service_type: str, description: str, property_size: str
) -> ServiceQuote:
    """Generate service quote based on job description."""

@tool("track_technician")
def track_technician(
    booking_id: str
) -> TechnicianLocation:
    """Track assigned technician's location and ETA."""

@tool("handle_complaint")
def handle_complaint(
    booking_id: str, complaint_type: str, description: str
) -> ComplaintResolution:
    """Handle service complaint with warranty check and escalation."""

```

Sample Call Script (Emergency Plumbing)

AI: "Hello! [Service Company] emergency line. How can I help?"

CUSTOMER: "My kitchen pipe is leaking badly. Water everywhere!"

AI: "I understand this is urgent. Let me find the nearest available plumber for you right away. Can you confirm your location -- are you in Indiranagar, Bangalore?"

CUSTOMER: "Yes, 100 Feet Road."

AI: "I have a certified plumber, Ramesh, available who can reach you in approximately 35 minutes. He specializes in pipe repairs. For emergency calls, there's a visit charge of INR 350 plus repair costs which depend on the issue. Shall I dispatch him?"

CUSTOMER: "Yes, please hurry!"

AI: "Ramesh is on his way. I'm sending his photo, phone number, and live location to your WhatsApp. In the meantime, if you can locate the main water valve and shut it off, that will help minimize the damage. It's usually near the water meter. Ramesh will call you when he's 5 minutes away."

Expected Call Volume and Cost

- Monthly call volume per service company: 2,000-10,000 minutes
- Average call duration: 3-4 minutes
- Cost per call: INR 6-8 (\$0.07-0.10)
- Same-day booking rate target: 70-80%

Key Metrics

- Emergency response booking: 95%+ (vs. 60% with human after-hours)
- Average booking time: 2 minutes
- Customer satisfaction: target 4.2+/5.0
- Repeat customer rate: +25%

Section 8: AgentCore as a Standalone Product (Go-to-Market)

8.1 Three Revenue Models

Model A: Platform API (B2B2B)

Target: CRM builders across ALL industries who want to add voice AI to their products.

How it works: CRM companies integrate AgentCore's API. They register custom tools for their industry, configure voice persona. AgentCore handles infrastructure. Whether the CRM is for healthcare, real estate, automotive, or education -- the integration pattern is identical.

Pricing: INR 10,000 - 1,00,000/month based on volume tier

Tier	Monthly Minutes	Monthly Price (INR)	Per-Minute Rate
Starter	Up to 5,000	10,000	2.00
Growth	Up to 20,000	30,000	1.50
Scale	Up to 67,000	75,000	1.12
Enterprise	Up to 200,000	1,00,000	0.50
Custom	200,000+	Custom	Negotiated

Model B: White-Label SaaS

Target: Digital marketing agencies, BPO companies, resellers, system integrators.

Pricing: INR 25,000 base + INR 2,000 per tenant per month

Component	Monthly Price (INR)
Base platform fee	25,000
Per tenant (first 10)	2,000/tenant
Per tenant (11-50)	1,500/tenant
Per tenant (51+)	1,000/tenant
Voice minutes	1.50/minute (pooled)
WhatsApp messages	0.50/message
Custom voice persona	5,000 one-time
White-label setup	15,000 one-time

Model C: Vertical SaaS Direct

Target: End customers in specific verticals who subscribe directly.

Plan	Monthly Price (INR)	Included Minutes	Additional Minutes
Basic	5,000	500	3.00/min
Professional	12,000	2,000	2.00/min
Enterprise	25,000	5,000	1.50/min

8.2 Who Buys AgentCore (All 15 Industries)

AgentCore's buyer universe spans every customer-facing industry. Here are all 15 target segments:

1. Tour & Travel Agencies, DMCs, Tour Operators

Use case: Lead qualification, itinerary proposals, booking, visa assistance

Products: ItinerarySuite, Ytrawise

TAM India: 50,000+ travel agencies, 5,000+ DMCs

2. Photographers, Photo Studios, Event Photographers

Use case: Studio booking, package quoting, trial session scheduling

Products: PhotoAdda

TAM India: 200,000+ professional photographers, 50,000+ studios

3. Financial Services (BFSI): Banks, NBFCs, Insurance, Mutual Funds, Wealth Management

Use case: Policy renewal, claim status, loan status, KYC, portfolio queries

TAM India: 100+ banks, 10,000+ NBFCs, 50+ insurance companies, 40,000+ MF distributors

4. Healthcare: Clinics, Hospitals, Diagnostic Centers, Pharmacies, Telemedicine

Use case: Appointment booking, lab results, prescription refills, follow-ups

TAM India: 70,000+ hospitals, 1,500,000+ clinics, 800,000+ pharmacies

5. Real Estate: Agents, Brokers, Developers, Property Management

Use case: Property enquiry, site visit scheduling, EMI calculation, tenant management

TAM India: 200,000+ real estate agents, 10,000+ developers, 50,000+ property managers

6. Legal: Law Firms, Advocates, Corporate Legal Teams

Use case: Consultation scheduling, case status, document collection, fee estimates

TAM India: 1,700,000+ advocates, 100,000+ law firms

7. Education & Coaching: Coaching Centers, Tutors, EdTech, Schools, Colleges

Use case: Admission enquiry, fee reminders, demo class booking, result queries

TAM India: 100,000+ coaching centers, 1,500,000+ schools, 50,000+ colleges

8. Fitness & Wellness: Gyms, Yoga Studios, Spas, Salons, Personal Trainers

Use case: Trial booking, membership renewal, class scheduling, re-engagement

TAM India: 30,000+ gyms, 50,000+ yoga studios, 300,000+ salons

9. Automotive: Car Dealers, Service Centers, Fleet Management

Use case: Test drive scheduling, service reminders, vehicle queries, finance calculator

TAM India: 25,000+ car dealerships, 50,000+ authorized service centers

10. Hospitality & Food: Hotels, Resorts, Restaurants, Cafes, Homestays

Use case: Reservation management, room upsell, menu queries, guest feedback

TAM India: 200,000+ hotels, 7,500,000+ restaurants

11. Retail & E-Commerce: D2C Brands, Physical Stores, Online Marketplaces

Use case: Order tracking, returns, product recommendations, abandoned cart recovery

TAM India: 100,000+ D2C brands, 14,000,000+ retail establishments

12. Logistics & Supply Chain: Courier Companies, Freight, Last-mile Delivery

Use case: Shipment tracking, pickup scheduling, complaint handling, B2B documentation

TAM India: 1,000+ logistics companies, 50,000+ courier service points

13. Home Services: HVAC, Plumbing, Electricians, Cleaning, Pest Control

Use case: Service booking, technician scheduling, quotes, complaint management

TAM India: 500,000+ home service providers (organized + platform-listed)

14. Event Management: Wedding Planners, Corporate Events, Venue Booking

Use case: Event enquiry, proposal generation, vendor coordination, booking management

TAM India: 50,000+ wedding planners, 10,000+ event companies

15. NGOs & Government: Citizen Services, Social Organizations

Use case: Grievance registration, scheme information, document processing, query handling

TAM India: 3,500,000+ registered NGOs, 250,000+ government service points

8.3 TAM by Industry

Industry	Estimated Entities (India)	Avg. Monthly Minutes	Avg. Revenue/Entity (INR/month)	Annual TAM (INR Cr)
BFSI	50,000	10,000	25,000	1,500
Healthcare	100,000	3,000	12,000	1,440
Retail & E-Commerce	50,000	5,000	15,000	900
Education & Coaching	50,000	3,000	12,000	720
Real Estate	30,000	2,000	10,000	360
Travel & Tourism	20,000	5,000	15,000	360
Hospitality & Food	30,000	2,000	10,000	360
Automotive	15,000	3,000	12,000	216
Logistics	5,000	10,000	25,000	150
Home Services	20,000	2,000	8,000	192
Legal	10,000	1,000	5,000	60
Fitness & Wellness	15,000	1,000	5,000	90
Photography	10,000	1,000	5,000	60
Event Management	5,000	2,000	10,000	60
NGOs & Government	5,000	5,000	15,000	90
Total TAM	415,000			6,558

Conservative serviceable addressable market (SAM): 5% of TAM = INR 328 Cr annually

Conservative serviceable obtainable market (SOM): 1% of TAM = INR 66 Cr annually (Year 3 target)

8.4 Detailed Pricing Table

Platform API Pricing (Model A)

Component	Starter	Growth	Scale	Enterprise
Monthly base	INR 10,000	INR 30,000	INR 75,000	INR 1,00,000
Included minutes	5,000	20,000	67,000	200,000
Overage rate	INR 2.50/min	INR 2.00/min	INR 1.50/min	INR 1.00/min
Concurrent calls	5	20	50	Unlimited
Custom tools	5	15	Unlimited	Unlimited

Component	Starter	Growth	Scale	Enterprise
Languages	3	6	All	All
Memory layers	2	4	4	4
WhatsApp	Add-on	Included	Included	Included
Industry templates	1	3	All	All + custom
Dedicated support	Email	Email + Chat	Dedicated CSM	Dedicated CSM + SLA
SLA	99.5%	99.9%	99.95%	99.99%

White-Label Pricing (Model B)

Component	Price
Platform setup (one-time)	INR 50,000
Monthly platform fee	INR 25,000
Per-tenant fee (1-10 tenants)	INR 2,000/tenant/month
Per-tenant fee (11-50 tenants)	INR 1,500/tenant/month
Per-tenant fee (51+ tenants)	INR 1,000/tenant/month
Voice minutes (pooled)	INR 1.50/minute
WhatsApp messages	INR 0.50/message
Custom domain setup	INR 5,000 one-time
Custom voice persona	INR 5,000 one-time
Industry template customization	INR 10,000 one-time per industry

8.5 Revenue Projections (Multi-Industry)

Year 1: Foundation (Months 1-12)

Revenue Stream	Q1	Q2	Q3	Q4	Annual
Model A (Platform API)	0	2L	6L	12L	20L
Model B (White-Label)	0	0	4L	8L	12L
Model C (Vertical SaaS - Travel + Photo)	3L	4.5L	6L	9L	22.5L
Model C (Vertical SaaS - BFSI + Healthcare)	0	0	2L	5L	7L
Total Revenue	3L	6.5L	18L	34L	61.5L

Year 2: Growth (Months 13-24)

Revenue Stream	Annual
Model A (Platform API) -- 5 industries	1.5 Cr
Model B (White-Label) -- 3 partners	80L
Model C (Vertical SaaS) -- 8 industries	2 Cr
Total Revenue	4.3 Cr

Year 3: Scale (Months 25-36)

Revenue Stream	Annual
Model A (Platform API) -- 12 industries	6 Cr
Model B (White-Label) -- 10 partners	3 Cr
Model C (Vertical SaaS) -- 15 industries	5.5 Cr
Total Revenue	14.5 Cr

8.6 The Moat (Why Hard to Replicate)

Technical Moat

1. **Hybrid Pipeline Architecture:** Cascaded + S2S with mid-call switching. 6-12 months to replicate.
2. **4-Layer Memory System:** Novel architecture. Competitors offer zero layers.
3. **Multi-Agent Orchestration:** 7 parallel agents per call via LangGraph.
4. **Self-Hosted Infrastructure:** Orders of magnitude harder than cloud-only.

Data Moat

5. **Cross-Vertical Learning:** Every industry deployment improves the platform. A qualification pattern learned from travel applies to real estate. A compliance rule from banking strengthens healthcare. Data across 15 industries creates network effects no single-vertical player can match.
6. **Indian Language Corpus:** Proprietary Hinglish and regional language data grows with every interaction across every industry.

Network Moat

7. **Multi-Product Ecosystem:** ItinerarySuite + PhotoAdda + Aayiq + SpendPilot + Ytrawise = ecosystem lock-in.
8. **White-Label Partner Network:** Each partner brings 10-100 tenants across multiple industries.

9. **Industry Template Library:** Each deployed industry template makes it easier to onboard the next customer in that industry, creating a flywheel.

Business Model Moat

10. **Self-Hosted Cost Advantage:** \$0.039-0.077/min vs. \$0.10-0.31/min for competitors.

11. **Compliance-First Design:** DPDP, TRAI, GDPR, HIPAA, BFSI, RERA compliance built in, not bolted on. Competitors would need 12+ months to build equivalent compliance frameworks.

Section 9: Complete Cost Model (V2 Updated)

9.1 Per-Minute Cost for Each Language Path (4 Paths)

Path 1: English/Hinglish

Deepgram Nova-3 STT:	\$0.0077/min
GPT-4.1-mini LLM:	\$0.006/min
ElevenLabs Flash API:	\$0.045/min (via direct API, not platform)
LiveKit self-hosted:	\$0.002/min
Exotel:	\$0.005/min
TOTAL:	\$0.066/min (INR 5.5/min)

Path 2: Regional Indian (Sarvam Full Stack)

Sarvam Saaras V3 STT:	\$0.006/min (INR 0.50/min -- cheapest!)
Sarvam-M LLM:	\$0.000/min (FREE!)
Sarvam Bulbul v3 TTS:	\$0.027/min (INR 2.25/min)
LiveKit self-hosted:	\$0.002/min
Exotel:	\$0.005/min
TOTAL:	\$0.040/min (INR 3.3/min) -- CHEAPEST

Path 3: Premium Regional (Sarvam STT + ElevenLabs TTS)

Sarvam STT:	\$0.006/min
GPT-4.1-mini:	\$0.006/min
ElevenLabs Flash API:	\$0.045/min
LiveKit:	\$0.002/min
Exotel:	\$0.005/min
TOTAL:	\$0.064/min (INR 5.3/min)

Path 4: Speech-to-Speech (Natural Conversation)

OpenAI Realtime mini: \$0.070/min
LiveKit: \$0.002/min
Exotel: \$0.005/min
TOTAL: \$0.077/min (INR 6.4/min)

9.2 Weighted Average

PATH	COST/MIN	INR/MIN	USE CASE

Regional Indian (Sarvam)	\$0.040	INR 3.3	Pure Hindi/Tamil/Telugu
English/Hinglish	\$0.066	INR 5.5	Mixed conversations
Global languages	\$0.066	INR 5.5	Arabic, Chinese, etc.
Speech-to-Speech	\$0.077	INR 6.4	Fast natural conversation

WEIGHTED AVERAGE (India-centric product):
60% English/Hinglish: 0.60 x \$0.066 = \$0.040
30% Regional Indian: 0.30 x \$0.040 = \$0.012
10% S2S: 0.10 x \$0.077 = \$0.008
WEIGHTED AVERAGE: \$0.060/min (INR 5.0/min)

VS RETELL AI: \$0.087/min (INR 7.2/min)
SAVING vs RETELL: 31% cheaper

9.3 Monthly Costs at 5 Scale Tiers

Tier 1: 5,000 Minutes/Month (Single SMB Tenant)

Component	Monthly Cost
EC2 - LiveKit Server (shared)	\$12
ECS - Agent Workers (1 instance)	\$49
RDS PostgreSQL (shared)	\$35
ElastiCache Redis (shared)	\$25
S3 (50GB)	\$2
CloudWatch + ALB	\$20
Total Infrastructure	\$143
AI API costs (5,000 min x \$0.060)	\$300
Total Monthly Cost	\$443

Tier 2: 20,000 Minutes/Month (Growth Stage)

Component	Monthly Cost
EC2 - LiveKit Server (dedicated)	\$49
ECS - Agent Workers (2 instances)	\$98
RDS PostgreSQL (db.r6g.large)	\$180
ElastiCache Redis (cache.r6g.large)	\$130
S3 (200GB)	\$5
CloudWatch + ALB	\$35
Total Infrastructure	\$497
AI API costs (20,000 min x \$0.055)	\$1,100
Total Monthly Cost	\$1,597

Tier 3: 67,000 Minutes/Month (Scale Stage)

VOICE + CHAT APIS AT 67,000 MIN/MONTH:

English/Hinglish (40,200 min):

Deepgram STT: \$310
GPT-4.1-mini: \$241
ElevenLabs Flash API: \$1,809
Subtotal: \$2,360

Regional Indian (20,100 min):

Sarvam STT: \$121 (INR 10,050)
Sarvam LLM: \$0 (FREE)
Sarvam Bulbul TTS: \$543 (INR 45,225)
Subtotal: \$664

Speech-to-Speech (6,700 min):

OpenAI Realtime: \$469
Subtotal: \$469

VOICE APIS TOTAL: \$3,493/month

Fixed AWS infra: \$147/month
LiveKit (in fixed): included
Exotel (67k x \$0.005): \$335
WhatsApp (50k msgs): \$550
Storage (S3): \$150
Monitoring: \$50

TOTAL MONTHLY: \$4,872/month (INR 4.05 lakhs)

Tier 4: 200,000 Minutes/Month (Enterprise)

Component	Monthly Cost
EC2 - LiveKit Server (3 instances, HA)	\$147
ECS - Agent Workers (8 instances)	\$392
RDS PostgreSQL (db.r6g.2xlarge, Multi-AZ)	\$650
ElastiCache Redis (cluster, 6 nodes)	\$400
S3 (2TB)	\$46
CloudWatch + ALB + WAF + Shield	\$120
Total Infrastructure	\$1,755
AI API costs (200,000 min x \$0.050)	\$10,000
Total Monthly Cost	\$11,755

Tier 5: 500,000 Minutes/Month (Large Enterprise)

Component	Monthly Cost
EC2 - LiveKit Server (5 instances)	\$245
ECS - Agent Workers (15 instances)	\$735
RDS PostgreSQL (db.r6g.4xlarge, Multi-AZ)	\$1,200
ElastiCache Redis (cluster, 9 nodes)	\$600
S3 (5TB)	\$115
CloudWatch + ALB + WAF + Shield + support	\$200
Total Infrastructure	\$3,095
AI API costs (500,000 min x \$0.045)	\$22,500
Total Monthly Cost	\$25,595

9.4 AWS Hosting Costs Breakdown

Service	Purpose	Specification	Monthly Cost
RDS PostgreSQL	Episodic + procedural memory	db.r6g.large, Multi-AZ	~\$180
RDS PostgreSQL (pgvector)	Semantic memory	db.r6g.large	~\$180
ElastiCache Redis	Short-term memory	cache.r6g.large	~\$130
EC2 c6i.xlarge	LiveKit Server	Reserved 1yr	~\$49
ECS EC2 Cluster	Agent Workers	Per instance	~\$49
S3	Recordings, documents	Standard	~\$10-50

Service	Purpose	Specification	Monthly Cost
SQS	Job queuing	Standard	~\$5
CloudWatch	Monitoring	Standard + custom	~\$30
ALB	Load balancing	Application LB	~\$25
Route 53	DNS	Hosted zone	~\$5
Base total			~\$660-750

9.5 Cost Per Tenant (at 100 Tenants)

Assuming 100 tenants with average 2,000 min/month each (200,000 total):

Cost Category	Monthly Total	Per Tenant
AWS Infrastructure	\$1,755	\$17.55
AI API Costs	\$10,000	\$100.00
Platform Operations (est.)	\$500	\$5.00
Total	\$12,255	\$122.55

At INR 2,000/tenant/month + INR 1.50/minute:

Revenue Category	Monthly Total	Per Tenant
Tenant fees (100 x INR 2,000)	INR 2,00,000	INR 2,000
Usage fees (200K min x INR 1.50)	INR 3,00,000	INR 3,000
Total Revenue	INR 5,00,000	INR 5,000

9.6 Gross Margin Analysis

Scale (Monthly Minutes)	Revenue per Minute (INR)	Cost per Minute (INR)	Gross Margin
5,000	3.50	7.40	-111%
20,000	3.00	6.65	-122%
67,000	2.50	6.08	-143%
200,000 (with tenant fees)	2.50	5.10	-104%
200,000 (Platform API INR 4.20/min)	4.20	5.10	-21%
500,000 (Platform API INR 3.80/min)	3.80	4.28	-13%

Note: The business reaches profitability through platform subscription fees + usage charges combined. Pure per-minute revenue must be supplemented with base subscription fees to achieve positive margins. The infrastructure costs are largely fixed while revenue scales linearly.

9.7 Sarvam Savings vs V1 Estimates

V1 ESTIMATED TOTAL (67k min/month): \$5,030/month
V2 ACTUAL TOTAL (67k min/month): \$4,872/month
SAVING WITH SARVAM: \$158/month at this scale

V1 ESTIMATED (200k min/month): ~\$15,000/month
V2 ACTUAL (200k min/month): ~\$11,755/month
SAVING AT 200k MIN: ~\$3,245/month (INR 2.7 lakhs)

KEY SAVINGS DRIVERS:

1. Sarvam-M LLM is FREE (saves \$0.006/min on 30% of calls)
2. Sarvam STT cheaper than Deepgram for regional (\$0.006 vs \$0.0077)
3. Sarvam Bulbul TTS cheaper than ElevenLabs for Indian (\$0.027 vs \$0.045)
4. Direct ElevenLabs API vs Platform (\$0.045 vs \$0.080-0.100)

9.8 Cost by Industry Use Case

Average call durations differ significantly by industry, affecting per-interaction cost:

Industry	Avg Call Duration	Cost per Interaction (INR)	Cost per Interaction (USD)	Primary Use Case
Healthcare (appointment)	2 min	4-6	\$0.05-0.07	Appointment booking
Fitness (trial booking)	3 min	6-8	\$0.07-0.10	Trial class booking
Retail (order status)	2 min	4-6	\$0.05-0.07	Order tracking
Logistics (delivery status)	2 min	4-6	\$0.05-0.07	Shipment tracking
Automotive (service booking)	3 min	6-8	\$0.07-0.10	Service scheduling
Travel (package quote)	5 min	10-12	\$0.12-0.15	Package enquiry
Hospitality (reservation)	4 min	8-10	\$0.10-0.12	Room booking
BFSI (policy renewal)	4 min	8-10	\$0.10-0.12	Policy renewal call
Education (admission)	4 min	8-10	\$0.10-0.12	Admission enquiry
Real Estate (property)	6 min	12-14	\$0.15-0.17	Property enquiry
Legal (intake)	8 min	16-20	\$0.20-0.25	Consultation intake
Home Services (booking)	3 min	6-8	\$0.07-0.10	Service booking
Photography (quote)	4 min	8-10	\$0.10-0.12	Photography quote

Industry	Avg Call Duration	Cost per Interaction (INR)	Cost per Interaction (USD)	Primary Use Case
Event Management (enquiry)	6 min	12-14	\$0.15-0.17	Event planning

Key insight: The shortest calls (healthcare, retail, logistics at 2 minutes) have the highest automation ROI because they are highest volume and lowest complexity. Legal intake calls (8 minutes) are lowest volume but highest per-call value.

Section 10: Compliance & Governance

10.1 DPDP Act 2023 Requirements

The DPDP Act is India's comprehensive data protection legislation. AgentCore is designed for full compliance.

Consent Flows

Voice Call Consent:

AI: "This call may be recorded for quality assurance and to provide you with better service. Your data will be processed in accordance with our privacy policy. Do you consent to continue?"

[Wait for affirmative response]

AI: "Thank you. How can I help you today?"

WhatsApp Consent:

- Opt-in collected before first message via WhatsApp Flows
- Double opt-in for marketing messages
- Easy opt-out via "STOP" keyword

Purpose Limitation

- Primary purpose: Providing the requested service
- Secondary purpose: Improving service quality (only with explicit consent)
- Prohibited: Selling data to third parties

Right to Erasure

1. All episodic memories deleted
2. All semantic memory embeddings removed
3. Call recordings deleted from S3
4. Documents permanently deleted

5. Audit log entries retained but PII anonymized
6. Redis session data auto-expires
7. Confirmation sent to customer
8. Completed within 72 hours of request

10.2 TRAI Compliance

Call Recording Disclosure

- Every AI call must disclose automated nature and recording within first 10 seconds
- Customer must have option to opt out of recording

DND Registry

- Real-time check before outbound calls (not cached)
- Violation penalties: up to INR 25 lakh per incident

Time-of-Day Restrictions

- Outbound calls only between 9:00 AM and 9:00 PM (customer's local time)
- No calls on national holidays
- Frequency cap: maximum 3 call attempts per customer per week

Caller ID Requirements

- All outbound calls must display valid, registered caller ID
- CLI must not be spoofed or masked

10.3 Data Residency

- Primary: ap-south-1 (Mumbai)
- DR: ap-south-2 (Hyderabad)
- All data stays within India
- LLM inference via OpenAI API (data in transit to US) -- future roadmap supports local LLMs
- Database replicas within India only
- S3 buckets with region restriction policies

10.4 Kill-Switch Framework

Condition	Action	Scope
AI generating harmful content	Immediate call termination + review	Per-call
Compliance violation detected	Pause tenant + alert	Per-tenant

Condition	Action	Scope
Data breach suspected	Suspend all outbound + lock recordings	Platform-wide
Regulatory order received	Full platform shutdown	Platform-wide
Cost anomaly (10x normal)	Pause new calls + alert	Per-tenant
API provider outage	Fallback to secondary + alert	Per-service

```
class KillSwitch:
    LEVELS = {
        'CALL': 'Terminate specific call',
        'TENANT': 'Suspend all operations for a tenant',
        'SERVICE': 'Disable a specific service',
        'PLATFORM': 'Emergency shutdown of entire platform'
    }

    def activate(self, level, reason, activated_by):
        # 1. Immediate action based on level
        # 2. Log to immutable audit trail
        # 3. Notify operations team (PagerDuty)
        # 4. Notify affected tenants
        # 5. Create incident record
```

Recovery Process

1. Activation logged with timestamp, reason, activator
2. Active calls gracefully terminated with apology
3. New calls rejected with maintenance message
4. Root cause analysis initiated
5. Recovery requires dual authorization
6. Post-recovery verification before restoration

10.5 Audit Trail

What is logged:

- Every AI decision (which pipeline, agents, tools)
- Every tool invocation (input, output, latency, success/failure)
- Every data access (which customer data, by which agent)
- Every consent event (collection, withdrawal, expiry)
- Every escalation (reason, context, outcome)
- Every compliance check
- Every configuration change
- Every kill-switch activation and recovery

Retention:

- 7 years (configurable per tenant)
- Separate database partition
- Append-only (enforced by triggers)
- Encrypted at rest and in transit
- Exportable in JSON, CSV formats

10.6 HIPAA Eligibility

For healthcare CRM deployments:

- AES-256 encryption at rest, TLS 1.3 in transit
- Role-based access with MFA
- Comprehensive audit logging
- BAA readiness (AWS BAA + separate BAAs for OpenAI, Deepgram, ElevenLabs)
- PHI flagging, encryption, and access logging
- Minimum necessary data access principle
- Requires dedicated infrastructure (not shared) and additional audit procedures

10.7 Industry-Specific Compliance Notes

BFSI Compliance

- RBI guidelines for automated financial communication
- SEBI compliance for investment-related conversations
- IRDAI guidelines for insurance sales and renewal calls
- Financial advice disclaimer required in every interaction
- Recording retention: 7 years minimum per RBI mandate
- Customer identity verification before sharing account details

Healthcare Compliance

- HIPAA readiness for international deployments
- ABDM (Ayushman Bharat Digital Mission) data sharing protocols
- AI never provides diagnosis -- only shares doctor-confirmed results
- Prescription data handled with restricted access
- Patient consent required for each data type (voice recording, health data, sharing)
- Telemedicine regulations for remote consultation facilitation

Real Estate Compliance

- RERA (Real Estate Regulatory Authority) disclosure requirements
- All project details must match RERA-registered information
- Anti-money laundering (AML) checks for high-value transactions
- Agreement terms communicated clearly per RERA mandate

Legal Compliance

- Bar Council of India rules for client communication
- Attorney-client privilege protections from first interaction
- Conflict of interest checking before new client intake
- No legal advice from AI -- only scheduling, information, and intake
- Communication encryption and restricted access

Education Compliance

- Fee transparency as per UGC/AICTE guidelines
- No misleading placement or result claims
- Refund policy communication per regulations
- Student data protection under DPDP Act

Section 11: Implementation Phases (2.5-Month Timeline)

11.1 Phase 1: Core Voice + CRM + Proposal Engine (Month 1, Weeks 1-4)

Week 1: Infrastructure Setup

- Provision AWS: EC2 for LiveKit, ECS cluster, RDS PostgreSQL, ElastiCache Redis
- Deploy LiveKit Server with SSL via ALB
- Configure security groups (UDP 50000-60000 critical)
- Set up S3, CloudWatch, CI/CD pipeline

Milestone: LiveKit Server accepting WebRTC connections.

Week 2: Cascaded Pipeline

- Silero VAD + Deepgram Nova-3 STT + GPT-4.1-mini + ElevenLabs Flash TTS
- Sentence buffer, turn detection, pipeline latency monitoring
- Sarvam STT integration for Indian regional

Milestone: End-to-end voice conversation under 1 second latency.

Week 3: CRM Integration + Memory + Proposal Engine

- Tenant configuration, tool registration framework
- Redis session memory (Layer 1) + PostgreSQL episodic (Layer 2)
- Lead creation from voice, call session tracking
- [NEW] Proposal Engine: AI generation, PDF rendering, shareable link, tracking

Milestone: Voice agent qualifies leads, generates proposals.

Week 4: Testing + First Vertical

- End-to-end testing with real phone numbers
- Error handling, analytics dashboard
- Deploy ItinerarySuite voice agent to production

Milestone: ItinerarySuite live with lead qualification, proposal generation, and package search.

11.2 Phase 2: Intelligence + S2S + Chat Multilingual + Invoice Engine (Month 2, Weeks 5-8)

Week 5: Multi-Agent Orchestration

- LangGraph Supervisor/Worker architecture
- Qualification, Memory, Handoff agents
- Parallel execution, audit logging

Week 6: Semantic Memory + S2S Pipeline + Multilingual Chat

- pgvector deployment, embedding generation
- Semantic write/read pipeline
- OpenAI Realtime API integration, pipeline router
- [NEW] Chat multilingual: Sarvam language detection, translation routing

Milestone: Returning customers get personalized greetings. Chat works in 30+ languages.

Week 7: WhatsApp Integration + Invoice Engine

- WhatsApp Business API, ChatAgent with shared LLM/memory
- WhatsApp Flows for lead capture and booking
- [NEW] Invoice Engine: GST invoicing, payment tracking, WhatsApp delivery

Week 8: Quote Engine + Second Vertical

- Quote Engine with PDF, WhatsApp delivery, tracking
- Deploy second vertical (PhotoAdda or Aayiq)

Milestone: Quotes delivered via WhatsApp. Invoices generated. Two verticals live.

11.3 Phase 3: Full Platform + All New Engines + Load Test (Month 3, Weeks 9-12)

Week 9: Remaining Agents + New Engines

- Document Intelligence, Disruption Monitor, Compliance agents
- [NEW] Approval Engine, Refund/Cancel Engine, NPS/Feedback Engine
- [NEW] Supplier Management Engine, Campaign Engine

Week 10: Compliance + Tenant Management

- DPDP consent flows, TRAI compliance
- Kill-switch framework, tenant provisioning, billing metering
- White-label support

Week 11: Remaining Verticals + Load Testing

- Deploy Ytrawise, Aayiq, SpendPilot, HelpRevX
- [NEW] Itinerary Builder Engine, Expense Capture Engine
- Load test: 50 concurrent calls, failover testing, security audit

Milestone: All 6 verticals live. All 24 engines operational. 50 concurrent calls.

Week 12: Documentation + Launch

- API documentation (OpenAPI spec), SDK, quickstart guides
- Operations runbook, monitoring fine-tuning
- Demo environment, pricing infrastructure

Milestone: Platform ready for external customers.

Section 12: Appendix -- Data Tables

12.1 Channel x Object Matrix (Universal -- All Industries)

This matrix applies across ALL industries. "Lead" means any enquiry (trip, appointment, property, policy, consultation). "Proposal" means any structured offering document. "Booking" means any confirmed transaction (reservation, appointment, enrollment, membership). "Case" means any complaint or support request.

Object	Phone (Voice)	Website Chat	WhatsApp	Email	Web UI
Lead - Create	[YES]	[YES]	[YES]	[YES]	[YES]
Lead - Qualify	[YES]	[YES]	[YES]	[NO]	[YES]
Lead - Score	[YES]	[YES]	[YES]	Auto	Auto
Proposal - Generate	[YES]	[YES]	[YES]	[NO]	[YES]
Proposal - Send	WhatsApp	Email	In-chat	Email	WhatsApp/Email
Proposal - Approve	Voice confirm	Button	Button	Link	Button
Proposal - Modify	Voice request	In-chat	In-chat	Email	Form
Case - Create	[YES]	[YES]	[YES]	[YES]	[YES]
Case - Update	[YES]	[YES]	[YES]	[YES]	[YES]
Case - Escalate	[YES]	[YES]	[YES]	Auto	[YES]
Quote - Generate	[YES]	[YES]	[YES]	[NO]	[YES]
Quote - Send	WhatsApp	Email	In-chat	Email	WhatsApp/Email
Quote - Accept	Voice confirm	Button	Button	Link	Button
Booking - Create	[YES]	[YES]	[YES]	[NO]	[YES]
Booking - Pay	Transfer to link	In-widget	WhatsApp Pay	Link	Gateway
Invoice - Send	WhatsApp	Email	In-chat	Email	Download
Document - Upload	[NO]	[YES]	[YES]	[YES]	[YES]
Document - Verify	Auto (OCR)	Auto (OCR)	Auto (OCR)	Auto (OCR)	Manual
Feedback - Collect	[YES]	[YES]	[YES]	[YES]	[YES]
Proactive - Send	Outbound call	N/A	Template msg	Email	N/A

12.2 All 24 Engines Comparison Table (V1 vs V2)

#	Engine	V1 Status	V2 Status	What Changed
1	VoiceAgent	Existed	Updated	Sarvam STT + ElevenLabs API routing
2	ChatAgent	Existed	Updated	Full multilingual with Sarvam
3	WhatsApp Commerce	Existed	Same	No changes
4	Email Intelligence	Existed	Same	Minor update
5	Memory Service	Existed	Same	No changes
6	Multi-Agent Orchestrator	Existed	Same	No changes
7	Proactive Outbound	Existed	Same	NPS split to own engine
8	Lead Engine	Existed	Same	No changes
9	Proposal Engine	Missing	[NEW]	Complete proposal lifecycle

#	Engine	V1 Status	V2 Status	What Changed
10	Quote Engine	Existed	Updated	Linked to Proposal Engine
11	Case Engine	Existed	Same	No changes
12	Approval Engine	Missing	[NEW]	Multi-step approval workflows
13	Booking Engine	Existed	Enhanced	Partial payment, installment
14	Invoice Engine	Missing	[NEW]	GST invoicing, payment tracking
15	Refund/Cancel Engine	Missing	[NEW]	Post-booking lifecycle
16	Document Intelligence	Existed	Enhanced	E-signature integration
17	Analytics Service	Existed	Same	No changes
18	Supplier Mgmt Engine	Missing	[NEW]	Catalog, rates, contracts
19	Campaign Engine	Missing	[NEW]	Marketing automation
20	NPS/Feedback Engine	In Proactive	[NEW]	Standalone with insights
21	Compliance Service	Existed	Same	No changes
22	Tenant Management	Existed	Same	No changes
23	Itinerary Builder	Missing	[NEW]	Travel-specific planner
24	Expense Capture	Missing	[NEW]	Voice expense + receipt OCR

12.3 Language Support Matrix (All Providers, All Languages)

Language	Deepgram Nova-3 STT	Sarvam Saaras V3 STT	ElevenLabs Flash TTS	Sarvam Bulbul TTS	OpenAI S2S
English	[YES]	[YES]	[YES]	[NO]	[YES]
Hindi	[YES]	[YES]	[YES]	[YES]	[YES]
Hinglish	[YES]	[YES]	[YES]	[YES]	Partial
Tamil	Partial	[YES]	[YES]	[YES]	[NO]
Telugu	Partial	[YES]	[YES]	[YES]	[NO]
Kannada	[NO]	[YES]	[YES]	[YES]	[NO]
Malayalam	[NO]	[YES]	[YES]	[YES]	[NO]
Bengali	Partial	[YES]	[YES]	[YES]	[NO]
Marathi	[NO]	[YES]	[YES]	[YES]	[NO]
Gujarati	[NO]	[YES]	[YES]	[YES]	[NO]
Punjabi	[NO]	[YES]	[YES]	[YES]	[NO]
Odia	[NO]	[YES]	[YES]	[YES]	[NO]
Assamese	[NO]	[YES]	Partial	[NO]	[NO]

Language	Deepgram Nova-3 STT	Sarvam Saaras V3 STT	ElevenLabs Flash TTS	Sarvam Bulbul TTS	OpenAI S2S
Urdu	Partial	[YES]	[YES]	[YES]	[NO]
Spanish	[YES]	[NO]	[YES]	[NO]	[YES]
French	[YES]	[NO]	[YES]	[NO]	[YES]
German	[YES]	[NO]	[YES]	[NO]	[YES]
Portuguese	[YES]	[NO]	[YES]	[NO]	[YES]
Japanese	[YES]	[NO]	[YES]	[NO]	[YES]
Korean	[YES]	[NO]	[YES]	[NO]	[YES]
Mandarin	[YES]	[NO]	[YES]	[NO]	[YES]
Arabic	[YES]	[NO]	[YES]	[NO]	[YES]

12.4 Component Cost Summary

Component	Provider	Model/Tier	Unit Cost	Unit
STT	Deepgram	Nova-3 PAYG	\$0.0077	per minute
STT	Sarvam	Saaras V3	\$0.006	per minute
STT	ElevenLabs	Scribe v2	\$0.008	per minute
LLM	OpenAI	GPT-4.1-mini	\$0.006	per minute (est.)
LLM	Sarvam	Sarvam-M	\$0.000	per minute (FREE)
LLM	OpenAI	Realtime mini	\$0.070	per minute
TTS	ElevenLabs	Flash v2.5 API	\$0.045	per minute
TTS	Sarvam	Bulbul v3	\$0.027	per minute
Translation	Sarvam	Translate	INR 20	per 10K chars
Language ID	Sarvam	LangID	INR 3.5	per 10K chars
Infra	AWS	EC2 c6i.xlarge	\$49	per month
Infra	AWS	RDS db.r6g.large	\$180	per month
Infra	AWS	ElastiCache r6g.large	\$130	per month
Telephony	Exotel	India	\$0.005	per minute
Telephony	Twilio	Global	\$0.0085	per minute
WhatsApp	Meta	Cloud API	\$0.011	per message

12.5 V1 vs V2 Changes Log

Change	Type	Impact
Added Proposal Engine	New Engine	Critical – fills biggest gap in customer journey
Added Invoice Engine	New Engine	Required for complete billing lifecycle
Added Refund/Cancel Engine	New Engine	Required for post-booking management
Added Approval Engine	New Engine	Enables multi-step business workflows
Added Campaign Engine	New Engine	Marketing automation for tenants
Added NPS/Feedback Engine	New Engine	Standalone feedback with insights
Added Supplier Mgmt Engine	New Engine	Required for travel package building
Added Itinerary Builder	New Engine	Hero feature for travel vertical
Added Expense Capture	New Engine	Voice-first expense for Aaiyiq
Updated ChatAgent multilingual	Enhancement	Chat now supports 30+ languages
Updated STT/TTS to API-direct	Architecture	Lower cost, more control
Added Sarvam-M LLM	New Provider	FREE LLM for Indian languages
Updated cost model	Financial	Weighted avg \$0.060/min (was \$0.023-0.057)
Updated customer journey	Process	Full lifecycle from Lead to NPS

12.6 API Response Time Targets

Operation	Target Latency	Max Acceptable
First voice response (cascaded)	800ms	1,200ms
First voice response (S2S)	300ms	500ms
Pre-call memory retrieval	150ms	300ms
Tool execution (simple)	200ms	500ms
Tool execution (complex)	500ms	1,000ms
Proposal AI generation	5,000ms	15,000ms
Quote PDF generation	2,000ms	5,000ms
WhatsApp message delivery	500ms	2,000ms
Document OCR processing	3,000ms	10,000ms
Semantic search (pgvector)	50ms	200ms
Lead creation	100ms	300ms
Consent verification	50ms	100ms

12.7 Scaling Reference

Concurrent Calls	Agent Workers Needed	EC2 Instances	Est. Monthly Cost
5	1-2	1	\$443
10	2-3	2	\$800
25	4-6	3	\$1,800
50	8-10	5	\$4,000
100	15-20	8	\$8,000
250	35-50	15	\$18,000
500	70-100	30	\$35,000

Section 13: Claude Implementation Prompts

13.1 How to Use These Prompts

Each prompt below is a complete, self-contained instruction for Claude (in Cursor, [Claude.ai](#), or any AI coding tool).

RULES FOR CLAUDE:

1. Generate COMPLETE, PRODUCTION-READY code -- no placeholders, no TODOs, no "add your logic here"
2. Every function must be fully implemented with real logic
3. All error handling must be included
4. TypeScript strict mode throughout
5. Every file must include proper imports
6. Database queries must use parameterized inputs (no SQL injection)
7. Environment variables must be loaded via process.env with validation
8. Logging must be included (use pino or winston)
9. Never truncate code blocks -- output the entire file

BEFORE EACH PHASE: Run audit command at top of prompt first.

AFTER EACH PHASE: Run the validation command at bottom of prompt.

13.2 Phase 0 -- Audit Existing Codebase (Run First)

```
You are auditing the HelpRevX turborepo to understand what already exists before building.  
  
Run these commands and report what you find:  
  
STEP 1: Repository structure
```

```

ls -la apps/ packages/
cat package.json
cat pnpm-workspace.yaml

STEP 2: Existing packages
ls packages/
cat packages/db/package.json
cat packages/db/prisma/schema.prisma

STEP 3: Existing apps
ls apps/
cat apps/api/package.json

STEP 4: Technology check
cat package.json | grep -E "(next|react|node|typescript|prisma|redis|express|fastify|trpc)"
grep -r "DATABASE_URL|REDIS_URL" .env.example 2>/dev/null || echo "No .env.example found"

STEP 5: Existing API structure
ls apps/api/src/ 2>/dev/null || ls apps/api/app/ 2>/dev/null || ls apps/api/pages/api/ 2>/dev/null

STEP 6: Check for any existing voice/AI code
grep -r "retell\\|livekit\\|deepgram\\|elevenlabs\\|openai\\|voice" packages/ apps/ --include="*.ts"

STEP 7: Check existing DB schema for tenant structure
grep -A5 "model Tenant\\|table tenants" packages/db/prisma/schema.prisma 2>/dev/null

STEP 8: Check existing auth/middleware
find apps/api -name "*.ts" | xargs grep -l "tenant_id\\|tenantId" 2>/dev/null | head -10

REPORT FORMAT:
1. Package manager: [pnpm/npm/yarn]
2. API framework: [Next.js API routes / tRPC / Express / Fastify]
3. DB ORM: [Prisma / Drizzle / raw SQL]
4. Existing tenant table: [Yes/No, if yes list fields]
5. Existing Redis usage: [Yes/No]
6. Existing auth: [what kind]
7. Gaps found: [list]
8. Conflicts to watch: [list]

After audit, create: /home/user/workspace/codebase-audit-report.md

```

13.3 Phase 1 -- Database Foundation

CONTEXT: You are building AgentCore voice AI infrastructure into the HelpRevX turborepo. The codebase uses [FILL FROM AUDIT: package manager, ORM, API framework]. Existing DB schema: [PASTE relevant tenant/user tables from audit].

TASK: Add all AgentCore database tables to packages/db/prisma/schema.prisma. Also create raw SQL migration for pgvector (Prisma doesn't support vector type).

REQUIREMENTS:

- Do NOT modify existing tables/models
- ADD these new models to the end of schema.prisma
- All models must have tenant_id for multi-tenant isolation

- Use cuid() for all primary keys
- Include proper @index for all frequently queried fields
- Include @relation where appropriate to existing models

ADD THESE MODELS:

```
// 1. TenantVoiceConfig
model TenantVoiceConfig {
  [implement fully with all fields:
    retellAgentId, liveKitAgentName, personaName, voiceId,
    primaryLanguage, supportedLanguages, businessHoursJson,
    maxCallDurationMs, autoQuoteEnabled, qualificationQuestionsJson,
    escalationKeywords, tenantId (unique FK), timestamps]
}

// 2. AiCallSession
model AiCallSession {
  [implement fully: callSid, channel, direction, callerMobile,
    agentId, pathType, languageDetected, languageSelected,
    startedAt, endedAt, durationSeconds, recordingUrl,
    transcriptText, transcriptObject, aiSummary, extractedFields,
    emotionDetected, sentimentScore, urgencyFlag, outcome,
    callCostUsd, linkedRecordType, linkedRecordId,
    killSwitchTriggered, tenantId, timestamps, indexes]
}

// 3. CustomerMemoryEpisodic
model CustomerMemoryEpisodic {
  [implement: customerId, customerMobile, eventType, eventSummary,
    structuredData, sentimentScore, emotion, channel, occurredAt,
    tenantId, timestamps, indexes]
}

// 4. AgentAuditLog
model AgentAuditLog {
  [implement fully: sessionId, workerAgent, actionType,
    actionDescription, inputData, outputData, mcpToolCalls,
    confidenceScore, latencyMs, costTokens, costUsd,
    humanApproved, killSwitchTriggered, killSwitchReason,
    customerId, linkedRecordType, linkedRecordId,
    tenantId, occurredAt, indexes]
}

// 5. ProactiveMessage
model ProactiveMessage {
  [implement: triggerType, channel, templateName, messageContent,
    sentAt, delivered, opened, clicked, conversionOutcome,
    abVariant, customerId, customerMobile, linkedRecordType,
    linkedRecordId, tenantId, timestamps]
}

// 6. BookingDocument
model BookingDocument {
  [implement: bookingId, travellerId, docType, fileUrl,
    ocrExtracted, fraudCheckResult, validityCheck, status,
    uploadedVia, uploadedAt, tenantId, timestamps]
```

```

}

// 7. Interaction
model Interaction {
  [implement: customerId, customerMobile, channel, direction,
    agentId, aiHandled, subject, bodyText, summary,
    linkedRecordType, linkedRecordId, sessionId,
    occurredAt, tenantId, timestamps, indexes]
}

ALSO CREATE: packages/db/prisma/migrations/add_pgvector.sql
Content:
CREATE EXTENSION IF NOT EXISTS vector;
CREATE TABLE IF NOT EXISTS customer_memory_semantic (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id TEXT NOT NULL,
  customer_mobile TEXT NOT NULL,
  customer_id TEXT,
  content_type TEXT,
  content_text TEXT NOT NULL,
  embedding vector(1536),
  source_session_id TEXT,
  source_channel TEXT,
  occurred_at TIMESTAMPTZ,
  created_at TIMESTAMPTZ DEFAULT now()
);
CREATE INDEX IF NOT EXISTS customer_memory_semantic_tenant_mobile_idx
  ON customer_memory_semantic(tenant_id, customer_mobile);
CREATE INDEX IF NOT EXISTS customer_memory_semantic_embedding_idx
  ON customer_memory_semantic
  USING ivfflat (embedding vector_cosine_ops)
  WITH (lists = 100);

VALIDATION:
cd packages/db && npx prisma validate
cd packages/db && npx prisma generate
Check output for zero errors.

```

13.4 Phase 2 -- LiveKit Voice Agent Service

CONTEXT: Building the core voice AI engine.
 Location: apps/voice-agent/ (new Python service in the turborepo)
 This service connects to LiveKit Server and handles ALL inbound phone calls.

TASK: Create the complete voice agent service. Every file must be complete and working.

CREATE: apps/voice-agent/requirements.txt
 CREATE: apps/voice-agent/config.py (Settings with pydantic-settings)
 CREATE: apps/voice-agent/logger.py (structlog, JSON output)
 CREATE: apps/voice-agent/db.py (asyncpg with all query functions)
 CREATE: apps/voice-agent/memory.py (4-layer memory: Redis + asyncpg + pgvector)
 CREATE: apps/voice-agent/prompts.py (Dynamic system prompt builder)
 CREATE: apps/voice-agent/tools.py (CRM tool functions + vertical tool registry)
 CREATE: apps/voice-agent/pipeline.py (Intelligent hybrid router + AgentCoreVoiceAgent)

```
CREATE: apps/voice-agent/agent.py (Main entrypoint - LiveKit agent server)
CREATE: apps/voice-agent/post_call.py (Post-call processing - 8 steps)
CREATE: apps/voice-agent/Dockerfile
CREATE: apps/voice-agent/.env.example
```

All files must be complete, production-ready, with full error handling.

VALIDATION:

```
python -m py_compile agent.py pipeline.py tools.py memory.py db.py post_call.py prompt
```

All must compile with zero errors.

13.5 Phase 3 -- Inbound Call Dispatch API (Next.js)

CONTEXT: Adding voice webhook routes to apps/api/.

This receives calls from Exotel and dispatches to LiveKit.

TASK: Add these routes. Every file complete.

```
CREATE: apps/api/src/routes/telephony/inbound.ts (Exotel handler)
CREATE: apps/api/src/routes/webhooks/livekit.ts (LiveKit webhook)
CREATE: apps/api/src/routes/internal/quote/trigger.ts (Post-call quote)
CREATE: apps/api/src/routes/internal/proactive/schedule.ts (BullMQ scheduling)
CREATE: apps/api/src/routes/voice/config.ts (Tenant voice config CRUD)
CREATE: apps/api/src/middleware/livekit-auth.ts (Webhook signature verification)
```

VALIDATION:

```
cd apps/api && npx tsc --noEmit
```

Zero type errors required.

13.6 Phase 4 -- WhatsApp Engine

CONTEXT: Building WhatsApp integration.

Location: packages/agentcore-whatsapp/

```
CREATE: packages/agentcore-whatsapp/src/client.ts (Meta Cloud API client)
CREATE: packages/agentcore-whatsapp/src/templates.ts (Quote, notification, NPS templates)
CREATE: packages/agentcore-whatsapp/src/webhook-handler.ts (Inbound message processor)
CREATE: packages/agentcore-whatsapp/src/payment.ts (Razorpay/Stripe link generation)
CREATE: packages/agentcore-whatsapp/src/index.ts (Barrel exports)
CREATE: packages/agentcore-whatsapp/package.json
```

VALIDATION:

```
cd packages/agentcore-whatsapp && npx tsc --noEmit
```

Zero errors.

13.7 Phase 5 -- Memory Service & Lead Engine

```
CONTEXT: Building CRM intelligence layer.
Location: packages/agentcore-memory/ and packages/agentcore-lead/

CREATE: packages/agentcore-memory/src/memory-service.ts
(4-layer memory: preloadContext, writeInteraction, semanticSearch, recallForCall)

CREATE: packages/agentcore-lead/src/lead-engine.ts
(createOrUpdate, calculateScore, updateStage, deduplicate, assignToAgent)

VALIDATION:
cd packages/agentcore-memory && npx tsc --noEmit
cd packages/agentcore-lead && npx tsc --noEmit
Zero errors.
```

13.8 Phase 6 -- Proactive Outbound Engine

```
CONTEXT: Building proactive AI.
Location: packages/agentcore-proactive/

CREATE: packages/agentcore-proactive/src/scheduler.ts
(BullMQ queues: proactive-messages, disruption-checks, price-alerts)
(All job processors: followup_4hr, re_engage_7d, pre_trip_7d, post_trip_nps, etc.)

CREATE: packages/agentcore-proactive/src/price-monitor.ts
(Price drop detection, WhatsApp alerts)

CREATE: packages/agentcore-proactive/src/disruption-monitor.ts
(Flight monitoring, rebooking options, urgent case creation)

VALIDATION:
cd packages/agentcore-proactive && npx tsc --noEmit
```

13.9 Phase 7 -- Real-Time Agent Dashboard (Next.js)

```
CONTEXT: Live agent dashboard.
Location: apps/voice-dashboard/

CREATE: apps/voice-dashboard/src/components/ActiveCallPanel.tsx
(Real-time transcript, auto-filling form, AI co-pilot sidebar)

CREATE: apps/voice-dashboard/src/components/IncomingCallNotification.tsx
(Screen-pop with customer context)

CREATE: apps/voice-dashboard/src/hooks/useCallStream.ts
(Supabase Realtime subscription hook)

VALIDATION:
```

```
cd apps/voice-dashboard && npx tsc --noEmit
cd apps/voice-dashboard && npm run build
```

13.10 Phase 8 -- Document Intelligence

CONTEXT: Passport OCR, visa checking, document collection.
Location: packages/agentcore-document/

CREATE: packages/agentcore-document/src/ocr-service.ts
(AZAPI.ai passport OCR, validation, S3 storage)

CREATE: packages/agentcore-document/src/visa-checker.ts
(VisaHQ/Sherpa API, Redis caching, WhatsApp formatting)

CREATE: packages/agentcore-document/src/collection-flow.ts
(WhatsApp document collection, upload handling, routing)

VALIDATION:
cd packages/agentcore-document && npx tsc --noEmit

13.11 Phase 9 -- AWS Infrastructure as Code

CONTEXT: Deploying all services to AWS.

CREATE: infrastructure/voice/livekit-server.tf
(EC2 c6i.xlarge, security group, user data, Elastic IP)

CREATE: infrastructure/voice/livekit-config.yaml
(Complete LiveKit server configuration)

CREATE: infrastructure/voice/ecs-voice-agents.tf
(ECS EC2 cluster, auto-scaling, 600s stop timeout)

CREATE: infrastructure/voice/security-groups.tf
(UDP 50000-60000 CRITICAL for WebRTC audio)

CREATE: infrastructure/voice/ecs-task-definition.json
(Complete task definition for voice worker)

CREATE: scripts/deploy-voice.sh
(Build, push to ECR, update ECS, verify)

VALIDATION:
terraform validate infrastructure/voice/ OR npx cdk synth
docker build -t agentcore-voice-test apps/voice-agent/

13.12 Phase 10 -- Integration & End-to-End Testing

CONTEXT: Final integration testing.

CREATE: tests/voice/e2e/full-call-flow.test.ts

SCENARIO 1: New customer, travel enquiry, full AI flow

SCENARIO 2: Returning customer with context

SCENARIO 3: Escalation trigger

SCENARIO 4: WhatsApp inbound handling

CREATE: tests/voice/integration/memory.test.ts

CREATE: tests/voice/integration/lead-engine.test.ts

CREATE: tests/voice/integration/whatsapp.test.ts

VALIDATION CHECKLIST:

- [] npx vitest run tests/voice/ -- all pass
- [] TypeScript: zero errors across all packages
- [] Docker build: voice agent builds successfully
- [] Local: LiveKit server starts, agent connects
- [] Exotel test call: end-to-end works
- [] WhatsApp: test message sent and received
- [] Lead: created in DB after test call
- [] Memory: episodic written, RAG returns relevant results

13.13 Phase 11 -- Proposal Engine [NEW]

CONTEXT: Building the Proposal Engine -- the most important V2 addition.

This engine fills the gap between lead qualification and quote generation.

Location: packages/agentcore-proposal/

TASK: Create the complete Proposal Engine. Every file must be complete and working.

CREATE: packages/agentcore-proposal/src/proposal-engine.ts

[Complete Proposal Engine service:

```
class ProposalEngine {
  constructor(
    private db: PrismaClient,
    private openai: OpenAI,
    private whatsapp: WhatsAppClient,
    private storage: S3Client,
    private memory: MemoryService
  )

  async generateProposal(tenantId: string, leadId: string): Promise<Proposal>
    - Load lead data from DB
    - Load customer memory context
    - Load tenant catalog (supplier data, templates)
    - Call AI generator (see ai-generator.ts)
    - Generate shareable link (unique cuid-based URL)
    - Generate PDF version
    - Store in DB with status=draft
    - Return complete proposal object
```

```

async sendProposal(proposalId: string, channels: string[]): Promise<void>
  - Load proposal from DB
  - If 'whatsapp': send shareable link via WhatsApp template
  - If 'email': send PDF attachment via email service
  - Update status to 'sent'
  - Schedule follow-up if not opened in 24h (via ProactiveEngine)
  - Write to Interaction timeline
  - Log to AgentAuditLog

async trackOpen(proposalId: string, metadata: TrackingData): Promise<void>
  - Update firstOpenedAt if first time
  - Update lastViewedAt
  - Increment totalViewTimeMs
  - Update sectionsViewed
  - If first open: notify assigned agent via dashboard

async handleModification(proposalId: string, modification: ModificationRequest): Promise<void>
  - Create new version (increment version number)
  - Apply modification to relevant section
  - Re-generate affected sections via AI
  - Store new version
  - Notify agent of customer modification
  - Return updated proposal

async approveProposal(proposalId: string, method: string): Promise<void>
  - Validate proposal is in approvable state (sent|viewed|modified)
  - Update status to 'approved'
  - Set approvedAt and approvedByMethod
  - Trigger Quote Engine to generate formal quote from proposal
  - Notify agent
  - Write to Interaction timeline

async rejectProposal(proposalId: string, reason: string): Promise<void>
  - Update status to 'rejected'
  - Store rejection reason
  - Notify agent
  - Create follow-up task

async getTimeline(proposalId: string): Promise<TimelineEvent[]>
  - Return chronological list of all events for this proposal
  - Created, sent, opened, viewed sections, modified, approved/rejected

Full TypeScript types, Zod validation, error handling, logging
}]

```

CREATE: packages/agentcore-proposal/src/ai-generator.ts
 [GPT-4.1 prompt for proposal generation:

```

class ProposalAIGenerator {
  async generateProposal(input: ProposalInput): Promise<GeneratedProposal>

```

The GPT-4.1 prompt must generate:

- a) coverTitle: personalized headline
- b) personalGreeting: warm, specific to customer context
- c) destinationStory: 2-3 paragraphs, why this destination

- d) itineraryNarrative: day-by-day story format (from Itinerary Builder data)
- e) packageOptions: 2-3 tiers with names, highlights, pricing
- f) inclusions/exclusions
- g) testimonials selection (from tenant's review database)
- h) agentNote: personal recommendation
- i) pricingSection: linked to Quote Engine data
- j) nextSteps: clear CTA with expiry

The prompt must be:

- Dynamic based on vertical (travel vs photography vs expense)
- Personalized with customer name, preferences, past history
- Written in the customer's preferred language
- Branded with tenant's tone of voice

Full implementation with error handling, retry logic, token tracking

}}

CREATE: packages/agentcore-proposal/src/pdf-renderer.ts

[PDF generation for proposals:

```
class ProposalPDFRenderer {
  async render(proposal: Proposal, tenantBranding: TenantBranding): Promise<Buffer>
```

Use: puppeteer or @react-pdf/renderer

Generate: A4 PDF with:

- Cover page with hero image and title
- Personal greeting section
- Destination story with images
- Day-by-day itinerary with icons
- Package comparison table
- Pricing section
- Testimonials
- Agent note
- Footer with agency branding and contact
- Page numbers

Full implementation, proper fonts, image handling

}}

CREATE: packages/agentcore-proposal/src/web-renderer.ts

[Shareable web page for proposals:

```
class ProposalWebRenderer {
  async renderHTML(proposal: Proposal, tenantBranding: TenantBranding): Promise<string>
```

Generate: responsive HTML page that:

- Works on mobile (WhatsApp opens this)
- Shows all proposal sections
- Has interactive elements: [Approve] [Modify Day X] [Select Package] buttons
- Includes tracking pixel for open detection
- Has scroll tracking for section views
- Customer can add inline comments
- Has sticky CTA bar at bottom
- Loads fast (inline critical CSS)

Full implementation with HTML/CSS

```
}}
```

CREATE: packages/agentcore-proposal/src/tracking.ts

[Proposal tracking service:

```
class ProposalTracker {
  async recordOpen(proposalId: string, ip: string, userAgent: string): Promise<void>
  async recordSectionView(proposalId: string, section: string, durationMs: number): P
  async recordTimeOnPage(proposalId: string, totalMs: number): Promise<void>
  async getTrackingData(proposalId: string): Promise<TrackingReport>

  Implementation:
  - 1x1 pixel tracking image for email opens
  - JavaScript beacon for web page tracking
  - Aggregate section-level engagement data
  - Return summary: total views, time spent, most viewed section, engagement score
}
```

CREATE: apps/api/src/routes/proposals/index.ts

[All Proposal API routes:

```
POST  /api/proposals/generate
GET    /api/proposals/:id
PUT    /api/proposals/:id
POST   /api/proposals/:id/send
GET    /api/proposals/:id/preview
GET    /api/proposals/:id/pdf
POST   /api/proposals/:id/version
GET    /api/proposals/:id/tracking
POST   /api/proposals/:id/approve
POST   /api/proposals/:id/modify
POST   /api/proposals/:id/reject
GET    /api/proposals/:id/timeline
```

Each route: full Zod validation, auth middleware, error handling, logging

Public routes (no auth): preview (via shareableLink token), approve, modify, reject

Protected routes: generate, send, tracking, timeline

]

DB MODEL (Full Prisma schema -- add to packages/db/prisma/schema.prisma):

```
model Proposal {
  id          String    @id @default(cuid())
  tenantId    String
  leadId      String
  customerId  String?
  status      String    @default("draft")
  version     Int       @default(1)
  coverTitle  String?
  personalGreeting String? @db.Text
  destinationStory String? @db.Text
  itineraryNarrative Json?
  packageOptions Json?
  selectedPackage String?
  pricingData  Json?
  mediaUrls    Json?
```

```

testimonials      Json?
agentNote         String?   @db.Text
expiresAt         DateTime?
shareableLink     String?   @unique
pdfUrl           String?
firstOpenedAt     DateTime?
lastViewedAt      DateTime?
totalViewTimeMs   Int?
sectionsViewed    Json?
modifications     Json?
customerComments  String?   @db.Text
approvedAt        DateTime?
approvedByMethod  String?
aiGeneratedBy     String?
aiPromptVersion   String?
generationTimeMs  Int?
createdAt         DateTime  @default(now())
updatedAt         DateTime  @updatedAt

@@index([tenantId, leadId])
@@index([shareableLink])
@@index([status, tenantId])
}

```

VALIDATION:

```

cd packages/agentcore-proposal && npx tsc --noEmit
cd packages/db && npx prisma validate
Zero errors.

```

13.14 Phase 12 -- Advanced Chat Agent (Live Agent + Callback)

The full Phase 12 implementation prompt and architecture is documented in the embedded section above: **AgentCore — Advanced Chat Agent Architecture** (lines ~1720-3897). That section contains the complete Claude implementation prompt covering all 4 live agent scenarios, callback queue system, cobrowse integration, multilingual interpreter, and all supporting database schemas, TypeScript code, and acceptance criteria. Reference that section directly when running Phase 12 in Claude/Cursor.

13.15 Master Environment Variables (.env.example)

```

# LiveKit
LIVEKIT_URL=wss://livekit.internal.yourdomain.com
LIVEKIT_API_KEY=
LIVEKIT_API_SECRET=

# AI Providers
DEEPGRAM_API_KEY=
OPENAI_API_KEY=
ELEVENLABS_API_KEY=
CARTESIA_API_KEY=
SARVAM_API_KEY=

```

```
# Telephony
EXOTEL_SID=
EXOTEL_TOKEN=
EXOTEL_VIRTUAL_NUMBER=
TWILIO_ACCOUNT_SID=
TWILIO_AUTH_TOKEN=

# WhatsApp
META_WA_ACCESS_TOKEN=
META_WA_PHONE_NUMBER_ID=
META_WA_VERIFY_TOKEN=
META_WA_APP_SECRET=

# Database
DATABASE_URL=postgresql://...
REDIS_URL=redis://...

# Internal Service
AGENTCORE_API_URL=https://api.yourdomain.com
AGENTCORE_INTERNAL_SECRET=

# Document Services
AZAPI_OCR_KEY=
VISAHQ_API_KEY=

# Flight Monitoring
FLIGHTAWARE_API_KEY=

# AWS
AWS_REGION=ap-south-1
AWS_ECR_REGISTRY=
S3_BUCKET_RECORDINGS=
S3_BUCKET_DOCUMENTS=

# Supabase (for realtime)
SUPABASE_URL=
SUPABASE_ANON_KEY=
SUPABASE_SERVICE_ROLE_KEY=

# Payment
RAZORPAY_KEY_ID=
RAZORPAY_KEY_SECRET=
STRIPE_SECRET_KEY=

# Monitoring
SENTRY_DSN=
```

13.16 File Tree of All New Files

```
helprevx-monorepo/
|-- apps/
|   |-- voice-agent/          PYTHON SERVICE
|   |   |-- agent.py
|   |   |-- pipeline.py
|   |   |-- tools.py
|   |   |-- memory.py
|   |   |-- db.py
|   |   |-- post_call.py
|   |   |-- prompts.py
|   |   |-- config.py
|   |   |-- logger.py
|   |   |-- Dockerfile
|   |   |-- requirements.txt
|   |   --- .env.example
|   |
|   |-- api/ (EXISTING -- ADD THESE):
|   |   |-- src/routes/telephony/inbound.ts
|   |   |-- src/routes/webhooks/livekit.ts
|   |   |-- src/routes/webhooks/whatsapp.ts
|   |   |-- src/routes/internal/quote/trigger.ts
|   |   |-- src/routes/internal/proactive/schedule.ts
|   |   |-- src/routes/voice/config.ts
|   |   |-- src/routes/proposals/index.ts      [NEW - Phase 11]
|   |   --- src/middleware/livekit-auth.ts
|   |
|   --- voice-dashboard/      NEW NEXT.JS APP
|       |-- src/components/ActiveCallPanel.tsx
|       |-- src/components/IncomingCallNotification.tsx
|       --- src/hooks/useCallStream.ts
|
|-- packages/
|   |-- db/ (EXISTING -- ADD MODELS):
|   |   |-- prisma/schema.prisma (add models including Proposal)
|   |   --- prisma/migrations/add_pgvector.sql
|   |
|   |-- agentcore-whatsapp/    NEW PACKAGE
|   |   --- src/client.ts, templates.ts, webhook-handler.ts, payment.ts
|   |
|   |-- agentcore-memory/      NEW PACKAGE
|   |   --- src/memory-service.ts
|   |
|   |-- agentcore-lead/        NEW PACKAGE
|   |   --- src/lead-engine.ts
|   |
|   |-- agentcore-proactive/    NEW PACKAGE
|   |   --- src/scheduler.ts, price-monitor.ts, disruption-monitor.ts
|   |
|   |-- agentcore-document/     NEW PACKAGE
|   |   --- src/ocr-service.ts, visa-checker.ts, collection-flow.ts
|   |
|   --- agentcore-proposal/     NEW PACKAGE [Phase 11]
|       --- src/proposal-engine.ts, ai-generator.ts, pdf-renderer.ts,
|           web-renderer.ts, tracking.ts
```

```

|
|-- infrastructure/
|   --- voice/
|       |-- livekit-server.tf
|       |-- livekit-config.yaml
|       |-- ecs-voice-agents.tf
|       |-- security-groups.tf
|       --- ecs-task-definition.json
|
--- tests/voice/
    |-- e2e/full-call-flow.test.ts
    |-- integration/memory.test.ts
    |-- integration/lead-engine.test.ts
    --- integration/whatsapp.test.ts

```

Section 14: Enterprise-Grade Implementation Phases (Addendum)

Phase 13: Enterprise Compliance & WEM Implementation

This phase covers all enterprise-grade features identified in the competitor analysis — SOC 2 alignment, workforce engagement management, supervisor tools, A/B testing, predictive routing, and knowledge base management. This phase runs after Phase 12 (Advanced Chat Agent) and targets enterprise customer acquisition.

Phase 13.0 — Vanta Security Compliance Automation

Objective: Set up automated SOC 2 evidence collection from Day 1 to reduce audit preparation burden and enable SOC 2 Type I in Month 9, SOC 2 Type II in Month 18.

PHASE 13.0 – VANTA INTEGRATION FOR SOC 2 COMPLIANCE AUTOMATION

You are implementing automated compliance evidence collection for SOC 2 Type II certification.

CONTEXT: AgentCore is on AWS. The SOC 2 audit covers: Security, Availability, Confidentiality.

REQUIRED INTEGRATIONS:

1. AWS integration: Vanta → CloudTrail (all API calls), IAM access reviews, S3 bucket pol
2. GitHub integration: Vanta → all commits, PRs, branch protection rules, required code
3. Application layer: implement these controls in AgentCore codebase:
 - a. MFA enforcement for all admin users (AWS Cognito or Auth0 MFA)
 - b. Role-based access control with least-privilege principle
 - c. Automated offboarding checklist (when employee leaves → access revoked within 24 h
 - d. Encryption key rotation (AWS KMS key rotation every 90 days)
 - e. Vulnerability scanning (AWS Inspector for EC2, ECR Image scanning for containers)
 - f. Dependency scanning (npm audit + snyk in GitHub Actions CI/CD)
 - g. SIEM integration: CloudWatch Logs → security event alerting

DELIVERABLES:

File: /infrastructure/compliance/vanta-controls-map.json – mapping of each SOC 2 control
File: /infrastructure/compliance/soc2-readiness-checklist.md – checklist of all controls
File: /src/middleware/compliance-logging.ts – application-level security event logging
File: /src/auth/mfa-enforcement.ts – MFA requirement enforcement for admin and supervisor

ACCEPTANCE CRITERIA:

- Vanta dashboard shows >85% controls passing (green) with no critical findings
- All AWS resources encrypted at rest (RDS, S3, EBS)
- All admin actions logged with user identity, timestamp, action, affected resource
- Zero open GitHub branch protection rule violations
- MFA enabled for all admin users
- Automated access review scheduled monthly

Phase 13.1 — HIPAA BAA Implementation

Objective: Build HIPAA-compliant tenant infrastructure and PHI handling procedures.

PHASE 13.1 – HIPAA COMPLIANT TENANT INFRASTRUCTURE

You are implementing HIPAA-compliant infrastructure for healthcare tenant deployments in

CONTEXT: Healthcare tenants (hospitals, clinics, diagnostic labs, telemedicine) handle P

HIPAA REQUIRED TECHNICAL SAFEGUARDS TO IMPLEMENT:

1. DEDICATED INFRASTRUCTURE: Healthcare tenants get isolated infrastructure
 - Separate RDS PostgreSQL instance (not shared with non-HIPAA tenants)
 - Separate S3 bucket for voice recordings with server-side encryption (SSE-KMS)
 - Separate Redis ElastiCache cluster
 - VPC isolation: healthcare tenant in separate VPC or subnet with strict security group
 - No data co-mingling with non-HIPAA tenants
2. PHI FIELD FLAGGING: Identify and protect PHI in conversation transcripts
 - Create: /src/compliance/phi-detector.ts
 - Detect PHI patterns in transcript: patient name, DOB, diagnosis, medication, insurance
 - Apply field-level encryption to PHI fields before storage (AES-256, KMS-managed key)
 - Minimum necessary principle: agents only receive PHI fields required for the current call
 - Example: appointment booking agent receives name + DOB + appointment history, NOT full transcript
3. AUDIT LOGGING FOR PHI ACCESS:
 - Every access to PHI data logged: who (agent/human/supervisor), what PHI field, at what time
 - Audit log is append-only, encrypted, retained for 7 years minimum
 - Automated PHI access report generated monthly for compliance officer
4. CALL RECORDING REDACTION:
 - Patient name, DOB, insurance ID, phone number automatically redacted from stored transcripts
 - Redacted format: "Patient [NAME REDACTED] called about appointment on [DOB REDACTED]"
 - Original encrypted recording retained for 7 years (per HIPAA retention)
 - Redacted transcript used for analytics, QA, and search
5. CONSENT MANAGEMENT:
 - Explicit voice consent required before recording: "This call may be recorded for quality improvement"
 - Consent event logged in audit trail with timestamp and call ID

- Consent withdrawal: patient can call back and withdraw consent → recordings deleted

6. BREACH NOTIFICATION:

- Automated detection of unauthorized PHI access (unusual query patterns, after-hours)
- Alert to HIPAA Security Officer within 1 hour of detected potential breach
- 72-hour notification to HHS (US) if breach confirmed (for international deployments)
- Breach log maintained with all incidents, investigations, and outcomes

DELIVERABLES:

File: /src/compliance/hipaa/phi-detector.ts – PHI detection and field-level encryption

File: /src/compliance/hipaa/hipaa-audit-logger.ts – PHI access audit logging

File: /src/compliance/hipaa/transcript-redactor.ts – automatic PHI redaction from transcripts

File: /src/compliance/hipaa/consent-manager.ts – voice consent collection and logging

File: /infrastructure/hipaa/hipaa-tenant-terraform.tf – dedicated HIPAA infrastructure

File: /docs/compliance/hipaa-baa-template.md – HIPAA Business Associate Agreement template

File: /docs/compliance/hipaa-implementation-guide.md – guide for customers deploying on AWS

ACCEPTANCE CRITERIA:

- PHI fields in transcript are field-encrypted at rest
- Every PHI access generates an audit log entry
- Redacted transcripts do not contain raw PHI
- Healthcare tenant infrastructure is in isolated VPC subnet
- Voice consent captured and logged for 100% of healthcare tenant calls
- Breach detection alerts fire within 1 minute of simulated unauthorized access

Phase 13.2 — PCI-DSS Compliance for Payment Flows

Objective: Ensure AgentCore never stores raw card data and all payment collection flows are PCI-DSS compliant.

PHASE 13.2 – PCI-DSS COMPLIANCE FOR VOICE PAYMENT FLOWS

You are implementing PCI-DSS compliant payment handling in AgentCore's voice and chat agents.

CONTEXT: BFSI, retail/e-commerce, hospitality, and travel tenants may need to collect payment information.

PCI-DSS RULES FOR VOICE AI:

1. NEVER store CVV/CVV2/CVC in any system – not in transcripts, not in logs, not in memory
2. NEVER store full PAN (card number) – only last 4 digits allowed for display
3. Call recordings must be PAUSED during card number collection (DTMF tones not recorded)
4. Preferred approach: redirect card data collection to PCI-DSS Level 1 certified processor

IMPLEMENTATION APPROACH – TOKENIZATION:

When customer needs to pay by card during a voice call:

1. AI agent says: "I'll connect you to our secure payment line for card details. Please hold."
2. LiveKit call transitions to DTMF-only mode (AI voice paused, recording paused)
3. Customer enters card number via keypad – DTMF tones go directly to Razorpay IVR API
4. Razorpay returns a payment token (no raw card data ever reaches AgentCore servers)
5. AI agent resumes with payment status from Razorpay webhook

CODE TO IMPLEMENT:

File: /src/engines/payment/pci-payment-handler.ts

- initiateIVRPayment(callId, amount, currency, tenant_id): pauses recording, connects to

- `handlePaymentComplete(token, status)`: resumes recording, updates conversation context
- `CardDataFilter` class: scans all outgoing data for card number patterns before logging/
 - Regex: 16-digit sequences, card number patterns → replace with `[CARD-REDACTED]`
 - Applied to: conversation transcript storage, tool call logging, memory storage

File: `/src/middleware/pqi-transcript-scanner.ts`

- Post-processing filter applied to ALL conversation transcripts before storage
- Pattern matching: credit card numbers, CVV patterns, expiry dates
- Replacement with `[PAYMENT-DATA-REDACTED]` tokens
- Alert if card data detected in pre-storage scan (indicates upstream bug)

File: `/infrastructure/compliance/pqi-scope-reduction.md`

- Document the architecture decisions that minimize PCI scope
- Show that AgentCore qualifies for SAQ-A (redirect model) not SAQ-D (storage model)

ACCEPTANCE CRITERIA:

- Zero card numbers in stored transcripts (validated by scanning 1,000 test transcripts)
- Zero CVV in any log file, memory record, or database record
- Call recording confirmed paused during DTMF payment entry (verified via recording tin
- Razorpay IVR integration tested end-to-end with test cards
- `CardDataFilter` passes all 50 test cases (valid cards, partial cards, formatted cards)

Phase 13.3 — Workforce Engagement Management (WEM) Module

Objective: Build the agent scheduling, quality scoring, and performance management features.

PHASE 13.3 – WORKFORCE ENGAGEMENT MANAGEMENT MODULE

You are building the WEM (Workforce Engagement Management) module for AgentCore's super

CONTEXT: Enterprise contact center customers need agent scheduling, quality scoring, coa

SCOPE:

1. Agent scheduling (demand forecast + schedule generation)
2. Quality scoring (auto-QA of AI conversations + human agent conversations)
3. Performance gamification (leaderboards, badges, recognition)
4. Coaching tools (supervisor notes, coaching tasks, 1:1 templates)

--- PART A: DEMAND FORECASTING ---

File: `/src/engines/wem/demand-forecaster.ts`

Algorithm: Prophet time-series model (Facebook's open-source forecasting library)

Input: Historical call volume per 15-minute interval (from AgentCore analytics DB)

Output: Forecasted call volume for next 7 days per 15-minute interval, per channel (voice

Features:

- Seasonal decomposition (weekly patterns, monthly patterns, holiday detection)
- Campaign injection: if a marketing campaign is scheduled, add estimated volume increa
- Confidence intervals: upper/lower bounds at 80% and 95% confidence
- Accuracy tracking: compare forecast vs actual, track MAPE (Mean Absolute Percentage Err

--- PART B: QUALITY SCORECARD ENGINE ---

File: `/src/engines/wem/qa-scorecard.ts`

Scorecard schema:

```

{
  scorecard_id: string,
  tenant_id: string,
  name: string,
  criteria: [
    {
      criterion_id: string,
      name: "Did agent greet customer professionally?",
      weight: 10, // points
      evaluation_method: "llm_auto" | "human_review",
      llm_prompt: "Did the agent greet the customer professionally? Score 1 for yes, 0 for no",
      auto_pass_patterns: ["hello", "good morning", "welcome"], // if any match, auto-pass
      auto_fail_patterns: ["shut up", "I don't know"] // if any match, auto-fail
    }
  ]
}

```

Auto-scoring flow:

1. Call ends → transcript + tool call log stored
2. Post-call evaluation queue receives job
3. For each criterion: LLM evaluator (Claude Haiku for cost efficiency) evaluates transcript
4. Score aggregated → stored as QAScore record
5. If score < tenant's quality threshold → flag for supervisor review

--- PART C: GAMIFICATION ---

File: /src/engines/wem/gamification.ts

Badge definitions (configurable per tenant):

- "Perfect Week": 5-day streak with quality score $\geq 90\%$
- "Speed Champion": handle 50 calls in one day with AHT < 4 minutes
- "Customer Hero": receive 10 CSAT 5-star ratings in one week
- "Escalation Avoidance Expert": 7-day streak with escalation rate < 10%
- "Polyglot": handle calls in 3+ languages in one day

Leaderboard engine:

- Daily, weekly, monthly leaderboards
- Configurable metrics: QA score, CSAT, calls handled, escalation rate
- Visibility: public (team sees) or private (individual + manager only)
- Configurable by tenant admin

--- PART D: SUPERVISOR DASHBOARD ADDITIONS ---

Extend /src/pages/supervisor/ with:

- /supervisor/workforce/forecast – demand forecast visualization (Chart.js)
- /supervisor/workforce/schedule – drag-and-drop agent schedule builder
- /supervisor/qa/queue – list of conversations awaiting QA review
- /supervisor/qa/scoring – QA scorecard review UI
- /supervisor/agents/performance – individual agent performance dashboard
- /supervisor/agents/leaderboard – team leaderboard with badges

ACCEPTANCE CRITERIA:

- Demand forecast MAPE < 15% on 30-day backtesting dataset
- QA auto-scoring agreement rate with human raters $\geq 85\%$ on test set of 100 conversations
- Schedule generation produces valid schedule in <5 seconds for 50-agent team
- Badge awards fire within 60 seconds of achieving qualifying action
- Leaderboard updates within 5 minutes of new data

Phase 13.4 — Conversation A/B Testing Framework

PHASE 13.4 – CONVERSATION A/B TESTING ENGINE

You are building the A/B testing framework for AgentCore conversation optimization.

CONTEXT: Teams need to test variations in AI prompts, greeting scripts, escalation thresholds

IMPLEMENTATION:

File: /src/engines/experiments/experiment-manager.ts

Core data model:

```
interface Experiment {
  experiment_id: string;
  tenant_id: string;
  name: string;
  status: 'draft' | 'running' | 'paused' | 'completed' | 'winner_declared';

  variants: ExperimentVariant[]; // Must sum to 100% traffic

  // Targeting
  target_channels: ('voice' | 'chat' | 'whatsapp')[];
  target_call_types: string[]; // e.g., ['inbound', 'outbound']
  target_industries: string[];

  // Success criteria
  primary_metric: 'containment_rate' | 'csat' | 'aht' | 'conversion_rate';
  minimum_sample_size: number; // Per variant before stat test
  significance_level: number; // Default 0.95

  // Dates
  started_at: Date;
  completed_at?: Date;
  winner_variant_id?: string;
}

interface ExperimentVariant {
  variant_id: string;
  name: string;
  traffic_weight: number; // 0.0 to 1.0, all variants must sum to 1.0

  // Overrides (only specified overrides apply; others use tenant default)
  system_prompt_override?: string;
  greeting_template_override?: string;
  escalation_sensitivity_override?: number; // 0-1 scale
  tools_enabled_override?: string[];
  llm_model_override?: string;
}
```

Variant assignment:

- When a call starts, check if tenant has any running experiments
- Apply targeting rules to check if this call qualifies
- $\text{Hash}(\text{customer_id} + \text{experiment_id}) \% 100 \rightarrow$ deterministic assignment (same customer always)
- Record assignment in conversation metadata

Statistical significance engine:

- For containment_rate (binary): Two-proportion z-test
- For CSAT (1-5 scale): Welch's t-test
- For AHT (continuous): Mann-Whitney U test (non-parametric, handles skew)
- Minimum detectable effect size: 5% relative lift (configurable)
- Auto-stop and declare winner when $p < (1 - \text{significance_level})$ and sample size met
- Alert if no winner after 2x minimum sample size (experiment may be inconclusive)

Dashboard pages:

File: /src/pages/experiments/

- /experiments/list – all experiments with status and top-line results
- /experiments/new – create experiment with variant configuration UI
- /experiments/[id]/results – detailed results: metric graphs, significance, sample size

ACCEPTANCE CRITERIA:

- Variant assignment is deterministic (same customer always same variant for same experiment)
- Statistical significance test produces correct p-value on 10 test datasets
- Winner declaration fires when $p < 0.05$ and minimum sample size reached
- Experiment results dashboard loads in <2 seconds
- Experiment creation form validates that variant weights sum to 1.0

Phase 13.5 — Predictive Routing Engine

PHASE 13.5 – PREDICTIVE ROUTING ENGINE FOR HUMAN AGENT ESCALATIONS

You are building the ML-based predictive routing system for AgentCore.

CONTEXT: When the AI agent decides to escalate a call to a human agent, the routing decision is made.

IMPLEMENTATION:

File: /src/engines/routing/predictive-router.ts

FEATURES TO EXTRACT PER ESCALATION:

Customer features:

- customer_sentiment_score: float (0-1, from real-time sentiment analysis)
- customer_tier: string ('standard' | 'premium' | 'vip')
- customer_language: string (detected language code)
- call_intent_category: string (e.g., 'billing_dispute', 'technical_support', 'sales_inquiry')
- prior_escalation_count: int (how many times this customer has escalated in last 90 days)
- prior_resolution_rate: float (fraction of prior calls resolved on first contact)
- call_duration_so_far: int (seconds)
- ai_confidence_score: float (AI's confidence in escalation decision)

Agent features (per available agent):

- agent_id: string
- agent_skill_tags: string[] (e.g., ['billing', 'tier1_support', 'hindi', 'sales'])
- agent_skill_match_score: float (cosine similarity between call_intent and agent_skills)
- agent_current_queue_depth: int (how many calls waiting for this agent)
- agent_avg_handle_time: float (rolling 7-day average)
- agent_csat_score: float (rolling 7-day average CSAT)
- agent_first_contact_resolution: float (rolling 7-day FCR rate)
- agent_language_match: bool (agent speaks customer's language)
- agent_seniority: string ('junior' | 'mid' | 'senior')

- prior_interaction_with_customer: bool (has this agent handled this customer before?)
- prior_interaction_outcome: float | null (CSAT from prior interaction if exists)

MODEL:

- Algorithm: Gradient Boosting (XGBoost or LightGBM)
- Training data: historical escalations with known outcomes (CSAT, FCR, AHT)
- Label: CSAT score from post-call survey
- Training cycle: retrain weekly on new data (MLflow experiment tracking)
- Cold start: before enough data for ML model, use weighted rule-based routing
 - language_match: 40% weight
 - skill_match_score: 30% weight
 - queue_depth (inverted): 20% weight
 - agent_seniority (for VIP customers): 10% weight
- Transition to ML model when: 1,000+ historical escalations per tenant

FALLBACK ROUTING:

If predictive model unavailable (cold start, model error): use rule-based fallback

Priority order:

1. Language match (must match if non-English call)
2. VIP customer → senior agent
3. Prior interaction with same agent → prefer continuity
4. Skill match score
5. Queue depth (least loaded agent)

IMPLEMENTATION FILES:

```
/src/engines/routing/predictive-router.ts – main routing class
/src/engines/routing/feature-extractor.ts – extract features from conversation context
/src/engines/routing/rule-based-router.ts – fallback rule-based router
/ml/routing/train-routing-model.py – training script (runs on SageMaker or local GPU)
/ml/routing/model-evaluation.py – evaluation metrics: NDCG, CSAT correlation
```

ACCEPTANCE CRITERIA:

- Predictive router returns ranked agent list in <50ms
- Rule-based fallback works correctly when ML model unavailable
- Features extracted correctly for 100 test conversations
- Model achieves CSAT correlation ≥ 0.4 on held-out test set (better than random routing)
- Language match constraint respected 100% of the time (never route non-English call to E

Phase 13.6 — Knowledge Base Management Engine

PHASE 13.6 – KNOWLEDGE BASE MANAGEMENT ENGINE

You are building the Knowledge Base (KB) engine for AgentCore – the system that creates,

CONTEXT: AI agents need a managed, searchable knowledge repository to answer customer q

CORE ARCHITECTURE:

File: /src/engines/knowledge/kb-article-manager.ts

KBArticle data model:

```
{
  article_id: uuid,
  tenant_id: uuid,
```

```

title: string,
content: string, // Markdown
category: string,
subcategory?: string,
tags: string[],
industry: string,

// Versioning
version: integer, // Auto-increments on edit
status: 'draft' | 'review' | 'published' | 'archived',
published_at: timestamp,
last_reviewed_at: timestamp,
review_due_at: timestamp, // published_at + review_interval (default 90 days)
reviewed_by: uuid,

// AI metadata
embedding: vector(1536), // pgvector embedding (text-embedding-3-small)
auto_summary: string, // Claude-generated 2-sentence summary
search_keywords: string[], // Extracted by Claude for hybrid search

// Performance
retrieval_count: integer,
helpful_count: integer,
unhelpful_count: integer,
helpfulness_rate: float, // helpful / (helpful + unhelpful), min 5 votes

// Source tracking
source_url?: string, // If imported from existing help center
source_type: 'manual' | 'imported' | 'ai-generated' | 'agent_flagged'
}

```

SEARCH IMPLEMENTATION (Hybrid BM25 + Semantic):

File: /src/engines/knowledge/kb-search.ts

// Stage 1: BM25 keyword search (fast, lexical)

```

const bm25Results = await pg.query(`
  SELECT article_id, ts_rank(search_vector, query) as rank
  FROM kb_articles
  WHERE tenant_id = $1 AND status = 'published'
  AND search_vector @@ plainto_tsquery('english', $2)
  ORDER BY rank DESC LIMIT 20
`, [tenant_id, query]);

```

// Stage 2: Semantic search (pgvector)

```

const queryEmbedding = await openai.embeddings.create({ input: query, model: 'text-emb
const semanticResults = await pg.query(`
  SELECT article_id, 1 - (embedding <=> $1) as similarity
  FROM kb_articles
  WHERE tenant_id = $2 AND status = 'published'
  ORDER BY similarity DESC LIMIT 20
`, [queryEmbedding, tenant_id]);

```

// Stage 3: Reciprocal Rank Fusion (RRF) – merge BM25 + semantic results

```

const merged = reciprocalRankFusion([bm25Results, semanticResults], k=60);
return merged.slice(0, 5); // Top 5 articles

```

GAP DETECTION ENGINE:

File: /src/engines/knowledge/kb-gap-detector.ts

Logic:

1. Track all AI agent queries to KB where no article was retrieved (empty results)
2. Track all retrievals where customer explicitly said "that doesn't help" or escalated
3. Cluster unanswered queries by semantic similarity (identify themes)
4. Weekly report: "Top 20 content gaps" – questions customers asked with no KB article

Gap report structure:

```
{
  gap_id: uuid,
  query_cluster_theme: string, // Claude-generated theme name
  example_queries: string[], // Up to 5 examples from the cluster
  frequency: integer, // How many times this gap was hit in last 7 days
  estimated_containment_lift: float, // Predicted % improvement in containment rate if
  suggested_article_title: string, // Claude-suggested article title
  suggested_article_outline: string // Claude-suggested article outline
}
```

CONTENT MANAGEMENT UI:

File: /src/pages/knowledge/

- /knowledge/articles – article list with search, filters, performance metrics
- /knowledge/articles/new – create article with rich text editor + AI draft assistant
- /knowledge/articles/[id] – view/edit article with version history
- /knowledge/categories – category tree management
- /knowledge/gaps – gap report with article creation wizard
- /knowledge/analytics – KB performance: top articles, helpfulness trends, gap trends
- /knowledge/import – bulk import from URL (help center scraping) or CSV

AI DRAFT ASSISTANT:

"I need an article about [topic]" → Claude generates:

1. Suggested title
2. Article structure (H2/H3 outline)
3. Full draft content
4. Suggested tags and categories

Human reviews, edits, and publishes. Draft stored with source_type = 'ai_generated'.

ACCEPTANCE CRITERIA:

- Hybrid search returns relevant article in top-3 results for 90% of test queries
- Gap detection correctly identifies top 10 content gaps from test conversation dataset
- Article creation + publishing flow works end-to-end
- AI draft generation completes in <10 seconds
- Knowledge base search integrated into VoiceAgent and ChatAgent as the kb_search tool
- Article performance metrics update within 1 hour of interaction
- Content freshness alerts fire when article is 90+ days without review

Updated Phase Summary Table (All 14 Implementation Phases)

Phase	Name	Timeline	Key Deliverables
Phase 0	Codebase Audit	Week 1	Full audit, migration plan

Phase	Name	Timeline	Key Deliverables
Phase 1	Database	Week 2-3	47-table schema, migrations, RLS
Phase 2	LiveKit Voice Agent	Week 4-5	Voice agent, 4 language paths, hybrid pipeline
Phase 3	Inbound Dispatch	Week 6	Multi-tenant routing, priority queues
Phase 4	WhatsApp Commerce	Week 7-8	Flows v2, UPI payments, carousels
Phase 5	Memory + Lead	Week 9-10	4-layer memory, lead engine, dedup
Phase 6	Proactive Outbound	Week 11	Trigger engine, segments, campaigns
Phase 7	Agent Dashboard	Week 12	Supervisor UI, live queue, analytics
Phase 8	Document Intelligence	Week 13-14	OCR, verification, compliance scan
Phase 9	AWS Infrastructure	Week 15-16	Terraform, ECS, LiveKit server, Vanta
Phase 10	E2E Testing	Week 17	Full test suite, load testing, chaos
Phase 11	Proposal Engine	Week 18-19	AI proposal, tracking, approval, invoice
Phase 12	Advanced Chat Agent	Week 20-22	4 live agent scenarios, callback, cobrowse
Phase 13	Enterprise Compliance + WEM	Month 6-12	SOC 2 alignment, HIPAA BAA, WEM module, A/B testing, predictive routing, KB engine

Document Information

Field	Value
Document Title	AgentCore -- Universal Agentic AI Platform
Version	3.0
Date	March 2026
Author	HelpRevX Technologies Pvt Ltd
Classification	Confidential -- Internal & Investor Use
Total Sections	14 (Sections 1-13 + Appendix)
Total Engines	24
Industries Covered	15
Status	Final Draft -- V3 (Enterprise Audit Update)
Previous Version	V2.0 (24 engines, 15 industries, chat agent update)

Industries Covered in This Document

1. Tour & Travel (ItinerarySuite, Ytrawise)
2. Photography (PhotoAdda)
3. Expense Management (Aayiq)
4. Spend Analytics (SpendPilot)
5. Financial Services / BFSI (Insurance, Banking, Lending, Mutual Funds, Wealth Management)
6. Healthcare (Clinics, Hospitals, Diagnostics, Pharmacies, Telemedicine)
7. Real Estate (Residential, Commercial, Rental, Property Management)
8. Legal Services (Law Firms, Advocates, Corporate Legal)
9. Education & Coaching (Coaching Centers, Schools, Colleges, EdTech)
10. Fitness & Wellness (Gyms, Yoga Studios, Spas, Salons, Trainers)
11. Automotive (Dealerships, Service Centers, Fleet Management)
12. Hospitality & Food (Hotels, Restaurants, Homestays, Catering)
13. Retail & E-Commerce (D2C, Physical Stores, Marketplaces)
14. Logistics & Delivery (Courier, Freight, Last-mile)
15. Home Services (HVAC, Plumbing, Electrical, Cleaning, Pest Control)

Additional markets mentioned: Event Management, NGOs & Government

END OF DOCUMENT